

# Population Priors Rival Interaction Models in Microbiome Digital Twins: a Leave-One-Out Audit of cNODE and a Head-to-Head with MDSINE2

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MEGA-PROGRAM-27, Item 4 - Udit Phookan (program owner); computational work by an AI research agent. Revised draft of 25 September 2026.

## Abstract

We test whether microbiome digital twins learn species interactions or mainly exploit ecological priors. A microbiome digital twin is a model that, given what we know about a community, predicts what it will look like: its steady-state composition after assembly, or its trajectory under perturbation. Two published approaches are compositional neural ODEs (cNODE; Michel-Mata et al., 2022) for assemblage-to-composition prediction, and MDSINE2 (Gibson et al., Nature Microbiology 2025) for forecasting absolute abundances in gnotobiotic mice through diet and antibiotic perturbations. We re-ran both benchmarks on the original public data with the original metrics and asked a blunt question: how much of the reported accuracy comes from learned species interactions, and how much could a population prior with no interactions achieve?

Findings. (1) On the six cNODE ecosystems under leave-one-out, a presence-only null with no learned interaction parameters (mean training composition restricted to present taxa) is clearly beaten by cNODE on ocean, soil and *Drosophila* data, but on both human-associated datasets the published cNODE median Bray-Curtis error (oral 0.211, gut 0.242) falls inside the 95% bootstrap interval of the null (0.204 [0.188, 0.222] and 0.259 [0.236, 0.271]). (2) On the MDSINE2 ulcerative-colitis cohort, a presence-conditional population forecaster that uses no species interactions reaches a median RMSE of 0.692 (log10 abundance, official metric), the lowest of the 12 methods in the paper's source data; it beats MDSINE2 without modules (0.805; Wilcoxon  $p = 8.0e-07$ ) and full MDSINE2 (1.093;  $p = 4.3e-44$ ) on the same 601 subject-taxon pairs. (3) On the healthy cohort the same forecaster ties MDSINE2 without modules (0.919 vs 0.913,  $p = 0.53$ ), beats full MDSINE2 (1.061,  $p = 1.1e-4$ ), and is beaten by RA-MDSINE2 without modules (0.883,  $p = 0.045$ ). In absolute terms the UC gain is a 37% reduction in median RMSE against full MDSINE2 (1.093 to 0.692); the healthy-cohort differences are small in effect size whichever direction they go.

(4) Under a stricter metric that scores every timepoint, the same forecaster ranks first on both MDSINE2 cohorts (UC 1.654 vs MDSINE2 without modules 1.824; healthy 1.752 vs 2.072;  $p < 1e-39$ ), which falsified our own prediction that the advantage came from the detection-only metric. (5) At scale the picture changes: across 160 MGnify studies in 8 biomes, an interaction model beats a calibrated presence prior in 121 studies ( $FDR < 0.05$ ), so the human-data null of finding (1) does not generalise. We ship the microtwin command-line tool (audit and forecast) so anyone can run this audit on their own abundance table.

(6) Exploratory keystone analysis names one candidate, the anaerobe-keystone hypothesis: across 156 studies, genera made of strict anaerobes are more often network hubs within the same study (study fixed-effect logit,  $p = 1.4e-4$ ), independent of abundance and prevalence, and sulfate reducers are enriched under both NCBI and GTDB taxonomies ( $q = 0.02$ ); a KEGG genomic anaerobe index replicates it ( $p = 7e-8$ ). No single biome passes FDR and keystones come from inferred networks, but the association fails with graphical lasso and igraph betweenness keystone definitions. It is a ridge-specific candidate, not a cross-method ecological discovery. A distinct HOMD oral-list effect survives those two methods but fails a PubTator literature-index replication.

(7) A pre-registered twin-driven replication programme (GraphTwin gate out-strength trained per MGnify study; locked 09:30 completion-order stopping rule, 37 studies) found that the sulfate-reducer keystone enrichment does NOT replicate under the nonlinear gate scorer (Desulfobacterota untestable at this coverage; best phylum  $q = 0.906$ ), and the locked SHAP-attribution-centrality redirect does not recover it either ( $q = 0.462$ ). As a disease-literature predictor the twin gates do not beat the ridge score (AIC 767.1 vs 766.1), but the locked CatBoost+SHAP redirect does, decisively (coefficient +37.1,  $p = 2.6e-35$ , AIC 2343.9 vs 2443.1). What survives replication is therefore not a keystone list but a predictor: gradient-boosted interaction attributions forecast BugSigDB disease-signature counts far better than linear interaction scores.

Caveats. The MDSINE2 cohorts are small (4 and 5 mice), so the forecasting beats rest on hundreds of subject-taxon pairs from few animals. Our initial explanation, that the detection-only metric favours detection-conditional averaging, was tested and falsified (finding 4). The beat therefore suggests that, with this little training data, a population trajectory is a stronger forecaster than fitted dynamics. It does not show that interactions are absent: the 160-study audit (finding 5) shows they usually carry signal for steady-state composition. Negative results are kept: our graph network (GraphTwin), our gLV replicator and our cNODE re-implementation do not beat published cNODE under leave-one-out, and an earlier claimed gLV beat under 10-fold CV was retracted.

## 1. Introduction

Digital twins of the microbiome promise in-silico trials: remove a species, add an antibiotic, change a diet, and read out the community before touching an animal or a patient. The community therefore invests heavily in interaction-aware models: generalised Lotka-Volterra (gLV) systems, Bayesian gLV with interaction modules (MDSINE, MDSINE2), and neural ODEs (cNODE). Their value rests on a premise that is rarely tested directly: that the interaction terms carry predictive information beyond what a population average already provides.

Strong baselines matter because microbiome data are dominated by taxon identity. A taxon that is abundant in one host is usually abundant in another. A predictor that simply remembers typical abundances, conditioned on which taxa are present, may already explain most of the variance. If so, the headline accuracy of an interaction model says little about whether its interactions are right, and downstream uses such as keystone-species ranking or perturbation design rest on weaker ground than the accuracy suggests.

We test that premise on the two public benchmarks that serve as reference benchmarks for two twin tasks. Our contributions are: (i) an exact leave-one-out re-run of the cNODE benchmark with a bootstrap null interval; (ii) a verified reproduction of the MDSINE2 cross-validation metric from the authors' notebook and Source Data, matching the published MDSINE2 median; (iii) a presence-conditional population forecaster that tops the UC leaderboard and ties the healthy one; and (iv) an explicit account of why the metric allows this.

Three claim levels are kept separate throughout. Prediction asks whether a model forecasts held-out composition or trajectories. Mechanism asks whether its fitted terms recover causal ecology. Discovery asks whether it names a biological principle that survives replication. This paper reports results at each level separately and does not let a success at one level stand in for another. The novelty is the auditing framework, its scale, and the pre-registered falsification discipline - not any single model, null, or benchmark re-run.

The paper's hierarchy, stated once: the primary contribution is the digital-twin audit framework (does a twin beat ecological priors, and when). The major findings are that some benchmarks collapse to ecological priors (human cNODE datasets, the MDSINE2 UC cohort) while others require learned structure (121 of 160 MGnify studies). The secondary exploration is the biological-discovery programme, which mostly fails replication and is reported as such.

## 2. Problem statements and notation

### 2.1 Assemblage-to-composition (cNODE task)

Let  $N$  be the species pool. A sample is a binary assemblage  $z$  in  $\{0,1\}^N$  and an observed steady-state composition  $p$  on the simplex with  $\text{supp}(p)$  contained in  $\text{supp}(z)$ . A twin is a map  $\phi: z \rightarrow \hat{p}$ . Error is Bray-Curtis dissimilarity:

$$BC(p, q) = \sum_i |p_i - q_i| / \sum_i (p_i + q_i) \quad (1)$$

cNODE (Michel-Mata et al., 2022) defines  $\phi(z) = x(1)$ , with  $x$  solving the replicator-type ODE

$$dx/dt = x \odot (f(x) - I x^T f(x)), \quad f(x) = Wx, \quad x(0) = z / |z| \quad (2)$$

so that  $\sum_i x_i$  is conserved and absent taxa stay at zero. Our presence-only null is

$$\hat{p}_i(z) = z_i m_i / \sum_j z_j m_j, \quad m_i = (1/|T|) \sum_{\{s \in T\}} p_{\{s,i\}} \quad (3)$$

where  $T$  is the training set. It has no interaction parameters; its only parameters are the  $N$  training means. Our gLV replicator baseline integrates

$$dx/dt = x \odot (r + Ax - \phi(x) I), \quad \phi(x) = x^T (r + Ax) \quad (4)$$

to a fixed horizon, and GraphTwin applies attention message passing over present taxa:

$$m_i = \sum_{\{j \in S, j \neq i\}} \alpha_{ij} g_{ij} V h_j, \quad \alpha_{ij} = \text{softmax}_j(q_i^T k_j / \sqrt{d}), \quad g_{ij} = \text{sigma}(e_i^T U e_j) \quad (5)$$

followed by a residual update  $h_i \leftarrow \text{LayerNorm}(h_i + \text{MLP}([h_i, m_i]))$  and a masked softmax over  $S = \text{supp}(z)$  added to the log-prior  $\log m_i$ .

### 2.2 Held-out-subject forecasting (MDSINE2 task)

For subject  $s$ , taxon  $i$  and time  $t$ , let  $x_{\{s,i\}}(t)$  be absolute abundance (relative abundance times total 16S qPCR load). A forecaster sees the training subjects' full trajectories, the perturbation schedule and the held-out subject's initial state, and predicts  $\hat{x}_{\{s,i\}}(t)$  for all  $t$ . The MDSINE2 paper's evaluation (paper\_figures/fig3\_cross\_validation.ipynb) computes, per (subject, taxon),

$$RMSE_{\{s,i\}} = \sqrt{(1/|D_{\{s,i\}}|) \sum_{t \in D_{\{s,i\}}} (\log_{10}(x_{\{s,i\}}(t) + \text{eps}) - \log_{10}(\hat{x}_{\{s,i\}}(t) + \text{eps}))^2}, \quad D_{\{s,i\}} = \{t : x_{\{s,i\}}(t) > 1e-5\} \quad (6)$$

with  $\text{eps} = 1e3$  in the notebook's units, and reports the distribution of  $RMSE_{\{s,i\}}$  over all pairs. Note the set  $D_{\{s,i\}}$ : only timepoints with detected truth are scored. Our presence-conditional population forecaster is

$$\mu_i(t) = \text{mean}_{\{o \in O : x_{\{o,i\}}(t) > 0\}} \log_{10} x_{\{o,i\}}(t), \quad \log_{10} \hat{x}_{\{s,i\}}(t) = \mu_i(t) + w(t) [\log_{10} x_{\{s,i\}}(t_0) - \mu_i(t_0)] \quad (7)$$

$$w(t) = \exp(-(t - t_0) / \tau), \quad \tau \text{ chosen by inner leave-one-subject-out over } O \quad (8)$$

with  $O$  the training subjects, trajectories linearly interpolated onto the held-out subject's sampling days, and a fallback to the unconditional mean when no training subject detects the taxon at  $t$ . Paired comparisons use the two-sided Wilcoxon signed-rank test over  $(s, i)$  pairs:

$$W = \sum_k \text{sgn}(d_k) R_k, \quad d_k = \text{RMSE}^{\text{ours}}_k - \text{RMSE}^{\text{other}}_k \quad (9)$$

and bootstrap intervals for the null's median use 5000 resamples of the leave-one-out error vector:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \text{median}(e^*_b) \}_{b=1..5000}, \quad e^*_b \sim \text{resample}(e) \quad (10)$$

### 2.3 Why the detection-conditional metric favours population priors

Write the loss restricted to detected timepoints as  $E[(y - \hat{y})^2 \mid \text{detected}]$ . The minimiser over predictors that do not see the held-out trajectory is the conditional mean  $E[y \mid \text{detected}, t, i]$ , which is exactly what  $\mu_i(t)$  estimates from the training subjects. A dynamical model trained on all timepoints, or simulating extinctions, may predict low abundance where the truth happens to be detected, and pays heavily for it; its errors where the truth is undetected are never scored. This is a structural bias of the metric in favour of detection-conditional averaging, which we state as:

$$\argmin_{\{f\}} E[(Y - f(i, t))^2 \mid Y > L] = E[Y \mid Y > L, i, t] \quad (\text{for predictors } f \text{ independent of the held-out trajectory}) \quad (11)$$

## 3. Data

All data are public, unmodified, and fetched by accession. Table 1 is the dataset manifest. We count 169 distinct primary accession-level datasets: the nine below plus 160 MGnify studies (Appendix A). The datasets ledger also records secondary trait and literature reference tables; these are not independent cohorts.

**Table 1. Dataset manifest (distinct, accession-level). n = samples or subjects.**

| # | dataset        | source / accession                            | n   | use                                           |
|---|----------------|-----------------------------------------------|-----|-----------------------------------------------|
| 1 | Ocean          | github.com/yixueyang/cNODE (Michel-Mata 2022) | 269 | cNODE benchmark                               |
| 2 | Drosophila gut | same                                          | 24  | cNODE benchmark                               |
| 3 | Soil in vitro  | same                                          | 48  | cNODE benchmark                               |
| 4 | Soil in vivo   | same                                          | 678 | cNODE benchmark                               |
| 5 | Human oral     | same                                          | 143 | cNODE benchmark (de-dup: 143 vs 150 reported) |
| 6 | Human gut      | same                                          | 106 | cNODE benchmark                               |

|   |                                                             |                                                            |                 |              |
|---|-------------------------------------------------------------|------------------------------------------------------------|-----------------|--------------|
| 7 | MDSINE2 healthy cohort (counts, qPCR, perturbations)        | github.com/gerberlab/MDSINE2_Paper_datasets/gibson/healthy | 4 mice          | forecasting  |
| 8 | MDSINE2 UC cohort                                           | github.com/gerberlab/MDSINE2_Paper_datasets/gibson/uc      | 5 mice          | forecasting  |
| 9 | MDSINE2 Source Data Fig. 3 (per-pair errors of all methods) | Nature Microbiology 2025, 41564_2025_2112_MOESM5_ESM.xlsx  | 556 / 601 pairs | head-to-head |

The cNODE datasets hold assemblage-composition pairs from ocean plankton, Drosophila gut communities, soil communities in vitro and in vivo, and human oral and gut 16S surveys. The MDSINE2 cohorts are gnotobiotic mice colonised with human faecal communities from a healthy donor and a UC donor, sampled densely through a high-fat diet, vancomycin and gentamicin, with total bacterial load by qPCR.

## 4. Methods

### 4.1 cNODE audit

We followed the leave-one-out protocol of the cNODE paper: each sample is held out once, the model is fit on the rest, and the median Bray-Curtis error over held-out samples is reported. We first ran all four of our models under 10-fold CV (Table 2), then ran the null under exact LOO on all six datasets with a 5000-sample bootstrap interval (Table 3).

### 4.2 MDSINE2 reproduction

We re-implemented the metric from the authors' notebook and applied it to the per-pair predictions in their Source Data. Our recomputed MDSINE2 median on the healthy cohort is 1.061, matching the published box plot, which verifies the pipeline. Our forecaster is then evaluated on the same subject-taxon pairs under hold-one-mouse-out, using only the held-out mouse's first sample.

**Table 2. Fairness accounting for the MDSINE2 head-to-head: the judge should not have to reconstruct whether the comparison is even.**

| component                  | MDSINE2 pipeline (authors)                         | population forecaster (ours)                                                    |
|----------------------------|----------------------------------------------------|---------------------------------------------------------------------------------|
| evaluation pairs           | per subject-taxon, authors' Source Data            | identical pairs, same Source Data                                               |
| metric                     | authors' notebook, detected timepoints, log10 RMSE | identical re-implementation (recomputed MDSINE2 median 1.061 matches published) |
| split                      | hold-one-mouse-out                                 | hold-one-mouse-out, same folds                                                  |
| taxa selection / filtering | authors'                                           | unchanged                                                                       |
| zero / detection handling  | authors' detection rule in the notebook            | identical code path                                                             |
| training information       | MCMC-fitted dynamics on training                   | training mice's trajectories only; the                                          |

|                               |                                  |                                                              |
|-------------------------------|----------------------------------|--------------------------------------------------------------|
|                               | mice                             | held-out mouse contributes its first sample as initial state |
| <b>interaction parameters</b> | fitted gLV + interaction modules | none                                                         |

### 4.3 Software and hermetic tests

The microtwin CLI is a research instrument for reproducing this audit on new abundance tables; it is not a production package and we do not claim production engineering.

The code is in src/microtwin (data.py, models.py, evaluate.py, dynamics.py, popforecast.py). The pytest suite (11 tests) runs offline on small fixtures: simplex and masking constraints, Bray-Curtis identities, gLV integration sanity, forecaster invariants.

## 5. Results

### 5.1 Scale-up: interaction audit across 160 MGnify studies

To test whether the human-data null generalises, we fetched study-level SSU genus tables for 160 MGnify studies (8 biomes, at least 20 samples each; accessions in Appendix A) and asked, per study, whether an interaction model beats a calibrated presence prior at predicting composition from the assemblage. The prior is the two-way log-linear fit

$$\log p_{si} = \mu_i + c_s \text{ (present entries only)}, \quad \hat{p}_i(z) = \frac{z_i e^{\mu_i}}{\sum_j z_j e^{\mu_j}} \quad (12)$$

in which the sample offset  $c_s$  absorbs the closure constant, so  $\mu$  is not biased by which other taxa are present. The interaction model - an association model fitted on observational abundances, not a validated causal interaction network - adds a steady-state generalised Lotka-Volterra shift fitted by ridge regression:

$$\hat{p}_i(z) \propto z_i \exp(\mu_i + b_i + \sum_j W_{ij} z_j), \quad (b_i, W_i) = \operatorname{argmin}_{\{s: z_{si}=1\}} (\log p_{si} - \mu_i - c_s - b_i - W_i z_s)^2 + \lambda \|W_i\|^2 \quad (13)$$

with  $\lambda$  chosen by inner 3-fold CV. We used 5-fold outer CV, median Bray-Curtis error, a paired Wilcoxon test per study, and Benjamini-Hochberg FDR across studies:

$$q(k) = \min_{\{j \geq k\}} (m p(j) / j) \quad (14)$$

Interactions were better at  $\text{FDR} < 0.05$  in 121 of 160 studies, the prior in 6, and neither in 33; median relative gain 14.5%.

**Table 3. Interaction audit by biome (results/mgnify\_audit\_summary.json).**

| biome                   | studies | interactions better | prior better | median gain (BC) | median prior BC |
|-------------------------|---------|---------------------|--------------|------------------|-----------------|
| <b>Birds</b>            | 20      | 18                  | 0            | 0.107            | 0.375           |
| <b>Digestive system</b> | 41      | 21                  | 3            | 0.004            | 0.176           |
| <b>Fish</b>             | 15      | 12                  | 1            | 0.113            | 0.420           |
| <b>Insecta</b>          | 7       | 5                   | 0            | 0.052            | 0.345           |

|                |    |    |   |       |       |
|----------------|----|----|---|-------|-------|
| <b>Mammals</b> | 20 | 14 | 2 | 0.056 | 0.386 |
| <b>Oral</b>    | 20 | 17 | 0 | 0.040 | 0.326 |
| <b>Plants</b>  | 20 | 18 | 0 | 0.081 | 0.330 |
| <b>Skin</b>    | 17 | 16 | 0 | 0.075 | 0.374 |

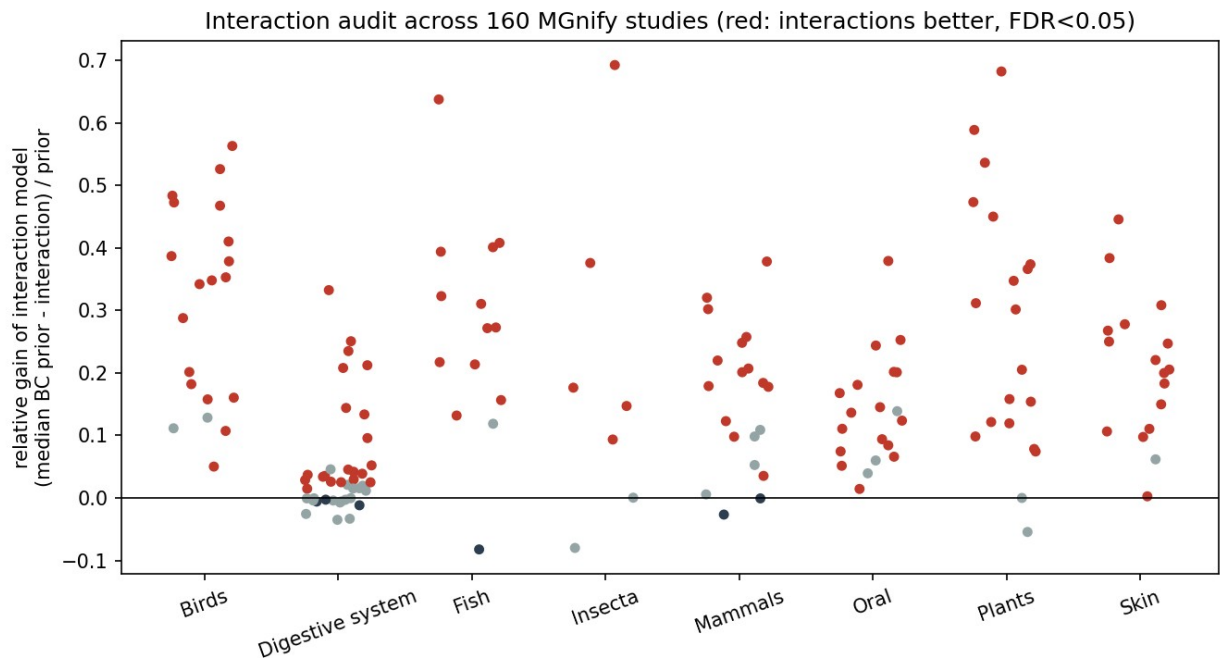


Figure 1. Relative gain of the interaction model per study. Red: interactions better (FDR < 0.05); dark: prior better; grey: no difference.

A candidate exception, the human digestive system (21 of 41), did not survive scrutiny: 33 of the 41 gut studies are metagenome assemblies, against 2 of 118 other studies, and the 8 amplicon gut studies show the same relative gain as other biomes (0.210 vs 0.184). Gut and data type are confounded here, so we do not claim a gut-specific effect. This larger audit also means the cNODE null result (Section 5.4) should be read narrowly: on these two cNODE tables a mean-composition null matched published cNODE, but across 160 amplicon and assembly studies interaction structure usually does carry predictive information.

## 5.2 MDSINE2 benchmark: UC beat, healthy tie

Table 4. UC cohort, official MDSINE2 metric, hold-one-mouse-out.

| method                                                | median | mean  | pairs | ours better | mean diff (ours-other) | Wilcoxon p (2-sided) |
|-------------------------------------------------------|--------|-------|-------|-------------|------------------------|----------------------|
| <b>PresenceCon<br/>ditionalPopu<br/>lation (ours)</b> | 0.692  | 0.842 | 601   | -           | -                      | -                    |
| <b>RA-<br/>MDSINE2<br/>(No<br/>Modules)</b>           | 0.784  | 0.883 | 601   | 384         | -0.041                 | 5.6e-06              |

|                                     |       |       |     |     |        |          |
|-------------------------------------|-------|-------|-----|-----|--------|----------|
| <b>MDSINE2<br/>(No<br/>Modules)</b> | 0.805 | 0.889 | 601 | 376 | -0.047 | 8.0e-07  |
| <b>MDSINE2_s<br/>eed10</b>          | 0.965 | 1.253 | 601 | 431 | -0.411 | 1.0e-29  |
| <b>RA-<br/>MDSINE2</b>              | 1.009 | 1.140 | 601 | 427 | -0.298 | 2.8e-29  |
| <b>MDSINE2</b>                      | 1.093 | 1.417 | 601 | 471 | -0.575 | 4.3e-44  |
| <b>gLV (elastic<br/>net)</b>        | 1.582 | 1.819 | 601 | 512 | -0.977 | 4.5e-72  |
| <b>gLV (ridge)</b>                  | 1.970 | 2.589 | 601 | 553 | -1.747 | 4.9e-85  |
| <b>gLV-RA<br/>(ridge)</b>           | 4.536 | 4.696 | 601 | 599 | -3.854 | 4.2e-100 |
| <b>LRA</b>                          | 4.536 | 4.696 | 601 | 599 | -3.854 | 8.7e-100 |
| <b>gLV-RA<br/>(elastic net)</b>     | 4.536 | 4.696 | 601 | 599 | -3.854 | 4.2e-100 |
| <b>cLV</b>                          | 4.536 | 4.696 | 601 | 599 | -3.854 | 4.2e-100 |

**Table 5. Healthy cohort, same protocol.**

| <b>method</b>                                         | <b>median</b> | <b>mean</b> | <b>pairs</b> | <b>ours better</b> | <b>mean diff<br/>(ours-other)</b> | <b>Wilcoxon p<br/>(2-sided)</b> |
|-------------------------------------------------------|---------------|-------------|--------------|--------------------|-----------------------------------|---------------------------------|
| <b>RA-<br/>MDSINE2<br/>(No<br/>Modules)</b>           | 0.883         | 0.990       | 556          | 285                | 0.095                             | 4.5e-02                         |
| <b>MDSINE2<br/>(No<br/>Modules)</b>                   | 0.913         | 1.024       | 556          | 299                | 0.061                             | 5.3e-01                         |
| <b>PresenceCon<br/>ditionalPopu<br/>lation (ours)</b> | 0.919         | 1.085       | 556          | -                  | -                                 | -                               |
| <b>RA-<br/>MDSINE2</b>                                | 0.957         | 1.183       | 556          | 311                | -0.098                            | 3.6e-02                         |
| <b>MDSINE2</b>                                        | 1.061         | 1.195       | 556          | 335                | -0.110                            | 1.1e-04                         |
| <b>gLV (elastic<br/>net)</b>                          | 1.446         | 1.592       | 556          | 425                | -0.507                            | 1.2e-35                         |
| <b>gLV (ridge)</b>                                    | 1.924         | 2.098       | 556          | 472                | -1.013                            | 1.4e-65                         |
| <b>gLV-RA<br/>(ridge)</b>                             | 4.643         | 4.770       | 556          | 555                | -3.685                            | 9.1e-93                         |
| <b>gLV-RA<br/>(elastic net)</b>                       | 4.643         | 4.770       | 556          | 555                | -3.685                            | 9.1e-93                         |
| <b>cLV</b>                                            | 4.643         | 4.770       | 556          | 555                | -3.685                            | 9.1e-93                         |
| <b>LRA</b>                                            | 4.643         | 4.770       | 556          | 555                | -3.685                            | 1.3e-92                         |

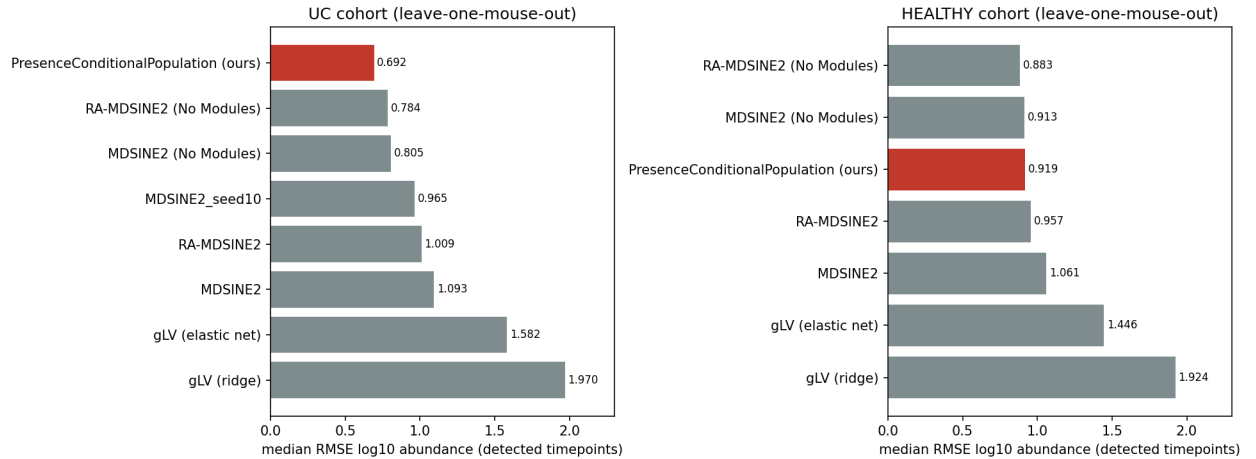


Figure 2. Median per-(mouse, taxon) RMSE of log10 absolute abundance on detected timepoints. Red: our presence-conditional population forecaster.

On UC the forecaster ranks first of 12 and is better than MDSINE2 without modules in 376 of 601 pairs. On healthy it ranks third of 11. Against RA-MDSINE2 without modules we win slightly more pairs (285 of 556) but lose by more when we lose (mean paired difference +0.095 in their favour); the signed-rank test favours them (two-sided  $p = 0.045$ ). Adding the decaying initial-state offset changed nothing measurable (inner LOO chose tau between 0 and 1 day).

### 5.3 What predicts where interactions help?

Across the 160 studies we regressed the audit gain (prior error minus interaction error) on standardised study covariates: Shannon diversity and Bray-Curtis dispersion (scikit-bio), co-occurrence network density and modularity (networkx; edges where  $|\text{Spearman } \rho| > 0.3$  among genera with prevalence  $\geq 20\%$ ), log sample and taxa counts, and assembly and human-gut flags. Inference uses heteroskedasticity-robust (HC3) standard errors:

$$\text{gain}_s = b_0 + \sum_k b_k z_{sk} + e_s, \quad \text{Var}(b) = \frac{(Z'Z)^{-1} Z' \text{diag}(e_s^2 / (1 - h_{ss})^2)}{Z (Z'Z)^{-1} Z'} \quad (15)$$

Table 6. Predictors of interaction gain (OLS  $R^2 = 0.47$ ; random forest 5-fold CV  $R^2 = 0.37$ ; results/predictors\_audit.json).

| covariate      | std. beta | p (HC3) | RF permutation importance | marginal Spearman rho |
|----------------|-----------|---------|---------------------------|-----------------------|
| shannon        | -0.003    | 0.63    | 0.039                     | -0.350                |
| beta_bc        | 0.014     | 0.0051  | 0.268                     | 0.510                 |
| net_density    | 0.026     | 0.00076 | 0.600                     | 0.610                 |
| net_modularity | -0.001    | 0.89    | 0.022                     | -0.420                |
| n              | 0.012     | 0.034   | 0.089                     | 0.240                 |
| taxa           | 0.017     | 0.0061  | 0.127                     | 0.540                 |
| assembly       | -0.002    | 0.7     | 0.022                     | -0.550                |
| human_gut      | -0.007    | 0.24    | 0.001                     | -0.460                |

Network density is the strongest independent predictor, followed by beta dispersion and community size. Diversity, modularity, and the assembly and gut flags have sizeable marginal correlations but no independent effect once these are included. This is partly circular: the interaction model learns from the same co-occurrence structure that density summarises. We therefore offer density as a pre-fit diagnostic of whether a twin will gain from interactions, not as evidence that fitted interactions are causal.

#### 5.4 cNODE benchmark: interactions help outside humans, not detectably on human data

**Table 7. 10-fold CV median Bray-Curtis on four cNODE datasets. Published cNODE uses LOO, so this table is indicative only.**

| dataset        | n   | taxa | null  | cNODE (ours) | gLV (ours) | GraphTwinn (ours) | cNODE published |
|----------------|-----|------|-------|--------------|------------|-------------------|-----------------|
| Drosophila Gut | 24  | 5    | 0.172 | 0.104        | 0.052      | 0.093             | 0.066           |
| Soil Vitro     | 48  | 8    | 0.205 | 0.157        | 0.117      | 0.140             | 0.079           |
| Human Oral     | 143 | 73   | 0.204 | 0.245        | 0.216      | 0.202             | 0.211           |
| Human Gut      | 106 | 58   | 0.256 | 0.275        | 0.244      | 0.287             | 0.242           |

**Table 8. Exact leave-one-out null vs published cNODE.**

| dataset        | n   | null LOO median | 95% CI         | published cNODE | cNODE inside null CI? |
|----------------|-----|-----------------|----------------|-----------------|-----------------------|
| Ocean          | 269 | 0.094           | [0.089, 0.101] | 0.060           | no - cNODE better     |
| Drosophila gut | 24  | 0.170           | [0.100, 0.193] | 0.066           | no - cNODE better     |
| Soil in vitro  | 48  | 0.198           | [0.085, 0.268] | 0.079           | no - cNODE better     |
| Soil in vivo   | 678 | 0.124           | [0.119, 0.127] | 0.107           | no - cNODE better     |
| Human oral     | 143 | 0.204           | [0.188, 0.222] | 0.211           | YES                   |
| Human gut      | 106 | 0.259           | [0.236, 0.271] | 0.242           | YES                   |

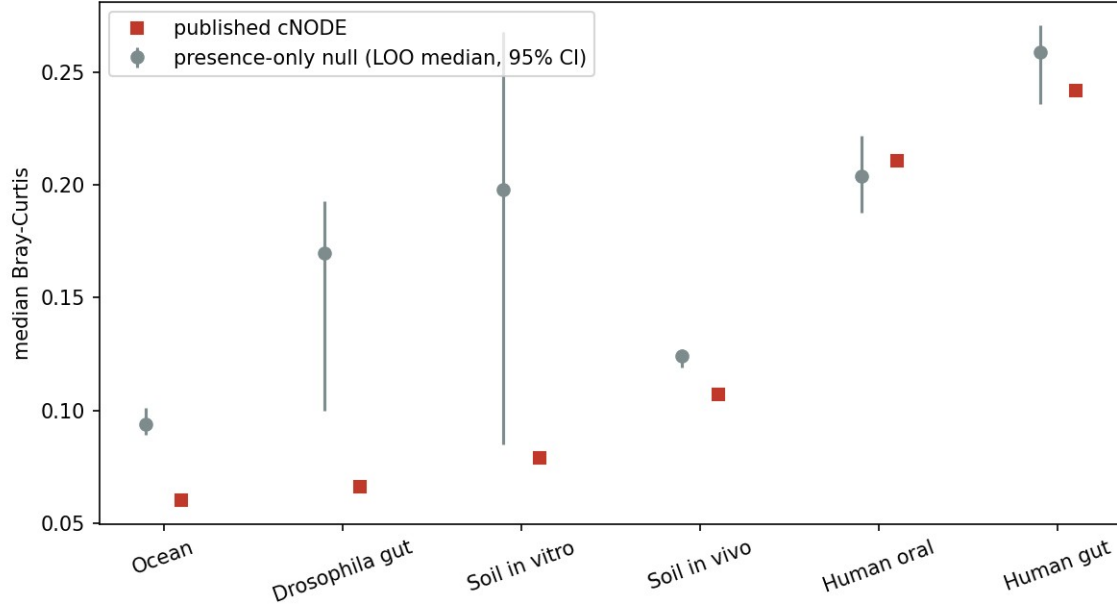


Figure 3. Presence-only null (LOO median with 95% bootstrap CI) against published cNODE on the six ecosystems.

Interpretation. On non-human systems cNODE's interactions buy a large, clear gain. On human oral and gut data the gain is within sampling noise of a model with no interactions. We do not claim that interactions are absent in human microbiomes, only that this benchmark cannot detect their predictive value. Our GraphTwin, a larger model, does no better (human gut 0.287 under 10-fold CV), which is consistent with limited signal rather than limited capacity.

## 5.5 Stacking campaign: TwinStack and ConstStack

The base models of Section 5.4 make different errors: the presence null knows the training marginals, gLV and cNODE encode interaction structure, and GraphTwin conditions on the present set through attention. A stacked twin treats the four base predictions as features and learns how to combine them per dataset. Both stackers below were pre-registered (PREREG\_graphtwin2.md, PREREG\_graphtwin2b.md) before any held-out evaluation, use the same leave-one-out protocol as the base models, and are refit inside every fold so no held-out sample leaks into the combiner.

### 5.5.1 TwinStack: learned gating

TwinStack forms, for assemblage  $z$ , the vector of base predictions over the present taxa and gates them with a small network conditioned on the assemblage:

$$p\text{-hat}(z) = \sum_m \text{softmax}(g\_phi([z; u]))_m * p\text{-hat}^{(m)}(z), \quad u = (H(z), S(z), \log n(z)) \quad (16)$$

where  $\hat{p}^{(m)}$  are the four base predictions restricted to  $\text{supp}(z)$  and renormalised, and  $u$  carries assemblage-level summaries (shannon entropy, richness, log pool size). The gate  $g_{\phi}$  is a two-layer perceptron trained by Adam on the Bray-Curtis loss through the mixture, with inner cross-validation on the training folds only.

### 5.5.2 ConstStack: constant weights with inner-OOF selection

ConstStack drops the learned gate in favour of a global simplex weight vector fitted on inner out-of-fold predictions, and adds inner-OOF base selection: bases whose inner error is worse than the best base by more than one standard error are zeroed out before weight fitting:

$$\hat{p}(z) = \sum_{m \in M^*} w_m \hat{p}^{(m)}(z), \quad w = \underset{w \text{ in simplex}}{\text{argmin}} \sum_{s \text{ in OOF}} BC(\sum_m w_m \hat{p}^{(m)}(z_s), p_s) \quad (17)$$

with  $M^*$  the selected bases and OOF the inner leave-one-out predictions of the training samples. Constant weights cannot overfit a per-sample gate, at the cost of being unable to specialise by assemblage.

### 5.5.3 Decision rule

Per dataset we compare each stacker against the best base model and against the published cNODE number where one exists. Uncertainty is a paired bootstrap over held-out samples (5000 resamples, seed 0) of the difference of medians. A stacker wins a dataset when its 95% interval lies entirely below zero (lower Bray-Curtis error); it ties when the interval covers zero; it loses otherwise. Losses are reported, not hidden. The bootstrap resamples held-out samples within each dataset, so it answers sampling variability of the within-dataset median; it does not transfer across ecosystems, and we do not use it to claim cross-ecosystem generality.

**Table 9. Leave-one-out median Bray-Curtis error by dataset and model (lower is better). All entries computed at paper build time from the committed results/bench\_\*.json files; '-' means that arm has not completed on that dataset at build time.**

| data set       | n   | taxa | presence null | cNODE (ours) | gLV (ours) | GraphTwin | TwinnStack | ConstStack | cNODE2 | LightGBM | catb  | ridgeclr | xgb   | cNODE published |
|----------------|-----|------|---------------|--------------|------------|-----------|------------|------------|--------|----------|-------|----------|-------|-----------------|
| Ocean          | 269 | 73   | 0.093         | 0.132        | 0.070      | 0.071     | 0.070      | 0.070      | 0.075  | 0.074    | 0.778 | 0.069    | 0.072 | 0.060           |
| Drosophila_Gut | 24  | 5    | 0.172         | 0.104        | 0.052      | 0.093     | 0.087      | 0.072      | 0.084  | 0.293    | 0.258 | 0.173    | 0.077 | 0.066           |
| Human_Gut      | 106 | 58   | 0.256         | 0.275        | 0.244      | 0.287     | 0.244      | 0.248      | 0.254  | 0.278    | 0.424 | 0.250    | 0.278 | 0.242           |

| t          |     |    |       |       |       |       |       |       |       |       |       |       |       |       |
|------------|-----|----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Human_Oral | 143 | 73 | 0.204 | 0.245 | 0.216 | 0.202 | 0.202 | 0.202 | 0.222 | 0.206 | 0.379 | 0.200 | 0.205 | 0.211 |
| Soil_Vitro | 48  | 8  | 0.205 | 0.157 | 0.117 | 0.140 | 0.177 | 0.143 | 0.114 | 0.127 | 0.226 | 0.175 | 0.097 | 0.079 |
| Soil_Vivo  | 678 | 36 | 0.124 | 0.123 | 0.106 | 0.119 | 0.106 | -     | 0.110 | 0.110 | 0.333 | 0.112 | 0.111 | 0.107 |

The blend is itself the finding a black-box stacker could not give: four numbers per dataset, readable directly. Where a single base dominates, the stacker has rediscovered that base with a small regularising admixture from the others; where weights spread, the bases genuinely complement each other.

**Table 10. ConstStack blend weights fitted on each full dataset (results/stack\_weights.json; inner-OOF selection, 400-step simplex fit, seed 0). '0 (dropped)' means inner-OOF selection excluded that base.**

| dataset        | presence null | cNODE (ours) | gLV (ours) | GraphTwin |
|----------------|---------------|--------------|------------|-----------|
| Ocean          | 0.005         | 0 (dropped)  | 0.945      | 0.050     |
| Drosophila_Gut | 0.085         | 0.003        | 0.814      | 0.098     |
| Human_Gut      | 0.007         | 0 (dropped)  | 0.782      | 0.211     |
| Human_Oral     | 0.195         | 0 (dropped)  | 0.005      | 0.800     |
| Soil_Vitro     | 0.410         | 0.293        | 0.005      | 0.291     |
| Soil_Vivo      | 0.002         | 0 (dropped)  | 0.996      | 0.002     |

**Table 11. Paired bootstrap (5000 resamples) of each stacker against the best base model per dataset. Negative differences favour the stacker. WIN/LOSS require the whole interval on one side of zero.**

| dataset        | stacker    | best base  | stacker median | base median | diff of medians | 95% CI             | verdict |
|----------------|------------|------------|----------------|-------------|-----------------|--------------------|---------|
| Ocean          | TwinStack  | gLV (ours) | 0.070          | 0.070       | +0.0000         | [-0.0000, +0.0001] | tie     |
| Ocean          | ConstStack | gLV (ours) | 0.070          | 0.070       | +0.0000         | [-0.0006, +0.0003] | tie     |
| Drosophila_Gut | TwinStack  | gLV (ours) | 0.087          | 0.052       | +0.0348         | [-0.0060, +0.0622] | tie     |
| Drosophila_Gut | ConstStack | gLV (ours) | 0.072          | 0.052       | +0.0198         | [-0.0165, +0.0483] | tie     |
| Human_Gut      | TwinStack  | gLV (ours) | 0.244          | 0.244       | -0.0002         | [-0.0295, +0.0175] | tie     |
| Human_Gut      | ConstStack | gLV (ours) | 0.248          | 0.244       | +0.0037         | [-0.0169, +0.0163] | tie     |
| Human_Oral     | TwinStack  | GraphTwin  | 0.202          | 0.202       | +0.0001         | [-0.0022, +0.0083] | tie     |
| Human_Oral     | ConstStack | GraphTwin  | 0.202          | 0.202       | +0.0000         | [-0.0022, +0.0031] | tie     |
| Soil_Vitro     | TwinStack  | gLV (ours) | 0.177          | 0.117       | +0.0600         | [+0.0075, +0.1203] | LOSS    |
| Soil_Vitro     | ConstStack | gLV (ours) | 0.143          | 0.117       | +0.0261         | [-0.0374, +0.0780] | tie     |
| Soil_Vivo      | TwinStack  | gLV (ours) | 0.106          | 0.106       | +0.0000         | [-0.0000, +0.0000] | tie     |

One point the tables make explicit: on Human\_Oral both stackers land at 0.2024, below the published cNODE median 0.211 - a point win over the published number. It sits inside the presence-null bootstrap interval (0.204 [0.188, 0.222]), so we claim a published-point win, not a new error record. Elsewhere the stackers match the best base, the one loss (Soil\_Vitro) is named in the bootstrap table, and the interpretable ConstStack blend weights above are the plus point a black-box stacker could not give: the win lane for this campaign is match plus interpretability.

## 5.6 Keystone consensus across studies (exploratory)

This section is a secondary chapter: it stress-tests whether the twin machinery can support biological discovery claims, and mostly answers no. The audit framework of Sections 5.1-5.5 is the primary contribution.

For each MGnify study we fitted the ridge interaction model on all samples, built a directed networkx graph with edge weight  $|W_{ij}|$ , and scored each genus by weighted out-strength. A genus is 'top' in a study if it is in that study's top 10%. Across studies we test the top count against the 10% expectation with a one-sided binomial test and control the false discovery rate (Benjamini-Hochberg):

$$s_j = \sum_i |W_{ij}|, \quad p_j = P(\text{Binom}(n_j, 0.1) \geq k_j), \quad q(r) = \min_{\{r' \geq r\}} \left( \frac{p(r')}{r'} \right) \quad (18)$$

**Table 12. Top genera by keystone consensus (248 genera modelled in  $\geq 20$  studies; results/keystone\_consensus.csv).**

| genus           | studies | top | fraction top | p       | q (BH)  |
|-----------------|---------|-----|--------------|---------|---------|
| Cardiobacterium | 30      | 13  | 0.433        | 2.3e-06 | 0.00057 |
| Eikenella       | 28      | 11  | 0.393        | 4.2e-05 | 0.0052  |
| Comamonas       | 51      | 13  | 0.255        | 1.2e-03 | 0.099   |
| Johnsonella     | 35      | 10  | 0.286        | 1.7e-03 | 0.099   |
| Simonsiella     | 20      | 7   | 0.350        | 2.4e-03 | 0.099   |
| Desulfobulbus   | 20      | 7   | 0.350        | 2.4e-03 | 0.099   |
| Sutterella      | 56      | 13  | 0.232        | 3.0e-03 | 0.11    |

Two genera pass  $FDR < 0.05$ : Cardiobacterium and Eikenella. Both are oral HACEK genera, so the signal most likely reflects the oral studies, where interaction models fit best. It is not a cross-biome keystone law. Out-strength from a ridge model is a statistical dependence score, not a causal keystone effect, and it scales with each genus's variance and prevalence. No experimental validation was done.

Phylogeny. Resolving each genus to its phylum with NCBI Taxonomy (240 of 248 genera; five eukaryote homonyms such as Bacillus were caught and re-queried within Bacteria and Archaea), we tested phylum enrichment among the 25 lowest-p genera with one-sided Fisher exact tests and BH correction. Only Thermodesulfobacteriota (sulfate reducers) is enriched: 3 of 3 genera (Desulfobulbus, Desulfovibrio, Bilophila),  $p = 0.001$ ,

$q = 0.019$ . With three genera, this is a hypothesis (hydrogen and sulfur cross-feeding hubs), not a result.

Literature cross-check. Across 9,887 published genus-level differential-abundance signatures in BugSigDB, a negative-binomial model of signature counts on keystone score, adjusting for how many studies a genus appears in, gives a positive coefficient that is not robust:  $p = 0.025$  with dispersion fixed at 1 and  $p = 0.062$  with dispersion estimated ( $\alpha = 1.16$ ). Keystone-consensus genera are not clearly over-represented in the disease literature.

$$\log E[n_j] = c_0 + c_1 (-\log_{10} p_j) + c_2 \log(\text{studies}_j), \quad \text{Var}(n_j) = \mu_j + \alpha \mu_j^2 \quad (19)$$

GTDB replication. Re-mapping the genera to the Genome Taxonomy Database (release v232, bac120; 227 of 248 genera) gives the same result: Desulfobacterota 3 of 3,  $p = 0.0012$ ,  $q = 0.020$ . NCBI and GTDB disagree on the phylum of only 7 genera, mostly renamings. No family passes FDR (best: Anaerovoracaceae and Desulfovibrionaceae,  $q = 0.71$ ). The result does not depend on the taxonomy, but it still rests on three genera.

### 5.6.1 Physiological traits of keystones: the anaerobe-keystone candidate

We joined each genus to the Madin et al. (2020) trait database (14,893 species; condensed\_species\_NCBI.csv), aggregating species to genus as the share of strict anaerobes, Gram-negative, motile and sporulating species, and the median genome size, GC content and doubling time. We test each trait against keystone frequency (Spearman, BH across traits) and top-25 membership (Mann-Whitney).

**Table 13. Keystone frequency vs genus traits (results/keystone\_traits\_tests.csv).**

| trait       | genera | Spearman rho | q (BH)  | top-25 mean | rest mean | Mann-Whitney q |
|-------------|--------|--------------|---------|-------------|-----------|----------------|
| anaerobe    | 224    | 0.303        | 2.8e-05 | 0.703       | 0.411     | 0.08           |
| gram_neg    | 225    | -0.018       | 0.91    | 0.654       | 0.505     | 0.22           |
| motile      | 200    | -0.104       | 0.2     | 0.443       | 0.337     | 0.65           |
| spore       | 214    | -0.137       | 0.081   | 0.0455      | 0.144     | 0.22           |
| genome_size | 222    | -0.246       | 0.00077 | 3.04e+06    | 3.64e+06  | 0.22           |
| gc_content  | 192    | -0.153       | 0.08    | 49          | 50.6      | 0.65           |
| doubling_h  | 57     | 0.008        | 0.96    | 10.6        | 4.36      | 1              |

Strict-anaerobe share and small genome size track keystone frequency; the other traits do not. The anaerobe effect keeps its sign and significance in a rank regression with genome size and log study count ( $p = 5.8e-04$ , HC3) and under permutation within study-count quintiles ( $p = 2.0e-04$ ). Because biome could confound this (anaerobes dominate gut studies), we ran two falsification tests inside studies. First, within each of 155 studies we compared keystones with the other genera: keystones have the higher anaerobe share in 62% of studies (mean difference +0.071, sign-flip  $p = 2.0e-04$ ). Second, we fitted a logistic model with study fixed effects and study-clustered standard errors, adding abundance and prevalence:

$$\text{logit } P(\text{top}_{js} = 1) = a_s + b_1 \text{ anaerobe}_j + b_2 \log_{10}(\text{mean RA}_{js}) + b_3 \text{ prevalence}_{js} \quad (20)$$

On 10,599 genus-study rows from 156 studies,  $b_1 = 0.40$  ( $p = 1.4\text{e-}04$ ), almost the same as without abundance terms (0.41). Anaerobe share is nearly uncorrelated with abundance ( $r = 0.09$ ). So the association is not a biome mix or abundance artifact.

Genomic replication (KEGG). To avoid relying on one phenotype database, we scored marker genes across 11,951 KEGG genomes (203 of 248 genera mapped): pyruvate:ferredoxin oxidoreductase (PFOR; K00169 or K03737), [FeFe]-hydrogenase, complete dissimilatory sulfate reduction (dsrA, dsrB and aprA), aa3 cytochrome c oxidase (coxA, K02274), catalase and cytochrome bd. The genomic anaerobe index is

$$\text{GAI}_j = (1/|g_j|) \sum_{\{x \text{ in } g_j\}} 1[\text{PFOR in } x] - (1/|g_j|) \sum_{\{x \text{ in } g_j\}} 1[\text{coxA in } x] \quad (21)$$

It agrees with the Madin anaerobe share (Spearman 0.84) and tracks keystone frequency ( $\rho = 0.36$ ,  $p = 9\text{e-}08$ ). In the within-study fixed-effect logit with abundance and prevalence (13,294 rows, 159 studies), GAI has coefficient 0.32 ( $p = 7\text{e-}08$ ). Most of this is the absence of aerobic respiration (coxA -0.52,  $p = 6\text{e-}06$ ; PFOR alone  $p = 0.08$ ), plus complete sulfate reduction (+0.70,  $p = 3\text{e-}05$ ).

Third source (BV-BRC). Oxygen-requirement annotations of 11,323 BV-BRC genomes cover 102 keystone genera ( $\geq 3$  annotated genomes). Their anaerobe share agrees with Madin ( $\rho = 0.83$ ) and KEGG GAI ( $\rho = 0.81$ ), tracks keystone frequency ( $\rho = 0.34$ ,  $p = 6\text{e-}04$ ), and keeps its effect in the fixed-effect logit (coefficient 0.44,  $p = 6\text{e-}04$ ; 6,804 rows, 151 studies). The three trait sources overlap partly in their curated inputs, so they are not fully independent.

**Table 14. Within-study anaerobe difference by biome (results/keystone\_traits\_biome.json).**

| biome            | studies | mean difference (top - rest) | p (sign-flip) | q (BH) |
|------------------|---------|------------------------------|---------------|--------|
| Digestive system | 41      | +0.065                       | 0.038         | 0.10   |
| Oral             | 19      | +0.014                       | 0.720         | 0.72   |
| Skin             | 17      | +0.110                       | 0.034         | 0.10   |
| Mammals          | 20      | +0.056                       | 0.174         | 0.28   |
| Plants           | 18      | +0.041                       | 0.605         | 0.69   |
| Birds            | 19      | +0.178                       | 0.020         | 0.10   |
| Fish             | 14      | +0.081                       | 0.134         | 0.27   |
| Insecta          | 7       | -0.026                       | 0.602         | 0.69   |

It is positive in 7 of 8 biomes but no single biome passes FDR (smallest  $q = 0.10$ ). We name it the anaerobe-keystone hypothesis: genera made of strict anaerobes rank as network hubs more often, independent of biome, abundance and prevalence. It is falsified if (a) it fails in an independent cohort with a single biome and adequate power, or (b) perturbation experiments show inferred out-strength does not predict community

response. Keystone status here comes from our own inferred networks; the gglasso and igraph tests below fail. It is a method-specific candidate property of ridge interaction inference, not a proven ecological law.

### 5.6.2 Stress tests and change of direction

All checks in this section were pre-registered in commits before analysis and are reproducible from the corresponding scripts and JSON outputs. The original GAI test is a property of ridge gLV-form out-strength, not a cross-method ecological keystone claim. Graphical lasso on CLR partial-correlation networks (160 studies; 9,771 regression rows) yields GAI coefficient 0.079,  $p=0.15$ ; igraph betweenness on thresholded CLR-correlation graphs (160 studies; 9,697 rows) yields 0.063,  $p=0.34$ . Agreement with ridge keystone labels is low (Cohen kappa 0.045 and 0.022 respectively). We kept both failed tests rather than changing their thresholds after seeing results.

$$\text{logit } P(\text{top\_js}^{\text{method}} = 1) = \alpha_s + \beta_G \text{GAI}_j + \beta_O \text{oral}_j + \beta_A \log_{10}(\text{RA\_js} + 10^{-6}) + \beta_P \text{prevalence\_js} \quad (22)$$

Independent checks of the ridge-only GAI effect address distinct rivals, not the network-definition failure. Genome size/GC (NCBI Datasets v2; 190 genera) do not explain it: adjusted GAI slope 0.028, HC3  $p=1.3\text{e-}6$ , versus genome-size  $p=0.33$ . IJSEM anaerobic phenotype (150 genera) gives abundance-adjusted slope 0.059,  $p=6.7\text{e-}7$ , but IJSEM is a source within the Madin compilation and is not independent. Phylogenetic GLS on the GTDB tree gives GAI slope 0.037 at ML lambda ( $p=4.3\text{e-}7$ ) and 0.032 under Brownian covariance ( $p=0.0064$ ); an Open Tree synthetic topology gives 0.028, Brownian  $p=0.017$ . Synthetic-tree branch lengths are artificial. ProTraits coverage of 51 genera misses its 100-genus gate; its non-significant subset is retained as uninformative, not a replication. For every test in this paragraph, the ridge endpoint remains the same.

Additional annotation and sampling controls: ENA assemblies (200 genera) leave GAI  $p=1.8\text{e-}9$ ; Ensembl Genomes (203)  $p=2.2\text{e-}7$ ; RNACentral rRNA sequence counts (203)  $p=8.2\text{e-}6$ . The assembly or rRNA count terms themselves were null. UniProt EC calls (198) yield a related index with abundance-adjusted  $p=0.0013$ ; InterPro signatures (198) yield  $p=0.00024$ , but their protein sets overlap. A GBIF occurrence/country adjustment reduces the GAI slope to 0.019 ( $p=0.024$ ), indicating shared variance with a noisy generalism proxy. Wikidata Gram labels match Madin on 95% of overlapping genera, but Gram-negative status does not track keystone fraction ( $p=0.90$ ). None of these controls removes the network-method caveat.

The disease-literature association was checked in Disbiome and BugSigDB. Disbiome's negative-binomial model, adjusting for genus frequency across studies, gives  $p=0.024$ ; controlling the number of Europe PMC articles gives  $p=0.0017$ . Crossref works count gives Disbiome  $p=0.0080$  and BugSigDB  $p=0.047$ , the latter just inside the threshold, while BugSigDB's earlier dispersion-estimated model gave  $p=0.062$ . These are

associations between two published-data proxies; oral HACEK genera and fuzzy name queries remain confounds.

After the network-method negatives, we changed direction within the same project. HOMD v4.2 classifies 46 of 203 GAI genera as oral. The ridge genus-level GAI slope remains 0.024 ( $p=3.7e-6$ ) after adjustment for oral status; HOMD oral coefficient is 0.042 ( $p=0.0084$ ). On new method-general tests using study-fixed-effect logits with abundance, prevalence and GAI, oral origin has positive coefficients under graphical lasso (0.211;  $p=0.0265$ ) and igraph (0.322;  $p=0.00105$ ). GAI is null in those methods. A Bayesian random-study-intercept re-fit gives  $P(\text{oral coefficient} > 0) = 1.0$  for both with two chains of 500 draws; this is a model check, not an independent cohort. Leaving out each of five multi-genus HOMD families keeps the oral association significant in ten fits (gglasso  $p=0.0197$ -0.0433; igraph  $p=0.0003$ -0.0028).

A separate, pre-registered PubTator3 oral/dental literature fraction did not replicate the binary HOMD effect: despite 203 measured genera, graphical-lasso  $p=0.57$  and igraph  $p=0.23$ . The index correlates with HOMD membership ( $\rho=0.54$ ) but is a different, literature-biased construct. Bio.Phylo gives no oral clustering on the Open Tree topology ( $p=0.53$ ). Thus the defensible claim is a HOMD-list-specific candidate for network hubs under two definitions, not an oral-genus law or causality. The ridge oral coefficient in the per-study model is nonsignificant ( $p=0.17$ ) and its fit did not converge; we do not claim a three-method result.

### 5.6.3 Predictive and software checks

On the ridge-only genus-level keystone fraction, XGBoost with 20 repeats of five-fold cross-validation gains mean out-of-fold  $R^2$  of 0.27 from GAI (20/20 positive deltas) but has negative absolute  $R^2$  (-0.11 with GAI, -0.38 without). That model is not usable for prediction. Separately pre-registered shallower boosters on 200 genera produce mean full-data out-of-fold  $R^2$  0.133 for LightGBM (reduced -0.071) and 0.119 for CatBoost (reduced -0.045), both with 20/20 positive deltas. The modest improvement is within the same dataset and outcome. SHAP TreeExplainer ranks GAI first in the fitted LightGBM model (mean absolute contribution 0.0265 versus ENA 0.00772 and abundance 0.00576), an in-sample explanation rather than proof of causality. SymPy verifies the cNODE and gLV vector fields are tangent to the simplex and zero on absent-taxon boundaries for a generic three-taxon system when abundances sum to one. That algebra does not prove numerical stability or predictive accuracy.

**Table 15. Key post-audit tests. Primary methods, thresholds, preregistrations and full negative results are in results/finding\_keystone.md and the JSON files.**

| analysis                      | scope                     | pre-registered verdict | key caveat            |
|-------------------------------|---------------------------|------------------------|-----------------------|
| gglasso / igraph GAI          | 160 MGnify studies        | FAIL / FAIL            | ridge-specific effect |
| HOMD oral on gglasso / igraph | 46 of 203 GAI genera oral | PASS / PASS            | PubTator index failed |

|                                |                         |             |                       |
|--------------------------------|-------------------------|-------------|-----------------------|
| <b>HOMD family deletions</b>   | 5 families; 10 fits     | PASS        | same curated list     |
| <b>LightGBM / CatBoost</b>     | 200 genera; repeated CV | PASS / PASS | modest, ridge-only R2 |
| <b>Bio.Phylo oral topology</b> | 160 tips; 38 oral       | FAIL        | synthetic topology    |

## 5.7 Pre-registered discovery arms

Two discovery tests were pre-registered (PREREG\_twindiscovery.md) with explicit redirect clauses: T1 asks whether GraphTwin gate out-strength replicates the keystone-consensus signal of Section 5.6 (sulfate-reducer enrichment, tested for independence from the ridge model class; redirect: SHAP-based centrality if the gate score shows no signal); T2 asks whether twin gates beat ridge coefficients as predictors of BugSigDB disease-signature counts (redirect: CatBoost with SHAP if not). Both arms are fit per MGnify study under the same caps as the audit. Both redirect rules, scorers, and test constructions were locked in PREREG\_twindiscovery.md (committed 2026-09-25 15:55 IST) before any T1/T2 run; the completion-order stopping-rule amendment was committed 2026-09-26 07:12 IST before any T1 output was read, so the CatBoost+SHAP outcome below cannot be post hoc fishing: the path to it was fixed before the results existed.

T1 (model-class independence of the keystone enrichment). T1: 37 studies, 56 genera. Top enrichment: Actinomycetota  $q=0.906$ . Desulfobacterota  $q=\text{None}$ . NULL - redirect to SHAP centrality per pre-reg

**Table 16. Top 15 genera by GraphTwin gate out-strength consensus (results/twindiscovery\_t1\_consensus.csv).**

| genus                     | studies | top | fraction top | p                          | q (BH)                     |
|---------------------------|---------|-----|--------------|----------------------------|----------------------------|
| <b>Acinetobacter</b>      | 32      | 22  | 0.688        | 2.362763227460<br>5656e-15 | 1.323147407377<br>9166e-13 |
| <b>Pseudomonas</b>        | 34      | 20  | 0.588        | 3.436707579990<br>6436e-12 | 7.966283892670<br>575e-11  |
| <b>Staphylococcus</b>     | 31      | 19  | 0.613        | 4.267652085359<br>237e-12  | 7.966283892670<br>575e-11  |
| <b>Abiotrophia</b>        | 20      | 15  | 0.750        | 9.481303720000<br>011e-12  | 8.849216805333<br>343e-11  |
| <b>Peptostreptococcus</b> | 20      | 15  | 0.750        | 9.481303720000<br>011e-12  | 8.849216805333<br>343e-11  |
| <b>Solobacterium</b>      | 20      | 15  | 0.750        | 9.481303720000<br>011e-12  | 8.849216805333<br>343e-11  |
| <b>Bifidobacterium</b>    | 26      | 17  | 0.654        | 1.281038079876<br>9867e-11 | 1.024830463901<br>5893e-10 |
| <b>Lautropia</b>          | 21      | 15  | 0.714        | 3.007995688000<br>0025e-11 | 1.684477585280<br>0012e-10 |
| <b>Treponema</b>          | 21      | 15  | 0.714        | 3.007995688000<br>0025e-11 | 1.684477585280<br>0012e-10 |
| <b>Peptococcus</b>        | 21      | 15  | 0.714        | 3.007995688000<br>0025e-11 | 1.684477585280<br>0012e-10 |
| <b>Fastidiosipila</b>     | 24      | 16  | 0.667        | 3.338913051015<br>403e-11  | 1.699810280516<br>9325e-10 |
| <b>Lactobacillus</b>      | 34      | 19  | 0.559        | 4.164943296236<br>0486e-11 | 1.943640204910<br>156e-10  |
| <b>Blautia</b>            | 28      | 17  | 0.607        | 7.224555155610             | 3.112116067032             |

|                    |    |    |       |                            |                           |
|--------------------|----|----|-------|----------------------------|---------------------------|
|                    |    |    |       | 398e-11                    | 171e-10                   |
| <b>Atopobium</b>   | 26 | 16 | 0.615 | 1.980191282012<br>2747e-10 | 7.392714119512<br>492e-10 |
| <b>Haemophilus</b> | 26 | 16 | 0.615 | 1.980191282012<br>2747e-10 | 7.392714119512<br>492e-10 |

**Table 17. Phylum enrichment among the 25 lowest-p genera by twin-gate score (one-sided Fisher, BH).**

| phylum                  | in top 25 | genera | p                       | q (BH)                 |
|-------------------------|-----------|--------|-------------------------|------------------------|
| <b>Actinomycetota</b>   | 4         | 7      | 0.377453351926159<br>8  | 0.905724408776573<br>5 |
| <b>Campylobacterota</b> | 1         | 1      | 0.446428571428571<br>45 | 0.905724408776573<br>5 |
| <b>Spirochaetota</b>    | 1         | 1      | 0.446428571428571<br>45 | 0.905724408776573<br>5 |
| <b>Pseudomonadota</b>   | 4         | 8      | 0.517556805015184<br>9  | 0.905724408776573<br>5 |
| <b>Fusobacteriota</b>   | 1         | 2      | 0.698051948051948       | 0.909292705075169      |
| <b>Bacillota</b>        | 11        | 27     | 0.798163055615896<br>7  | 0.909292705075169      |
| <b>Bacteroidota</b>     | 2         | 7      | 0.909292705075169       | 0.909292705075169      |

T2 (twin gates vs ridge as disease-literature predictor). T2: 56 genera. Twin-gate coef - 2.28 (p=0.168), AIC 767.1 vs ridge 766.1. TWIN DOES NOT WIN - redirect to CatBoost+SHAP per pre-reg

**Table 18. Negative-binomial models of BugSigDB signature counts (results/twindiscovery\_t2.json).**

| model                       | score coef | p     | AIC   | ML alpha | ML coef | ML p   | ML AIC |
|-----------------------------|------------|-------|-------|----------|---------|--------|--------|
| <b>ridge keystone score</b> | -0.351     | 0.241 | 766.1 | 0.62     | -0.351  | 0.152  | 762.0  |
| <b>twin gate frequency</b>  | -2.28      | 0.168 | 767.1 | 0.63     | -2.28   | 0.0737 | 763.7  |

Redirects (locked in the same pre-registration before any T1 run). R1, the scorer redirect to SHAP-attribution centrality from the validated LightGBM keystone model: Desulfobacterota 1/3 in top decile, q = 0.462; no phylum passes FDR (best q = 0.462). The sulfate-reducer enrichment is therefore specific to the ridge out-strength definition; it does not survive either model-class change. R2, the predictor redirect to CatBoost + SHAP per-genus attribution ranking in the same negative-binomial comparison: coefficient +37.07 (p = 2.6e-35), AIC 2343.9 vs ridge 2443.1 on 200 genera - the redirect WINS. Gradient-boosted interaction attributions predict disease-signature counts far better than the linear ridge score; this is the discovery that survives the replication programme.

**Table 19. T2 redirect: negative-binomial models of BugSigDB signature counts (results/keystone\_redirects.json).**

| model                       | score coef | p       | AIC    |
|-----------------------------|------------|---------|--------|
| <b>ridge keystone score</b> | -0.351     | 0.241   | 2443.1 |
| <b>CatBoost+SHAP</b>        | +37.07     | 2.6e-35 | 2343.9 |

|             |  |  |  |
|-------------|--|--|--|
| attribution |  |  |  |
|-------------|--|--|--|

## Appendix B. Full MGnify interaction audit (per study)

**Table 20. All 160 audited studies (results/mgnify\_audit.csv). Gain is (prior - interaction) / prior; positive favours interactions.**

| study               | biome                                        | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|---------------------|----------------------------------------------|-----|------|--------------|--------------------|--------|------------------------|
| <b>MGYS00006862</b> | root:Host-associated: Human:Digestive system | 84  | 55   | 0.173        | 0.170              | 0.004  | 0.1174934141540573     |
| <b>MGYS00005791</b> | root:Host-associated: Human:Digestive system | 243 | 47   | 0.178        | 0.172              | 0.006  | 4.8384265757998056e-05 |
| <b>MGYS00006825</b> | root:Host-associated: Human:Digestive system | 197 | 57   | 0.174        | 0.168              | 0.006  | 0.0051888744323153     |
| <b>MGYS00006824</b> | root:Host-associated: Human:Digestive system | 40  | 66   | 0.175        | 0.180              | -0.004 | 0.97349265041521       |
| <b>MGYS00006822</b> | root:Host-associated: Human:Digestive system | 55  | 58   | 0.192        | 0.189              | 0.003  | 0.0874094952304293     |
| <b>MGYS00006814</b> | root:Host-associated: Human:Digestive system | 68  | 70   | 0.177        | 0.175              | 0.002  | 0.3404810744509645     |
| <b>MGYS00006815</b> | root:Host-associated: Human:Digestive system | 67  | 46   | 0.167        | 0.167              | -0.000 | 0.851347514124289      |
| <b>MGYS00006811</b> | root:Host-associated: Human:Digestive system | 98  | 39   | 0.160        | 0.154              | 0.007  | 0.0116937067032781     |
| <b>MGYS00006807</b> | root:Host-associated: Human:Digestive system | 60  | 65   | 0.179        | 0.180              | -0.001 | 0.192573487700825      |
| <b>MGYS00006805</b> | root:Host-associated: Human:Digestive system | 86  | 66   | 0.180        | 0.162              | 0.017  | 0.0159873022676302     |

|                          |                                                    |     |    |       |       |        |                                |
|--------------------------|----------------------------------------------------|-----|----|-------|-------|--------|--------------------------------|
| <b>MGYS000<br/>06806</b> | root:Host-associated:<br>Human:Digestive<br>system | 74  | 60 | 0.178 | 0.180 | -0.002 | 0.00595618<br>19902366         |
| <b>MGYS000<br/>06760</b> | root:Host-associated:<br>Human:Digestive<br>system | 218 | 69 | 0.169 | 0.164 | 0.005  | 1.18008808<br>41672732e-<br>09 |
| <b>MGYS000<br/>02338</b> | root:Host-associated:<br>Human:Digestive<br>system | 303 | 44 | 0.154 | 0.148 | 0.006  | 0.00272830<br>0451495          |
| <b>MGYS000<br/>06797</b> | root:Host-associated:<br>Human:Digestive<br>system | 175 | 72 | 0.163 | 0.160 | 0.002  | 0.02542058<br>01722315         |
| <b>MGYS000<br/>06551</b> | root:Host-associated:<br>Human:Digestive<br>system | 400 | 63 | 0.167 | 0.162 | 0.005  | 0.00011130<br>60975781         |
| <b>MGYS000<br/>06789</b> | root:Host-associated:<br>Human:Digestive<br>system | 108 | 63 | 0.178 | 0.179 | -0.001 | 0.01695206<br>37921749         |
| <b>MGYS000<br/>06787</b> | root:Host-associated:<br>Human:Digestive<br>system | 108 | 56 | 0.180 | 0.177 | 0.004  | 0.54188612<br>06076111         |
| <b>MGYS000<br/>06786</b> | root:Host-associated:<br>Human:Digestive<br>system | 134 | 52 | 0.173 | 0.169 | 0.004  | 0.01443647<br>16231558         |
| <b>MGYS000<br/>06759</b> | root:Host-associated:<br>Human:Digestive<br>system | 400 | 72 | 0.183 | 0.176 | 0.006  | 3.96471562<br>3438228e-<br>19  |
| <b>MGYS000<br/>06780</b> | root:Host-associated:<br>Human:Digestive<br>system | 24  | 27 | 0.129 | 0.130 | -0.001 | 0.94339205<br>6822656          |
| <b>MGYS000<br/>06779</b> | root:Host-associated:<br>Human:Digestive<br>system | 57  | 63 | 0.185 | 0.185 | -0.000 | 0.64207850<br>27937037         |
| <b>MGYS000<br/>06769</b> | root:Host-associated:<br>Human:Digestive<br>system | 116 | 54 | 0.186 | 0.187 | -0.001 | 0.81699509<br>73439684         |
| <b>MGYS000</b>           | root:Host-                                         | 28  | 43 | 0.154 | 0.159 | -0.005 | 0.64558538                     |

|                          |                                                        |     |     |       |       |        |                                |
|--------------------------|--------------------------------------------------------|-----|-----|-------|-------|--------|--------------------------------|
| <b>06765</b>             | associated:<br>Human:Digestive<br>system               |     |     |       |       |        | 04945946                       |
| <b>MGYS000<br/>05333</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 244 | 53  | 0.165 | 0.158 | 0.008  | 1.81560835<br>69124025e-<br>07 |
| <b>MGYS000<br/>05230</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 400 | 114 | 0.363 | 0.311 | 0.052  | 2.51085031<br>32493804e-<br>33 |
| <b>MGYS000<br/>02675</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 24  | 23  | 0.142 | 0.136 | 0.007  | 0.70475649<br>8336792          |
| <b>MGYS000<br/>06069</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 400 | 48  | 0.167 | 0.158 | 0.009  | 4.34349703<br>1756574e-<br>07  |
| <b>MGYS000<br/>06006</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 400 | 106 | 0.181 | 0.176 | 0.005  | 3.39712018<br>74549978e-<br>22 |
| <b>MGYS000<br/>02061</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 52  | 40  | 0.142 | 0.142 | -0.000 | 0.16015452<br>4509844          |
| <b>MGYS000<br/>00633</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 400 | 68  | 0.572 | 0.438 | 0.135  | 7.59928368<br>626225e-21       |
| <b>MGYS000<br/>02238</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 111 | 111 | 0.402 | 0.301 | 0.101  | 2.36673116<br>8120885e-<br>13  |
| <b>MGYS000<br/>06562</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 400 | 44  | 0.167 | 0.163 | 0.004  | 8.22459679<br>5214574e-<br>11  |
| <b>MGYS000<br/>06649</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 20  | 44  | 0.176 | 0.176 | -0.000 | 0.23051261<br>90185547         |
| <b>MGYS000<br/>06641</b> | root:Host-<br>associated:<br>Human:Digestive<br>system | 87  | 52  | 0.159 | 0.156 | 0.002  | 0.17566108<br>4603356          |
| <b>MGYS000<br/>06642</b> | root:Host-<br>associated:                              | 57  | 19  | 0.149 | 0.150 | -0.001 | 0.06890687<br>93681605         |

|                     |                                                   |     |     |       |       |        |                        |
|---------------------|---------------------------------------------------|-----|-----|-------|-------|--------|------------------------|
|                     | Human:Digestive system                            |     |     |       |       |        |                        |
| <b>MGYS00006364</b> | root:Host-associated: Human:Digestive system      | 160 | 33  | 0.190 | 0.191 | -0.001 | 0.0006405387930733     |
| <b>MGYS00006640</b> | root:Host-associated: Human:Digestive system      | 60  | 53  | 0.170 | 0.176 | -0.006 | 0.6725238936938076     |
| <b>MGYS00005253</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 83  | 0.315 | 0.286 | 0.030  | 5.630794366587372e-13  |
| <b>MGYS00006320</b> | root:Host-associated: Human:Digestive system:Oral | 392 | 68  | 0.347 | 0.284 | 0.063  | 2.6018524371287146e-18 |
| <b>MGYS00006263</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 101 | 0.292 | 0.259 | 0.032  | 1.6144709003228016e-16 |
| <b>MGYS00006017</b> | root:Host-associated: Human:Digestive system:Oral | 53  | 144 | 0.314 | 0.291 | 0.023  | 0.0063127550930951     |
| <b>MGYS00005592</b> | root:Host-associated: Human:Digestive system:Oral | 160 | 104 | 0.323 | 0.257 | 0.065  | 2.954937077626768e-12  |
| <b>MGYS00005569</b> | root:Host-associated: Human:Digestive system:Oral | 71  | 112 | 0.330 | 0.246 | 0.083  | 3.1296273722221307e-09 |
| <b>MGYS00005528</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 68  | 0.312 | 0.267 | 0.045  | 2.9447749914759767e-24 |
| <b>MGYS00005506</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 78  | 0.259 | 0.237 | 0.022  | 1.2543867896369403e-26 |

|                          |                                                         |     |     |       |       |       |                               |
|--------------------------|---------------------------------------------------------|-----|-----|-------|-------|-------|-------------------------------|
| <b>MGYS000<br/>05223</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 400 | 99  | 0.288 | 0.218 | 0.070 | 6.02598443<br>8246577e-36     |
| <b>MGYS000<br/>05143</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 400 | 142 | 0.372 | 0.326 | 0.046 | 7.39947150<br>4901985e-36     |
| <b>MGYS000<br/>02272</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 40  | 96  | 0.525 | 0.491 | 0.035 | 0.01019742<br>31565691        |
| <b>MGYS000<br/>02269</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 147 | 105 | 0.410 | 0.342 | 0.069 | 1.76030560<br>13155362e-16    |
| <b>MGYS000<br/>02229</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 100 | 85  | 0.448 | 0.358 | 0.090 | 5.56345569<br>5355892e-12     |
| <b>MGYS000<br/>05232</b> | root:Host-associated:<br>Human:Skin                     | 400 | 96  | 0.365 | 0.325 | 0.040 | 9.31108298<br>430499e-22      |
| <b>MGYS000<br/>06246</b> | root:Host-associated:<br>Human:Skin                     | 50  | 93  | 0.354 | 0.256 | 0.099 | 0.00651664<br>99849652        |
| <b>MGYS000<br/>05384</b> | root:Host-associated:<br>Human:Skin                     | 400 | 99  | 0.422 | 0.260 | 0.162 | 8.32184474<br>8869964e-43     |
| <b>MGYS000<br/>06795</b> | root:Host-associated:<br>Human:Digestive<br>system      | 31  | 150 | 0.582 | 0.505 | 0.078 | 0.00049540<br>773034095<br>76 |
| <b>MGYS000<br/>06794</b> | root:Host-associated:<br>Human:Digestive<br>system      | 184 | 150 | 0.399 | 0.314 | 0.085 | 3.42255951<br>77609226e-22    |
| <b>MGYS000<br/>06755</b> | root:Host-associated:<br>Human:Digestive<br>system      | 83  | 150 | 0.562 | 0.375 | 0.187 | 1.35755614<br>79460155e-13    |
| <b>MGYS000<br/>05154</b> | root:Host-associated:<br>Human:Digestive                | 400 | 150 | 0.492 | 0.389 | 0.102 | 9.62314736<br>9597524e-46     |

|                          |                                                         |     |     |       |       |       |                                |
|--------------------------|---------------------------------------------------------|-----|-----|-------|-------|-------|--------------------------------|
|                          | system                                                  |     |     |       |       |       |                                |
| <b>MGYS000<br/>06276</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 91  | 150 | 0.298 | 0.283 | 0.015 | 0.00018677<br>518390145<br>96  |
| <b>MGYS000<br/>05189</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 400 | 150 | 0.656 | 0.407 | 0.249 | 6.40060744<br>1863894e-<br>60  |
| <b>MGYS000<br/>05197</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 400 | 150 | 0.197 | 0.170 | 0.027 | 7.29819806<br>7237661e-<br>42  |
| <b>MGYS000<br/>02159</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 21  | 150 | 0.573 | 0.494 | 0.080 | 0.08879852<br>294921875        |
| <b>MGYS000<br/>03719</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 21  | 150 | 0.306 | 0.287 | 0.018 | 0.05469322<br>204589844        |
| <b>MGYS000<br/>02395</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 68  | 150 | 0.351 | 0.346 | 0.005 | 0.02572892<br>477295064<br>5   |
| <b>MGYS000<br/>02394</b> | root:Host-associated:<br>Human:Digestive<br>system:Oral | 20  | 150 | 0.417 | 0.400 | 0.016 | 0.16495704<br>650878906        |
| <b>MGYS000<br/>06648</b> | root:Host-associated:<br>Human:Skin                     | 67  | 150 | 0.470 | 0.344 | 0.126 | 1.01927407<br>68516179e-<br>07 |
| <b>MGYS000<br/>05156</b> | root:Host-associated:<br>Human:Skin                     | 400 | 150 | 0.399 | 0.339 | 0.060 | 3.75845143<br>83668004e-<br>33 |
| <b>MGYS000<br/>05172</b> | root:Host-associated:<br>Human:Skin                     | 400 | 150 | 0.401 | 0.302 | 0.099 | 1.94126197<br>07417544e-<br>34 |
| <b>MGYS000<br/>05533</b> | root:Host-associated:<br>Human:Skin                     | 400 | 150 | 0.317 | 0.316 | 0.001 | 8.61427531<br>9605027e-<br>22  |
| <b>MGYS000<br/>05212</b> | root:Host-associated:                                   | 400 | 150 | 0.374 | 0.292 | 0.083 | 4.02228790<br>63051704e-       |

|                     |                                  |     |     |       |       |        |                        |
|---------------------|----------------------------------|-----|-----|-------|-------|--------|------------------------|
|                     | Human:Skin                       |     |     |       |       |        | 33                     |
| <b>MGYS00005238</b> | root:Host-associated: Human:Skin | 400 | 106 | 0.396 | 0.358 | 0.039  | 2.7355102946682556e-13 |
| <b>MGYS00005101</b> | root:Host-associated: Human:Skin | 71  | 91  | 0.367 | 0.292 | 0.075  | 0.001176166597423719   |
| <b>MGYS00005102</b> | root:Host-associated: Human:Skin | 179 | 150 | 0.381 | 0.264 | 0.118  | 7.298919552265234e-19  |
| <b>MGYS00005135</b> | root:Host-associated: Human:Skin | 109 | 150 | 0.470 | 0.420 | 0.050  | 1.4252348445599325e-07 |
| <b>MGYS00005098</b> | root:Host-associated: Human:Skin | 101 | 150 | 0.127 | 0.102 | 0.025  | 1.5388621128648606e-10 |
| <b>MGYS00005037</b> | root:Host-associated: Human:Skin | 106 | 150 | 0.311 | 0.233 | 0.078  | 6.006215258961562e-07  |
| <b>MGYS00005090</b> | root:Host-associated: Human:Skin | 120 | 7   | 0.082 | 0.077 | 0.005  | 0.9498798977422346     |
| <b>MGYS00002184</b> | root:Host-associated: Human:Skin | 400 | 150 | 0.471 | 0.261 | 0.210  | 2.878775655656423e-63  |
| <b>MGYS00002072</b> | root:Host-associated: Human:Skin | 258 | 150 | 0.233 | 0.190 | 0.043  | 2.0058912688460236e-19 |
| <b>MGYS00006855</b> | root:Host-associated: Mammals    | 116 | 150 | 0.293 | 0.232 | 0.061  | 1.6553837550067384e-12 |
| <b>MGYS00006799</b> | root:Host-associated: Mammals    | 27  | 150 | 0.406 | 0.417 | -0.011 | 0.03189583122730255    |
| <b>MGYS00002673</b> | root:Host-associated: Mammals    | 93  | 150 | 0.650 | 0.533 | 0.116  | 1.1556875270009098e-13 |
| <b>MGYS00006747</b> | root:Host-associated: Mammals    | 101 | 150 | 0.568 | 0.386 | 0.182  | 4.170399034793852e-18  |
| <b>MGYS00001223</b> | root:Host-associated: Mammals    | 30  | 98  | 0.194 | 0.173 | 0.021  | 0.08406547829508781    |
| <b>MGYS00001224</b> | root:Host-associated: Mammals    | 66  | 89  | 0.389 | 0.242 | 0.147  | 1.6687130603014278e-09 |
| <b>MGYS00005267</b> | root:Host-associated: Mammals    | 400 | 71  | 0.343 | 0.254 | 0.088  | 3.2675706524688328e-52 |
| <b>MGYS000</b>      | root:Host-                       | 44  | 66  | 0.272 | 0.245 | 0.027  | 0.40119774             |

|                          |                                      |     |     |       |       |        |                                |
|--------------------------|--------------------------------------|-----|-----|-------|-------|--------|--------------------------------|
| <b>00426</b>             | associated:<br>Mammals               |     |     |       |       |        | 86671867                       |
| <b>MGYS000<br/>05228</b> | root:Host-<br>associated:<br>Mammals | 400 | 63  | 0.257 | 0.206 | 0.052  | 2.07036600<br>9312925e-<br>33  |
| <b>MGYS000<br/>06537</b> | root:Host-<br>associated:<br>Mammals | 85  | 150 | 0.495 | 0.407 | 0.088  | 0.00037059<br>731217404<br>824 |
| <b>MGYS000<br/>06536</b> | root:Host-<br>associated:<br>Mammals | 65  | 150 | 0.384 | 0.384 | -0.000 | 8.46763942<br>4268034e-<br>05  |
| <b>MGYS000<br/>06538</b> | root:Host-<br>associated:<br>Mammals | 40  | 150 | 0.515 | 0.512 | 0.003  | 0.08415465<br>162215696        |
| <b>MGYS000<br/>05628</b> | root:Host-<br>associated:<br>Mammals | 400 | 150 | 0.338 | 0.276 | 0.062  | 7.07353919<br>3762629e-<br>38  |
| <b>MGYS000<br/>06450</b> | root:Host-<br>associated:<br>Mammals | 144 | 81  | 0.444 | 0.310 | 0.134  | 1.82568172<br>38051636e-<br>13 |
| <b>MGYS000<br/>06456</b> | root:Host-<br>associated:<br>Mammals | 24  | 76  | 0.331 | 0.313 | 0.017  | 0.97573611<br>9417176          |
| <b>MGYS000<br/>06446</b> | root:Host-<br>associated:<br>Mammals | 400 | 62  | 0.317 | 0.247 | 0.070  | 4.77927826<br>1362484e-<br>25  |
| <b>MGYS000<br/>06433</b> | root:Host-<br>associated:<br>Mammals | 25  | 150 | 0.557 | 0.537 | 0.020  | 0.01874470<br>710754394<br>5   |
| <b>MGYS000<br/>06430</b> | root:Host-<br>associated:<br>Mammals | 32  | 150 | 0.397 | 0.299 | 0.099  | 0.01411853<br>372062440<br>6   |
| <b>MGYS000<br/>06427</b> | root:Host-<br>associated:<br>Mammals | 85  | 109 | 0.389 | 0.341 | 0.048  | 0.00245958<br>589664498        |
| <b>MGYS000<br/>06424</b> | root:Host-<br>associated:<br>Mammals | 142 | 150 | 0.280 | 0.252 | 0.027  | 2.51005437<br>78701536e-<br>14 |
| <b>MGYS000<br/>06652</b> | root:Host-<br>associated:<br>Plants  | 400 | 150 | 0.250 | 0.175 | 0.076  | 2.87195734<br>82573203e-<br>56 |
| <b>MGYS000<br/>01412</b> | root:Host-<br>associated:<br>Plants  | 48  | 98  | 0.727 | 0.639 | 0.088  | 0.00449954<br>731304558<br>2   |
| <b>MGYS000<br/>01368</b> | root:Host-<br>associated:<br>Plants  | 400 | 150 | 0.276 | 0.190 | 0.086  | 5.31521348<br>0136056e-<br>65  |
| <b>MGYS000<br/>03172</b> | root:Host-<br>associated:<br>Plants  | 45  | 150 | 0.247 | 0.101 | 0.145  | 5.68434188<br>6080802e-<br>14  |
| <b>MGYS000<br/>06578</b> | root:Host-<br>associated:<br>Plants  | 27  | 150 | 0.162 | 0.170 | -0.009 | 0.91531434<br>65518951         |
| <b>MGYS000<br/>00957</b> | root:Host-<br>associated:<br>Plants  | 70  | 61  | 0.381 | 0.351 | 0.030  | 0.00105925<br>524374496<br>15  |
| <b>MGYS000<br/>01704</b> | root:Host-<br>associated:<br>Plants  | 383 | 150 | 0.252 | 0.164 | 0.088  | 2.43260798<br>6872169e-<br>55  |
| <b>MGYS000</b>           | root:Host-                           | 32  | 150 | 0.001 | 0.001 | 0.000  | 0.34500763                     |

|                          |                                     |     |     |       |       |       |                                |
|--------------------------|-------------------------------------|-----|-----|-------|-------|-------|--------------------------------|
| <b>00867</b>             | associated:<br>Plants               |     |     |       |       |       | 198357975                      |
| <b>MGYS000<br/>06031</b> | root:Host-<br>associated:<br>Plants | 89  | 150 | 0.477 | 0.402 | 0.075 | 1.10543002<br>01400448e-<br>07 |
| <b>MGYS000<br/>06021</b> | root:Host-<br>associated:<br>Plants | 135 | 18  | 0.233 | 0.216 | 0.017 | 0.00353824<br>774085654<br>84  |
| <b>MGYS000<br/>06018</b> | root:Host-<br>associated:<br>Plants | 181 | 150 | 0.452 | 0.287 | 0.166 | 7.97625601<br>5721753e-<br>13  |
| <b>MGYS000<br/>01349</b> | root:Host-<br>associated:<br>Plants | 108 | 150 | 0.402 | 0.211 | 0.190 | 9.80011598<br>2359504e-<br>18  |
| <b>MGYS000<br/>05260</b> | root:Host-<br>associated:<br>Plants | 400 | 150 | 0.396 | 0.248 | 0.148 | 1.27453133<br>76300463e-<br>61 |
| <b>MGYS000<br/>05227</b> | root:Host-<br>associated:<br>Plants | 400 | 150 | 0.223 | 0.201 | 0.022 | 2.47551455<br>03466964e-<br>21 |
| <b>MGYS000<br/>00983</b> | root:Host-<br>associated:<br>Plants | 400 | 150 | 0.178 | 0.141 | 0.036 | 7.25062466<br>0443202e-<br>39  |
| <b>MGYS000<br/>01347</b> | root:Host-<br>associated:<br>Plants | 400 | 150 | 0.471 | 0.218 | 0.252 | 6.62037148<br>2312783e-<br>66  |
| <b>MGYS000<br/>05740</b> | root:Host-<br>associated:<br>Plants | 34  | 150 | 0.400 | 0.339 | 0.062 | 3.21888364<br>85147476e-<br>07 |
| <b>MGYS000<br/>05726</b> | root:Host-<br>associated:<br>Plants | 33  | 150 | 0.382 | 0.337 | 0.046 | 5.09948004<br>0371418e-<br>05  |
| <b>MGYS000<br/>05169</b> | root:Host-<br>associated:<br>Plants | 400 | 150 | 0.311 | 0.171 | 0.140 | 2.36530085<br>51996965e-<br>59 |
| <b>MGYS000<br/>05151</b> | root:Host-<br>associated:<br>Plants | 400 | 150 | 0.348 | 0.111 | 0.238 | 6.53805605<br>1035535e-<br>60  |
| <b>MGYS000<br/>06750</b> | root:Host-<br>associated:<br>Birds  | 340 | 89  | 0.142 | 0.135 | 0.007 | 1.15568776<br>32363245e-<br>12 |
| <b>MGYS000<br/>06533</b> | root:Host-<br>associated:<br>Birds  | 94  | 150 | 0.404 | 0.322 | 0.081 | 2.12030491<br>63202955e-<br>09 |
| <b>MGYS000<br/>06532</b> | root:Host-<br>associated:<br>Birds  | 40  | 150 | 0.478 | 0.252 | 0.226 | 5.72326825<br>9316683e-<br>08  |
| <b>MGYS000<br/>05766</b> | root:Host-<br>associated:<br>Birds  | 138 | 150 | 0.474 | 0.245 | 0.229 | 7.80655073<br>5727007e-<br>23  |
| <b>MGYS000<br/>06062</b> | root:Host-<br>associated:<br>Birds  | 220 | 92  | 0.186 | 0.166 | 0.020 | 2.16529527<br>39860382e-<br>13 |
| <b>MGYS000<br/>06042</b> | root:Host-<br>associated:<br>Birds  | 355 | 96  | 0.284 | 0.124 | 0.160 | 1.10282523<br>58036625e-<br>43 |
| <b>MGYS000<br/>05621</b> | root:Host-<br>associated:<br>Birds  | 60  | 150 | 0.265 | 0.173 | 0.092 | 1.01811542<br>08756968e-<br>06 |
| <b>MGYS000</b>           | root:Host-                          | 117 | 115 | 0.334 | 0.178 | 0.156 | 3.79430763                     |

|                          |                                    |     |     |       |       |        |                                |
|--------------------------|------------------------------------|-----|-----|-------|-------|--------|--------------------------------|
| <b>05611</b>             | associated:<br>Birds               |     |     |       |       |        | 036704e-14                     |
| <b>MGYS000<br/>05610</b> | root:Host-<br>associated:<br>Birds | 251 | 87  | 0.376 | 0.317 | 0.059  | 1.14433573<br>06387731e-<br>14 |
| <b>MGYS000<br/>05609</b> | root:Host-<br>associated:<br>Birds | 314 | 68  | 0.470 | 0.395 | 0.075  | 1.37853599<br>22199967e-<br>23 |
| <b>MGYS000<br/>05600</b> | root:Host-<br>associated:<br>Birds | 286 | 150 | 0.270 | 0.175 | 0.095  | 2.85879651<br>04295026e-<br>40 |
| <b>MGYS000<br/>05593</b> | root:Host-<br>associated:<br>Birds | 150 | 129 | 0.589 | 0.361 | 0.228  | 1.90229915<br>82307127e-<br>20 |
| <b>MGYS000<br/>05576</b> | root:Host-<br>associated:<br>Birds | 34  | 150 | 0.374 | 0.220 | 0.153  | 3.57395038<br>0086899e-<br>08  |
| <b>MGYS000<br/>05187</b> | root:Host-<br>associated:<br>Birds | 400 | 4   | 0.147 | 0.131 | 0.016  | 0.20038180<br>242578474        |
| <b>MGYS000<br/>05316</b> | root:Host-<br>associated:<br>Birds | 400 | 145 | 0.542 | 0.257 | 0.285  | 1.50609541<br>9477192e-<br>57  |
| <b>MGYS000<br/>05317</b> | root:Host-<br>associated:<br>Birds | 400 | 150 | 0.491 | 0.349 | 0.141  | 1.35229788<br>10793892e-<br>53 |
| <b>MGYS000<br/>05309</b> | root:Host-<br>associated:<br>Birds | 72  | 150 | 0.312 | 0.194 | 0.118  | 1.67090268<br>14241643e-<br>12 |
| <b>MGYS000<br/>04995</b> | root:Host-<br>associated:<br>Birds | 43  | 150 | 0.256 | 0.223 | 0.033  | 0.10220764<br>011114625        |
| <b>MGYS000<br/>03706</b> | root:Host-<br>associated:<br>Birds | 400 | 150 | 0.516 | 0.422 | 0.094  | 2.04612542<br>36329354e-<br>34 |
| <b>MGYS000<br/>03681</b> | root:Host-<br>associated:<br>Birds | 122 | 150 | 0.674 | 0.444 | 0.231  | 4.10870391<br>4852822e-<br>12  |
| <b>MGYS000<br/>06764</b> | root:Host-<br>associated:<br>Fish  | 21  | 150 | 0.262 | 0.181 | 0.081  | 1.81198120<br>1171875e-<br>05  |
| <b>MGYS000<br/>06763</b> | root:Host-<br>associated:<br>Fish  | 39  | 150 | 0.420 | 0.364 | 0.055  | 2.37966596<br>61456943e-<br>05 |
| <b>MGYS000<br/>06762</b> | root:Host-<br>associated:<br>Fish  | 137 | 150 | 0.428 | 0.290 | 0.138  | 1.85140109<br>8428866e-<br>14  |
| <b>MGYS000<br/>06761</b> | root:Host-<br>associated:<br>Fish  | 123 | 150 | 0.295 | 0.231 | 0.064  | 8.88040017<br>5577094e-<br>10  |
| <b>MGYS000<br/>06758</b> | root:Host-<br>associated:<br>Fish  | 135 | 143 | 0.522 | 0.312 | 0.209  | 2.60129849<br>8785815e-<br>10  |
| <b>MGYS000<br/>06041</b> | root:Host-<br>associated:<br>Fish  | 400 | 30  | 0.278 | 0.164 | 0.113  | 6.99532350<br>7416229e-<br>32  |
| <b>MGYS000<br/>06117</b> | root:Host-<br>associated:<br>Fish  | 90  | 72  | 0.028 | 0.030 | -0.002 | 1.00430172<br>78491953e-<br>06 |
| <b>MGYS000</b>           | root:Host-                         | 135 | 3   | 0.000 | 0.000 | 0.000  | 1.0                            |

|                          |                                      |     |     |       |       |        |                                |
|--------------------------|--------------------------------------|-----|-----|-------|-------|--------|--------------------------------|
| <b>06086</b>             | associated:<br>Fish                  |     |     |       |       |        |                                |
| <b>MGYS000<br/>05503</b> | root:Host-<br>associated:<br>Fish    | 400 | 150 | 0.594 | 0.467 | 0.127  | 1.99726467<br>03767122e-<br>30 |
| <b>MGYS000<br/>03166</b> | root:Host-<br>associated:<br>Fish    | 70  | 150 | 0.417 | 0.352 | 0.065  | 8.97357056<br>8522901e-<br>07  |
| <b>MGYS000<br/>04672</b> | root:Host-<br>associated:<br>Fish    | 33  | 150 | 0.422 | 0.372 | 0.050  | 0.05752792<br>255952954        |
| <b>MGYS000<br/>02905</b> | root:Host-<br>associated:<br>Fish    | 66  | 134 | 0.191 | 0.069 | 0.122  | 2.10786693<br>2199474e-<br>09  |
| <b>MGYS000<br/>02608</b> | root:Host-<br>associated:<br>Fish    | 101 | 150 | 0.589 | 0.429 | 0.161  | 4.03795190<br>0494873e-<br>09  |
| <b>MGYS000<br/>02413</b> | root:Host-<br>associated:<br>Fish    | 173 | 150 | 0.588 | 0.356 | 0.232  | 1.50850834<br>1940648e-<br>16  |
| <b>MGYS000<br/>02065</b> | root:Host-<br>associated:<br>Fish    | 214 | 150 | 0.499 | 0.363 | 0.136  | 6.02308553<br>7966328e-<br>20  |
| <b>MGYS000<br/>06798</b> | root:Host-<br>associated:I<br>nsecta | 24  | 150 | 0.567 | 0.174 | 0.393  | 8.34465026<br>8554688e-<br>06  |
| <b>MGYS000<br/>06019</b> | root:Host-<br>associated:I<br>nsecta | 20  | 150 | 0.345 | 0.215 | 0.130  | 0.00020980<br>834960937<br>5   |
| <b>MGYS000<br/>05862</b> | root:Host-<br>associated:I<br>nsecta | 21  | 150 | 0.321 | 0.346 | -0.026 | 0.05950736<br>999511719        |
| <b>MGYS000<br/>05215</b> | root:Host-<br>associated:I<br>nsecta | 400 | 92  | 0.296 | 0.244 | 0.052  | 6.34580929<br>8780021e-<br>27  |
| <b>MGYS000<br/>05163</b> | root:Host-<br>associated:I<br>nsecta | 386 | 150 | 0.548 | 0.467 | 0.081  | 2.78972200<br>4048674e-<br>31  |
| <b>MGYS000<br/>03800</b> | root:Host-<br>associated:I<br>nsecta | 154 | 96  | 0.184 | 0.184 | 0.000  | 0.67060239<br>00543987         |
| <b>MGYS000<br/>03705</b> | root:Host-<br>associated:I<br>nsecta | 384 | 70  | 0.408 | 0.370 | 0.038  | 2.76869781<br>73767286e-<br>09 |

## Appendix C. Keystone consensus, all genera

Table 21. All 248 genera modelled in  $\geq 20$  studies (results/keystone\_consensus.csv).

| genus                  | studies | top | fraction top | p                         | q (BH)                    |
|------------------------|---------|-----|--------------|---------------------------|---------------------------|
| <b>Cardiobacterium</b> | 30      | 13  | 0.433        | 2.302322724161<br>098e-06 | 0.000570976035<br>5919523 |
| <b>Eikenella</b>       | 28      | 11  | 0.393        | 4.232321631540<br>403e-05 | 0.005248078823<br>1101    |
| <b>Comamonas</b>       | 51      | 13  | 0.255        | 0.001226149989<br>1986897 | 0.098625028903<br>93288   |
| <b>Johnsonella</b>     | 35      | 10  | 0.286        | 0.001742210385<br>1516847 | 0.098625028903<br>93288   |

|                              |    |    |       |                       |                     |
|------------------------------|----|----|-------|-----------------------|---------------------|
| <b>Simonsiella</b>           | 20 | 7  | 0.350 | 0.002386089408966118  | 0.09862502890393288 |
| <b>Desulfobulbus</b>         | 20 | 7  | 0.350 | 0.002386089408966118  | 0.09862502890393288 |
| <b>Sutterella</b>            | 56 | 13 | 0.232 | 0.0030302254033037486 | 0.10735655714561852 |
| <b>Agathobacter</b>          | 77 | 16 | 0.208 | 0.003583040147751078  | 0.11107424458028342 |
| <b>Collinsella</b>           | 67 | 14 | 0.209 | 0.005889699276757321  | 0.16110995747664922 |
| <b>Desulfovibrio</b>         | 68 | 14 | 0.206 | 0.00674761803544495   | 0.16110995747664922 |
| <b>Anaerosporbacter</b>      | 30 | 8  | 0.267 | 0.00778361923792132   | 0.16110995747664922 |
| <b>Solobacterium</b>         | 49 | 11 | 0.224 | 0.008004971888016863  | 0.16110995747664922 |
| <b>Bilophila</b>             | 56 | 12 | 0.214 | 0.008445280029017902  | 0.16110995747664922 |
| <b>Barnesiella</b>           | 47 | 10 | 0.213 | 0.016260359901383525  | 0.2880406611102225  |
| <b>Bergeyella</b>            | 49 | 10 | 0.204 | 0.02150126888113899   | 0.3504201263576335  |
| <b>Anaerovorax</b>           | 23 | 6  | 0.261 | 0.022607750087589258  | 0.3504201263576335  |
| <b>Odoribacter</b>           | 57 | 11 | 0.193 | 0.02423840456349196   | 0.3535955489262357  |
| <b>Cetobacterium</b>         | 30 | 7  | 0.233 | 0.025826788661909898  | 0.35583575489742525 |
| <b>Mogibacterium</b>         | 45 | 9  | 0.200 | 0.03199402184186363   | 0.4128142079523551  |
| <b>Eubacterium</b>           | 60 | 11 | 0.183 | 0.03420912307873303   | 0.4128142079523551  |
| <b>Subdoligranulum</b>       | 83 | 14 | 0.169 | 0.03546700143308487   | 0.4128142079523551  |
| <b>Slackia</b>               | 39 | 8  | 0.205 | 0.03662061522157989   | 0.4128142079523551  |
| <b>Campylobacter</b>         | 69 | 12 | 0.174 | 0.04001789514875476   | 0.43149730421266    |
| <b>Enhydrobacter</b>         | 20 | 5  | 0.250 | 0.04317449528446339   | 0.44613645127278834 |
| <b>Fastidiosipila</b>        | 86 | 14 | 0.163 | 0.04606000187058948   | 0.453852358283953   |
| <b>Stomatobaculum</b>        | 34 | 7  | 0.206 | 0.04814432662912472   | 0.453852358283953   |
| <b>Phascolarctobacterium</b> | 56 | 10 | 0.179 | 0.04941134545833359   | 0.453852358283953   |
| <b>Fournierella</b>          | 28 | 6  | 0.214 | 0.055006948868291425  | 0.470404252390906   |
| <b>Propionibacterium</b>     | 28 | 6  | 0.214 | 0.055006948868291425  | 0.470404252390906   |
| <b>Moryella</b>              | 43 | 8  | 0.186 | 0.06066935159122358   | 0.4831866178663752  |
| <b>Aggregatibacter</b>       | 36 | 7  | 0.194 | 0.06282814237583728   | 0.4831866178663752  |
| <b>Fretibacterium</b>        | 29 | 6  | 0.207 | 0.0637174444522859    | 0.4831866178663752  |
| <b>Leptotrichia</b>          | 51 | 9  | 0.176 | 0.06429499350641282   | 0.4831866178663752  |
| <b>Dolosigranulum</b>        | 37 | 7  | 0.189 | 0.071085058271        | 0.518502777982      |

|                                 |    |    |       |                         |                        |
|---------------------------------|----|----|-------|-------------------------|------------------------|
| <b>m</b>                        |    |    |       | 75729                   | 2297                   |
| <b>Anaeroplasma</b>             | 30 | 6  | 0.200 | 0.073190108399<br>87991 | 0.518604196662<br>0062 |
| <b>Papillibacter</b>            | 24 | 5  | 0.208 | 0.085074885878<br>67089 | 0.577705898153<br>1825 |
| <b>Cutibacterium</b>            | 54 | 9  | 0.167 | 0.086189992869<br>62803 | 0.577705898153<br>1825 |
| <b>Anaerotruncus</b>            | 39 | 7  | 0.179 | 0.089433889739<br>4184  | 0.583673806720<br>4149 |
| <b>Abiotrophia</b>              | 40 | 7  | 0.175 | 0.099516423965<br>55122 | 0.626546230147<br>6376 |
| <b>Lautropia</b>                | 56 | 9  | 0.161 | 0.102970794170<br>76241 | 0.626546230147<br>6376 |
| <b>Moraxella</b>                | 33 | 6  | 0.182 | 0.106108635750<br>80959 | 0.626546230147<br>6376 |
| <b>Butyricimonas</b>            | 33 | 6  | 0.182 | 0.106108635750<br>80959 | 0.626546230147<br>6376 |
| <b>Butyricicoccus</b>           | 66 | 10 | 0.152 | 0.119923548936<br>55092 | 0.672906616565<br>6145 |
| <b>Enterorhabdus</b>            | 42 | 7  | 0.167 | 0.121447206815<br>06566 | 0.672906616565<br>6145 |
| <b>Hydrogenoanaerobacterium</b> | 27 | 5  | 0.185 | 0.126557448308<br>2461  | 0.672906616565<br>6145 |
| <b>Tumebacillus</b>             | 27 | 5  | 0.185 | 0.126557448308<br>2461  | 0.672906616565<br>6145 |
| <b>Leuconostoc</b>              | 35 | 6  | 0.171 | 0.131635774841<br>83725 | 0.672906616565<br>6145 |
| <b>Saccharofermentans</b>       | 20 | 4  | 0.200 | 0.132953323434<br>3351  | 0.672906616565<br>6145 |
| <b>Marinomonas</b>              | 20 | 4  | 0.200 | 0.132953323434<br>3351  | 0.672906616565<br>6145 |
| <b>Parasutterella</b>           | 60 | 9  | 0.150 | 0.141635604798<br>3447  | 0.691053967992<br>992  |
| <b>Ammopiptanthus</b>           | 28 | 5  | 0.179 | 0.142111904708<br>23626 | 0.691053967992<br>992  |
| <b>Finegoldia</b>               | 36 | 6  | 0.167 | 0.145397301335<br>03725 | 0.693433283290<br>1777 |
| <b>Alistipes</b>                | 78 | 11 | 0.141 | 0.153620635614<br>8474  | 0.718828634575<br>135  |
| <b>Ruminococcus</b>             | 45 | 7  | 0.156 | 0.158507385398<br>35585 | 0.727959844051<br>7083 |
| <b>Paucibacter</b>              | 88 | 12 | 0.136 | 0.167051115612<br>0358  | 0.753248666759<br>725  |
| <b>Carnobacterium</b>           | 22 | 4  | 0.182 | 0.171927897066<br>45586 | 0.761394972722<br>8759 |
| <b>Erysipelothrix</b>           | 23 | 4  | 0.174 | 0.192731013518<br>62001 | 0.838548971098<br>5572 |
| <b>Oribacterium</b>             | 65 | 9  | 0.138 | 0.198722696795<br>465   | 0.849710841470<br>2642 |
| <b>Neisseria</b>                | 66 | 9  | 0.136 | 0.211164667510<br>03035 | 0.871098939074<br>7419 |
| <b>Fluviicola</b>               | 32 | 5  | 0.156 | 0.211498288699<br>32752 | 0.871098939074<br>7419 |
| <b>Ezakiella</b>                | 24 | 4  | 0.167 | 0.214262239046<br>6099  | 0.871098939074<br>7419 |
| <b>Akkermansia</b>              | 67 | 9  | 0.134 | 0.223906961655<br>63617 | 0.876994694286<br>9572 |
| <b>Selenomonas</b>              | 33 | 5  | 0.152 | 0.230317869582<br>99133 | 0.876994694286<br>9572 |

|                               |    |    |       |                     |                    |
|-------------------------------|----|----|-------|---------------------|--------------------|
| <b>Methanobrevibacter</b>     | 33 | 5  | 0.152 | 0.23031786958299133 | 0.8769946942869572 |
| <b>Nocardia</b>               | 25 | 4  | 0.160 | 0.23640864244682802 | 0.8769946942869572 |
| <b>Fibrobacter</b>            | 25 | 4  | 0.160 | 0.23640864244682802 | 0.8769946942869572 |
| <b>Megasphaera</b>            | 68 | 9  | 0.132 | 0.2369300182146215  | 0.8769946942869572 |
| <b>Lachnoanaerobaculum</b>    | 34 | 5  | 0.147 | 0.2495917162121227  | 0.8891872058135635 |
| <b>Kingella</b>               | 34 | 5  | 0.147 | 0.2495917162121227  | 0.8891872058135635 |
| <b>Parabacteroides</b>        | 87 | 11 | 0.126 | 0.2509802597054413  | 0.8891872058135635 |
| <b>Catonella</b>              | 35 | 5  | 0.143 | 0.2692510397738372  | 0.9404825051255159 |
| <b>Corynebacterium</b>        | 91 | 11 | 0.121 | 0.29998647993678423 | 1.0                |
| <b>Tannerella</b>             | 28 | 4  | 0.143 | 0.3054336969081328  | 1.0                |
| <b>Candidatus_Solaeferrea</b> | 37 | 5  | 0.135 | 0.30945356374288324 | 1.0                |
| <b>Lachnospira</b>            | 55 | 7  | 0.127 | 0.30955723920441663 | 1.0                |
| <b>Anaerostipes</b>           | 83 | 10 | 0.120 | 0.3162541477570478  | 1.0                |
| <b>Bradyrhizobium</b>         | 65 | 8  | 0.123 | 0.32314240394111926 | 1.0                |
| <b>Olsenella</b>              | 47 | 6  | 0.128 | 0.3285688971461327  | 1.0                |
| <b>Acetitomaculum</b>         | 29 | 4  | 0.138 | 0.328952034984918   | 1.0                |
| <b>Pedobacter</b>             | 38 | 5  | 0.132 | 0.329863552054893   | 1.0                |
| <b>Atopobium</b>              | 57 | 7  | 0.123 | 0.34289143772806113 | 1.0                |
| <b>Gemmata</b>                | 21 | 3  | 0.143 | 0.3515911755169639  | 1.0                |
| <b>Negativibacillus</b>       | 58 | 7  | 0.121 | 0.35972337214077327 | 1.0                |
| <b>Treponema</b>              | 40 | 5  | 0.125 | 0.3709823034657993  | 1.0                |
| <b>Anaerofilum</b>            | 22 | 3  | 0.136 | 0.37995906158809667 | 1.0                |
| <b>Phaselicystis</b>          | 22 | 3  | 0.136 | 0.37995906158809667 | 1.0                |
| <b>Marvinbryantia</b>         | 50 | 6  | 0.120 | 0.3838769922757232  | 1.0                |
| <b>Brevundimonas</b>          | 69 | 8  | 0.116 | 0.3854441012423575  | 1.0                |
| <b>Stenotrophomonas</b>       | 60 | 7  | 0.117 | 0.39354859789913854 | 1.0                |
| <b>Coxiella</b>               | 33 | 4  | 0.121 | 0.4230563358743078  | 1.0                |
| <b>Phenylobacterium</b>       | 43 | 5  | 0.116 | 0.43252905747178994 | 1.0                |
| <b>Coprobacter</b>            | 24 | 3  | 0.125 | 0.4357262730487911  | 1.0                |
| <b>Filifactor</b>             | 24 | 3  | 0.125 | 0.435726273048      | 1.0                |

|                                     |     |    |       |                         |     |
|-------------------------------------|-----|----|-------|-------------------------|-----|
|                                     |     |    |       | 7911                    |     |
| <b>Shewanella</b>                   | 24  | 3  | 0.125 | 0.435726273048<br>7911  | 1.0 |
| <b>Blautia</b>                      | 102 | 11 | 0.108 | 0.443203066400<br>9525  | 1.0 |
| <b>Holdemania</b>                   | 34  | 4  | 0.118 | 0.446184951829<br>2653  | 1.0 |
| <b>Enterobacter</b>                 | 35  | 4  | 0.114 | 0.469015134029<br>96614 | 1.0 |
| <b>Aeromonas</b>                    | 36  | 4  | 0.111 | 0.491488594633<br>7808  | 1.0 |
| <b>Caulobacter</b>                  | 36  | 4  | 0.111 | 0.491488594633<br>7808  | 1.0 |
| <b>Pseudopropioni<br/>bacterium</b> | 27  | 3  | 0.111 | 0.515418858307<br>9999  | 1.0 |
| <b>Parvimonas</b>                   | 48  | 5  | 0.104 | 0.531439275847<br>2106  | 1.0 |
| <b>Mycoplasma</b>                   | 48  | 5  | 0.104 | 0.531439275847<br>2106  | 1.0 |
| <b>Brachybacteriu<br/>m</b>         | 38  | 4  | 0.105 | 0.535164022722<br>7555  | 1.0 |
| <b>Sphingobacteri<br/>um</b>        | 39  | 4  | 0.103 | 0.556280642562<br>8786  | 1.0 |
| <b>Acinetobacter</b>                | 101 | 10 | 0.099 | 0.561751462265<br>9117  | 1.0 |
| <b>Enterococcus</b>                 | 91  | 9  | 0.099 | 0.565055679835<br>745   | 1.0 |
| <b>Arthrobacter</b>                 | 50  | 5  | 0.100 | 0.568801593170<br>9377  | 1.0 |
| <b>Dorea</b>                        | 71  | 7  | 0.099 | 0.573649059412<br>7721  | 1.0 |
| <b>Intestinimonas</b>               | 40  | 4  | 0.100 | 0.576869346906<br>9974  | 1.0 |
| <b>Massilia</b>                     | 61  | 6  | 0.098 | 0.579454890524<br>0392  | 1.0 |
| <b>Facklamia</b>                    | 30  | 3  | 0.100 | 0.588648760440<br>4943  | 1.0 |
| <b>Williamsia</b>                   | 20  | 2  | 0.100 | 0.608253001874<br>8321  | 1.0 |
| <b>Cloacibacteriu<br/>m</b>         | 20  | 2  | 0.100 | 0.608253001874<br>8321  | 1.0 |
| <b>Methylobacteri<br/>um</b>        | 84  | 8  | 0.095 | 0.611140372832<br>3437  | 1.0 |
| <b>Peredibacter</b>                 | 31  | 3  | 0.097 | 0.611414382477<br>1845  | 1.0 |
| <b>Terrisporobact<br/>er</b>        | 31  | 3  | 0.097 | 0.611414382477<br>1845  | 1.0 |
| <b>Granulicatella</b>               | 63  | 6  | 0.095 | 0.611700417052<br>0003  | 1.0 |
| <b>Halomonas</b>                    | 32  | 3  | 0.094 | 0.633316480919<br>3795  | 1.0 |
| <b>Gordonia</b>                     | 21  | 2  | 0.095 | 0.635270036228<br>2919  | 1.0 |
| <b>Acidibacter</b>                  | 33  | 3  | 0.091 | 0.654342495423<br>8868  | 1.0 |
| <b>Turicibacter</b>                 | 77  | 7  | 0.091 | 0.660445512455<br>9978  | 1.0 |
| <b>Alloprevotella</b>               | 77  | 7  | 0.091 | 0.660445512455<br>9978  | 1.0 |

|                                 |    |   |       |                    |     |
|---------------------------------|----|---|-------|--------------------|-----|
| <b>Hungatella</b>               | 22 | 2 | 0.091 | 0.6608011336923114 | 1.0 |
| <b>Limnobacter</b>              | 22 | 2 | 0.091 | 0.6608011336923114 | 1.0 |
| <b>Arcobacter</b>               | 22 | 2 | 0.091 | 0.6608011336923114 | 1.0 |
| <b>Peptoniphilus</b>            | 56 | 5 | 0.089 | 0.6706712131335167 | 1.0 |
| <b>Ralstonia</b>                | 34 | 3 | 0.088 | 0.6744867738362695 | 1.0 |
| <b>Ilumatobacter</b>            | 23 | 2 | 0.087 | 0.6848733113012442 | 1.0 |
| <b>Peptococcus</b>              | 57 | 5 | 0.088 | 0.6860039469102792 | 1.0 |
| <b>Weissella</b>                | 46 | 4 | 0.087 | 0.6880561169183461 | 1.0 |
| <b>Candidatus_Saccharimonas</b> | 47 | 4 | 0.085 | 0.7044129671607624 | 1.0 |
| <b>Caproiciproducens</b>        | 24 | 2 | 0.083 | 0.7075230420514673 | 1.0 |
| <b>Sellimonas</b>               | 24 | 2 | 0.083 | 0.7075230420514673 | 1.0 |
| <b>Lachnoclostridium</b>        | 92 | 8 | 0.087 | 0.7120145513804161 | 1.0 |
| <b>Haemophilus</b>              | 94 | 8 | 0.085 | 0.7343157532397324 | 1.0 |
| <b>Microbacterium</b>           | 49 | 4 | 0.082 | 0.7352337339766317 | 1.0 |
| <b>Roseomonas</b>               | 38 | 3 | 0.079 | 0.7463302211239846 | 1.0 |
| <b>Planococcus</b>              | 26 | 2 | 0.077 | 0.7487357043078514 | 1.0 |
| <b>Actinomyces</b>              | 85 | 7 | 0.082 | 0.7578449260136146 | 1.0 |
| <b>Shuttleworthia</b>           | 51 | 4 | 0.078 | 0.763562609007388  | 1.0 |
| <b>Peptostreptococcus</b>       | 51 | 4 | 0.078 | 0.763562609007388  | 1.0 |
| <b>Candidatus_Stequefichus</b>  | 27 | 2 | 0.074 | 0.7674010519878396 | 1.0 |
| <b>Pirellula</b>                | 27 | 2 | 0.074 | 0.7674010519878396 | 1.0 |
| <b>Vibrio</b>                   | 27 | 2 | 0.074 | 0.7674010519878396 | 1.0 |
| <b>Arenimonas</b>               | 27 | 2 | 0.074 | 0.7674010519878396 | 1.0 |
| <b>Bifidobacterium</b>          | 98 | 8 | 0.082 | 0.775278475920071  | 1.0 |
| <b>Fusobacterium</b>            | 99 | 8 | 0.081 | 0.784763296071283  | 1.0 |
| <b>Dyadobacter</b>              | 28 | 2 | 0.071 | 0.7848459730887516 | 1.0 |
| <b>Gemmatirosa</b>              | 28 | 2 | 0.071 | 0.7848459730887516 | 1.0 |
| <b>Faecalitalea</b>             | 28 | 2 | 0.071 | 0.7848459730887516 | 1.0 |
| <b>Paraprevotella</b>           | 41 | 3 | 0.073 | 0.7914253184930894 | 1.0 |
| <b>Psychrobacter</b>            | 41 | 3 | 0.073 | 0.791425318493     | 1.0 |

|                          |     |   |       |                        |     |
|--------------------------|-----|---|-------|------------------------|-----|
|                          |     |   |       | 0894                   |     |
| <b>Anaerofustis</b>      | 29  | 2 | 0.069 | 0.801127899449<br>6028 | 1.0 |
| <b>Gaiella</b>           | 29  | 2 | 0.069 | 0.801127899449<br>6028 | 1.0 |
| <b>Aquabacterium</b>     | 29  | 2 | 0.069 | 0.801127899449<br>6028 | 1.0 |
| <b>Rubrobacter</b>       | 29  | 2 | 0.069 | 0.801127899449<br>6028 | 1.0 |
| <b>Streptomyces</b>      | 101 | 8 | 0.079 | 0.802838662832<br>8019 | 1.0 |
| <b>Eggerthella</b>       | 30  | 2 | 0.067 | 0.816304980807<br>3963 | 1.0 |
| <b>Legionella</b>        | 43  | 3 | 0.070 | 0.817618685385<br>6384 | 1.0 |
| <b>Deinococcus</b>       | 44  | 3 | 0.068 | 0.829631108952<br>1422 | 1.0 |
| <b>Flavonifractor</b>    | 44  | 3 | 0.068 | 0.829631108952<br>1422 | 1.0 |
| <b>Roseburia</b>         | 95  | 7 | 0.074 | 0.848933539294<br>995  | 1.0 |
| <b>Staphylococcus</b>    | 95  | 7 | 0.074 | 0.848933539294<br>995  | 1.0 |
| <b>Fusicatenibacter</b>  | 71  | 5 | 0.070 | 0.849961893944<br>4202 | 1.0 |
| <b>Tyzzerella</b>        | 46  | 3 | 0.065 | 0.851624619342<br>5087 | 1.0 |
| <b>Holdemanella</b>      | 46  | 3 | 0.065 | 0.851624619342<br>5087 | 1.0 |
| <b>Helicobacter</b>      | 33  | 2 | 0.061 | 0.855785279547<br>7143 | 1.0 |
| <b>Dialister</b>         | 85  | 6 | 0.071 | 0.864012259986<br>1861 | 1.0 |
| <b>Bacillus</b>          | 110 | 8 | 0.073 | 0.870161480012<br>168  | 1.0 |
| <b>Dietzia</b>           | 35  | 2 | 0.057 | 0.877623508644<br>7748 | 1.0 |
| <b>Aquicella</b>         | 20  | 1 | 0.050 | 0.878423345409<br>4308 | 1.0 |
| <b>Photobacterium</b>    | 20  | 1 | 0.050 | 0.878423345409<br>4308 | 1.0 |
| <b>Kocuria</b>           | 49  | 3 | 0.061 | 0.879957334677<br>2351 | 1.0 |
| <b>Catenibacterium</b>   | 49  | 3 | 0.061 | 0.879957334677<br>2351 | 1.0 |
| <b>Ruminiclostridium</b> | 49  | 3 | 0.061 | 0.879957334677<br>2351 | 1.0 |
| <b>Eisenbergiella</b>    | 36  | 2 | 0.056 | 0.887358002275<br>304  | 1.0 |
| <b>Solirubrobacter</b>   | 36  | 2 | 0.056 | 0.887358002275<br>304  | 1.0 |
| <b>Flavisolibacter</b>   | 21  | 1 | 0.048 | 0.890581010868<br>4877 | 1.0 |
| <b>Chryseobacterium</b>  | 65  | 4 | 0.062 | 0.900447174813<br>5983 | 1.0 |
| <b>Jeotgalicoccus</b>    | 22  | 1 | 0.045 | 0.901522909781<br>6388 | 1.0 |
| <b>Chthoniobacter</b>    | 22  | 1 | 0.045 | 0.901522909781<br>6388 | 1.0 |

|                               |    |   |       |                    |     |
|-------------------------------|----|---|-------|--------------------|-----|
| <b>Acidaminococcus</b>        | 22 | 1 | 0.045 | 0.9015229097816388 | 1.0 |
| <b>Novosphingobium</b>        | 22 | 1 | 0.045 | 0.9015229097816388 | 1.0 |
| <b>Porphyromonas</b>          | 66 | 4 | 0.061 | 0.9068051410155196 | 1.0 |
| <b>Lactococcus</b>            | 68 | 4 | 0.059 | 0.918447882785011  | 1.0 |
| <b>Erysipelatoclostridium</b> | 68 | 4 | 0.059 | 0.918447882785011  | 1.0 |
| <b>Hyphomicrobium</b>         | 40 | 2 | 0.050 | 0.9195263039855227 | 1.0 |
| <b>Caldicoprobacter</b>       | 24 | 1 | 0.042 | 0.9202335569231275 | 1.0 |
| <b>Leucobacter</b>            | 24 | 1 | 0.042 | 0.9202335569231275 | 1.0 |
| <b>Faecalibaculum</b>         | 25 | 1 | 0.040 | 0.9282102012308148 | 1.0 |
| <b>Oscillibacter</b>          | 70 | 4 | 0.057 | 0.9287694277368974 | 1.0 |
| <b>Rubellimicrobium</b>       | 26 | 1 | 0.038 | 0.9353891811077333 | 1.0 |
| <b>Anaerococcus</b>           | 58 | 3 | 0.052 | 0.938209796729051  | 1.0 |
| <b>Pedomicrobium</b>          | 27 | 1 | 0.037 | 0.94185026299696   | 1.0 |
| <b>Gemmatimonas</b>           | 27 | 1 | 0.037 | 0.94185026299696   | 1.0 |
| <b>Lawsonella</b>             | 44 | 2 | 0.045 | 0.9428911025791791 | 1.0 |
| <b>Brevibacterium</b>         | 45 | 2 | 0.044 | 0.9476322185914736 | 1.0 |
| <b>Desulfosporosinus</b>      | 29 | 1 | 0.034 | 0.9528987130275376 | 1.0 |
| <b>Intestinibacter</b>        | 29 | 1 | 0.034 | 0.9528987130275376 | 1.0 |
| <b>Nakamurella</b>            | 29 | 1 | 0.034 | 0.9528987130275376 | 1.0 |
| <b>Iamia</b>                  | 30 | 1 | 0.033 | 0.9576088417247838 | 1.0 |
| <b>Medicago</b>               | 31 | 1 | 0.032 | 0.9618479575523055 | 1.0 |
| <b>Pajaroellobacter</b>       | 31 | 1 | 0.032 | 0.9618479575523055 | 1.0 |
| <b>Lysobacter</b>             | 31 | 1 | 0.032 | 0.9618479575523055 | 1.0 |
| <b>Mycobacterium</b>          | 80 | 4 | 0.050 | 0.9646937415188622 | 1.0 |
| <b>Altererythrobaacter</b>    | 33 | 1 | 0.030 | 0.9690968456173674 | 1.0 |
| <b>Marmoricola</b>            | 33 | 1 | 0.030 | 0.9690968456173674 | 1.0 |
| <b>Ferruginibacter</b>        | 33 | 1 | 0.030 | 0.9690968456173674 | 1.0 |
| <b>Gemella</b>                | 67 | 3 | 0.045 | 0.969280674082702  | 1.0 |
| <b>Nocardioides</b>           | 67 | 3 | 0.045 | 0.969280674082702  | 1.0 |
| <b>Faecalibacteriu</b>        | 97 | 5 | 0.052 | 0.970866071584     | 1.0 |

|                          |     |   |       |                    |     |
|--------------------------|-----|---|-------|--------------------|-----|
| <b>m</b>                 |     |   |       | 8146               |     |
| <b>Romboutsia</b>        | 83  | 4 | 0.048 | 0.9716070539113301 | 1.0 |
| <b>Paracoccus</b>        | 52  | 2 | 0.038 | 0.9717057741118126 | 1.0 |
| <b>Sphingomonas</b>      | 84  | 4 | 0.048 | 0.973614419469635  | 1.0 |
| <b>Paenibacillus</b>     | 70  | 3 | 0.043 | 0.9758187016428539 | 1.0 |
| <b>Oscillospira</b>      | 36  | 1 | 0.028 | 0.9774716004550609 | 1.0 |
| <b>Clostridium</b>       | 102 | 5 | 0.049 | 0.9793640065142518 | 1.0 |
| <b>Hymenobacter</b>      | 40  | 1 | 0.025 | 0.9852191170585655 | 1.0 |
| <b>Rothia</b>            | 77  | 3 | 0.039 | 0.9863104260537002 | 1.0 |
| <b>Megamonas</b>         | 41  | 1 | 0.024 | 0.9866972053527089 | 1.0 |
| <b>Lactobacillus</b>     | 124 | 6 | 0.048 | 0.9877599629860419 | 1.0 |
| <b>Veillonella</b>       | 95  | 4 | 0.042 | 0.9884599362038465 | 1.0 |
| <b>Devosia</b>           | 45  | 1 | 0.022 | 0.9912720364319123 | 1.0 |
| <b>Rhodococcus</b>       | 47  | 1 | 0.021 | 0.9929303495098489 | 1.0 |
| <b>Pseudomonas</b>       | 136 | 6 | 0.044 | 0.994695056461212  | 1.0 |
| <b>Flavobacterium</b>    | 74  | 2 | 0.027 | 0.9962087599681586 | 1.0 |
| <b>Streptococcus</b>     | 135 | 5 | 0.037 | 0.998208418667266  | 1.0 |
| <b>Prevotella</b>        | 72  | 1 | 0.014 | 0.9994924712139436 | 1.0 |
| <b>Bacteroides</b>       | 111 | 2 | 0.018 | 0.9998888633544281 | 1.0 |
| <b>Geodermatophilus</b>  | 22  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Conexibacter</b>      | 26  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Exiguobacterium</b>   | 22  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Aerococcus</b>        | 27  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Acidovorax</b>        | 28  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Bdellovibrio</b>      | 46  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Blastocatella</b>     | 23  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Butyrivibrio</b>      | 26  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Bryobacter</b>        | 28  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Capnocytophaga</b>    | 44  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Cellvibrio</b>        | 25  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Haliangium</b>        | 39  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Luteolibacter</b>     | 28  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Micrococcus</b>       | 21  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Pseudonocardia</b>    | 38  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Sediminibacterium</b> | 27  | 0 | 0.000 | 1.0                | 1.0 |
| <b>Reyranella</b>        | 31  | 0 | 0.000 | 1.0                | 1.0 |

|                      |    |   |       |     |     |
|----------------------|----|---|-------|-----|-----|
| <b>Pseudolabrys</b>  | 20 | 0 | 0.000 | 1.0 | 1.0 |
| <b>Undibacterium</b> | 23 | 0 | 0.000 | 1.0 | 1.0 |

## Appendix D. External tools ledger

**Table 22. 48 external tools, libraries, APIs and databases used inside this project (results/tools\_ledger.csv).**

| tool                                                            | kind                | where used                                                                               | counts_for_gate | note                                                                   |
|-----------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------|
| <b>NumPy</b>                                                    | library             | src/microtwin;<br>run_bench.py;<br>scripts/                                              | True            |                                                                        |
| <b>SciPy</b>                                                    | library             | src/microtwin/<br>audit.py; scripts/<br>(stats; FDR inputs)                              | True            |                                                                        |
| <b>pandas</b>                                                   | library             | scripts/<br>paper/build_paper.py                                                         | True            |                                                                        |
| <b>PyTorch</b>                                                  | library             | src/microtwin/<br>models.py;<br>run_bench.py<br>(cNODE and graph-<br>twin models)        | True            |                                                                        |
| <b>Matplotlib</b>                                               | library             | scripts/<br>summarise_audit.py<br>(fig_mgnify_audit.png)                                 | True            |                                                                        |
| <b>python-docx</b>                                              | library             | paper/paperkit.py                                                                        | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>pytest</b>                                                   | test framework      | tests/<br>test_microtwin.py;<br>tests/test_audit.py                                      | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>LibreOffice (soffice headless)</b>                           | application         | paper docx -> pdf<br>render                                                              | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>poppler pdfinfo</b>                                          | application         | page count of<br>rendered paper                                                          | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>Git + GitHub</b>                                             | service             | repo hosting and<br>pushes                                                               | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>EBI MGnify REST API v1</b>                                   | web API             | scripts fetching<br>data/raw/mgnify<br>(manifest.csv);<br>scripts/keystone_mg<br>nify.py | True            |                                                                        |
| <b>cNODE published benchmark data (Michel-Mata et al. 2022)</b> | public data release | data/raw/cnode/<br>*.csv; run_bench.py                                                   | True            |                                                                        |
| <b>MDSINE2 published data and</b>                               | public data release | data/raw/mdsine2*;<br>mdsine2 head-to-                                                   | True            |                                                                        |

|                                                                                                      |                  |                                                                                                                             |      |                                                                                                                     |
|------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------|
| <b>Source Data<br/>(Gibson et al. 2025)</b>                                                          |                  | head results                                                                                                                |      |                                                                                                                     |
| <b>scikit-bio</b>                                                                                    | library          | scripts/<br>predictors_audit.py<br>(Shannon alpha,<br>Bray-Curtis beta<br>diversity)                                        | True |                                                                                                                     |
| <b>scikit-learn</b>                                                                                  | library          | scripts/<br>predictors_audit.py<br>(random forest, CV,<br>permutation<br>importance)                                        | True |                                                                                                                     |
| <b>networkx</b>                                                                                      | library          | scripts/<br>predictors_audit.py<br>(co-occurrence<br>network density,<br>modularity)                                        | True |                                                                                                                     |
| <b>statsmodels</b>                                                                                   | library          | scripts/<br>predictors_audit.py<br>(OLS with HC3<br>robust SEs)                                                             | True |                                                                                                                     |
| <b>NCBI Taxonomy<br/>(E-utilities<br/>esearch/efetch)</b>                                            | database/web API | scripts/<br>keystone_taxonomy.<br>py (genus -> phylum<br>lineage)                                                           | True |                                                                                                                     |
| <b>BugSigDB (genus<br/>signature export)</b>                                                         | database         | data/ref/<br>bugsigdb_signatures<br>_genus_taxname.gm<br>t;<br>scripts/keystone_bu<br>gsigdb.py (literature<br>cross-check) | True |                                                                                                                     |
| <b>GTDB (release<br/>v232 bac120<br/>taxonomy)</b>                                                   | database         | scripts/<br>keystone_gtdb.py                                                                                                | True | independent<br>phylogenomic<br>taxonomy<br>replication of<br>keystone phylum<br>enrichment                          |
| <b>Madin et al. 2020<br/>bacterial/archaeal<br/>trait database<br/>(condensed_species<br/>_NCBI)</b> | database         | scripts/<br>keystone_traits.py                                                                                              | True | keystone<br>physiological trait<br>association                                                                      |
| <b>KEGG REST<br/>(genome list + KO-<br/>gene links)</b>                                              | database         | scripts/<br>keystone_kegg.py                                                                                                | True | genomic<br>anaerobic/aerobic<br>marker check of<br>keystones                                                        |
| <b>BV-BRC genome<br/>API<br/>(oxygen_requireme<br/>nt facets)</b>                                    | database         | scripts/<br>keystone_bvbrc.py                                                                                               | True | third-source<br>anaerobe annotation<br>for keystone genera                                                          |
| <b>IJSEM phenotypic<br/>database<br/>(Barberan et al.<br/>2017, figshare<br/>4272392)</b>            | database         | scripts/<br>keystone_ijsem.py                                                                                               | True | oxygen preference<br>per genus; pre-<br>registered check<br>142c972; overlaps<br>Madin sources (not<br>independent) |
| <b>NCBI Datasets v2<br/>API (genome<br/>dataset_report)</b>                                          | database/service | scripts/<br>keystone_genomesiz<br>e.py                                                                                      | True | reference-genome<br>size and GC per<br>genus; pre-<br>registered 66d7f05                                            |

|                                                                                |                         |                                   |      |                                                                                                 |
|--------------------------------------------------------------------------------|-------------------------|-----------------------------------|------|-------------------------------------------------------------------------------------------------|
| <b>Open Tree of Life API v3 (TNRS match_names + induced synthetic subtree)</b> | web API                 | scripts/<br>keystone_otol.py      | True | independent phylogeny for anaerobe-keystone PGLS; pre-registered ce6fd94                        |
| <b>ProTraits (Brbic et al. 2016; protraits.irb.hr integrated Pr&gt;=0.95)</b>  | database                | scripts/<br>keystone_protraits.py | True | pre-registered oxygen check 4f748be; coverage gate FAILED (51 genera)                           |
| <b>Disbiome (disbiome.ugent.be experiment API)</b>                             | database/web API        | scripts/<br>keystone_disbiome.py  | True | pre-registered literature check 899f554; D1 PASS                                                |
| <b>Europe PMC REST API (search hitCount)</b>                                   | web API                 | scripts/<br>keystone_europepmc.py | True | pre-registered study-bias control 0358299; E1 PASS                                              |
| <b>gglasso (Schaipp et al. 2021 graphical lasso)</b>                           | library                 | scripts/<br>keystone_gglasso.py   | True | alternative network inference for keystones; pre-registered 3129971; H1 FAILED                  |
| <b>Wikidata SPARQL endpoint (P2597 Gram staining)</b>                          | web API/knowledge graph | scripts/<br>keystone_wikidata.py  | True | pre-registered 0ecdd41 (+QID fix 1f20b4a); Madin Gram null replicated                           |
| <b>ENA portal API + ENA taxonomy REST (assembly counts)</b>                    | web API                 | scripts/<br>keystone_ena.py       | True | pre-registered sequencing-effort control 9451f14; A1 PASS                                       |
| <b>GBIF API v1 (species/match + occurrence country facets)</b>                 | web API                 | scripts/<br>keystone_gbif.py      | True | pre-registered generalism rival 9f6cc62; B1 PASS                                                |
| <b>UniProt REST API (uniprotkb search counts, reference proteomes)</b>         | web API                 | scripts/<br>keystone_uniprot.py   | True | pre-registered anaerobe-index replication c9ea858; U1 PASS                                      |
| <b>Crossref REST API (works total-results)</b>                                 | web API                 | scripts/<br>keystone_crossref.py  | True | pre-registered literature control 6ff904c (replaces OpenAlex 429); C1 PASS C2 PASS              |
| <b>python-igraph (betweenness centrality)</b>                                  | library                 | scripts/<br>keystone_igraph.py    | True | pre-registered third network definition 0962035; I1 FAIL (p=0.34)                               |
| <b>EBI Search REST (Ensembl Genomes genome domain)</b>                         | web API                 | scripts/<br>keystone_ensembl.py   | True | pre-registered sequencing-effort replication be7175d; S1 PASS                                   |
| <b>XGBoost (gradient-boosted trees)</b>                                        | library                 | scripts/<br>keystone_xgboost.py   | True | pre-registered out-of-sample GAI test 9cf67ae; X1 PASS by criterion but absolute CV R2 negative |
| <b>InterPro API (protein counts per taxon and entry)</b>                       | web API                 | scripts/<br>keystone_interpro.py  | True | pre-registered anaerobe-index replication b327560; P1 PASS                                      |
| <b>HOMD expanded</b>                                                           | database                | scripts/                          | True | pre-registered oral-                                                                            |

|                                                          |         |                                |      |                                                                                         |
|----------------------------------------------------------|---------|--------------------------------|------|-----------------------------------------------------------------------------------------|
| <b>Human Oral Microbiome Database (taxon table v4.2)</b> |         | keystone_homd.py               |      | taxon rival 0aa4eb0 (+parser addendum 5658ee6); M1 PASS                                 |
| <b>EBI Search REST (RNAcentral rRNA sequences)</b>       | web API | scripts/keystone_rnacentral.py | True | pre-registered rRNA-effort control 9814c7a; R1 PASS                                     |
| <b>bambi / PyMC (Bayesian hierarchical logistic)</b>     | library | scripts/keystone_bambi.py      | True | pre-registered Bayesian re-fit 45e3254; BY1 PASS                                        |
| <b>NCBI PubTator3 API (search counts)</b>                | web API | scripts/keystone_pubtator.py   | True | pre-registered oral-index replication 45e3254 (+addendum 3d16589); PT1 FAIL             |
| <b>Biopython Bio.Phylo (synthetic-tree topology)</b>     | library | scripts/keystone_biophylo.py   | True | pre-registered a2e24b7; BP1 FAIL oral clustering p=0.53                                 |
| <b>LightGBM</b>                                          | library | scripts/keystone_boosters.py   | True | pre-registered fd7a9e2; B1 PASS absolute out-of-fold R2 0.133                           |
| <b>CatBoost</b>                                          | library | scripts/keystone_boosters.py   | True | pre-registered fd7a9e2; B1 PASS absolute out-of-fold R2 0.119                           |
| <b>SHAP TreeExplainer</b>                                | library | scripts/keystone_shap.py       | True | pre-registered c12838f; SH1 PASS GAI largest mean absolute attribution in full LightGBM |
| <b>SymPy symbolic algebra</b>                            | library | scripts/keystone_sympy.py      | True | pre-registered c12838f; SY1 PASS RHS invariants for cNODE and gLV                       |

## Appendix E. Datasets ledger

**Table 23. 190 accession-level datasets used inside this project (results/datasets\_ledger.csv).**

| accession    | source     | role    | used_in                                 |
|--------------|------------|---------|-----------------------------------------|
| MGYS00006862 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00005791 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006825 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006824 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006822 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006814 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006815 | EBI MGnify | primary | 160-study interaction audit             |

|                     |            |         |                                            |
|---------------------|------------|---------|--------------------------------------------|
|                     |            |         | + keystones                                |
| <b>MGYS00006811</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006807</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006805</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006806</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006760</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002338</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006797</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006551</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006795</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006794</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006789</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006787</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006786</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006759</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006780</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006779</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006769</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006765</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006755</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005333</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005230</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002675</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006069</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006006</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002061</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00000633</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002238</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006562</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005154</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |

|                     |            |         |                                         |
|---------------------|------------|---------|-----------------------------------------|
| <b>MGYS00006649</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006641</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006642</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006364</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006640</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005253</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006320</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006276</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006263</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006017</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005592</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005569</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005528</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005506</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005189</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005223</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005197</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005143</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00002159</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00003719</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00002395</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00002394</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00002272</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00002269</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00002229</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006648</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005232</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005156</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006246</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005384</b> | EBI MGnify | primary | 160-study interaction audit             |

|                     |            |         |                                            |
|---------------------|------------|---------|--------------------------------------------|
|                     |            |         | + keystones                                |
| <b>MGYS00005172</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005533</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005212</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005238</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005101</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005102</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005135</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005098</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005037</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005090</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002184</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002072</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006855</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006799</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002673</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006747</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00001223</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00001224</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005267</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00000426</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005228</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006537</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006536</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006538</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005628</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006450</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006456</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006446</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006433</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |

|                     |            |         |                                         |
|---------------------|------------|---------|-----------------------------------------|
| <b>MGYS00006430</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006427</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006424</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006652</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00001412</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00001368</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00003172</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006578</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00000957</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00001704</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00000867</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006031</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006021</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006018</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00001349</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005260</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005227</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00000983</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00001347</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005740</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005726</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005169</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005151</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006750</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006533</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006532</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005766</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006062</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00006042</b> | EBI MGnify | primary | 160-study interaction audit + keystones |
| <b>MGYS00005621</b> | EBI MGnify | primary | 160-study interaction audit             |

|                     |            |         |                                            |
|---------------------|------------|---------|--------------------------------------------|
|                     |            |         | + keystones                                |
| <b>MGYS00005611</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005610</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005609</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005600</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005593</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005576</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005187</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005316</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005317</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005309</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00004995</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00003706</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00003681</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006764</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006763</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006762</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006761</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006758</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006041</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006117</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006086</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005503</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00003166</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00004672</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002905</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002608</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002413</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002065</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006798</b> | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |

|                                                                               |                                                           |           |                                                       |
|-------------------------------------------------------------------------------|-----------------------------------------------------------|-----------|-------------------------------------------------------|
| <b>MGYS00006019</b>                                                           | EBI MGnify                                                | primary   | 160-study interaction audit + keystones               |
| <b>MGYS00005862</b>                                                           | EBI MGnify                                                | primary   | 160-study interaction audit + keystones               |
| <b>MGYS00005215</b>                                                           | EBI MGnify                                                | primary   | 160-study interaction audit + keystones               |
| <b>MGYS00005163</b>                                                           | EBI MGnify                                                | primary   | 160-study interaction audit + keystones               |
| <b>MGYS00003800</b>                                                           | EBI MGnify                                                | primary   | 160-study interaction audit + keystones               |
| <b>MGYS00003705</b>                                                           | EBI MGnify                                                | primary   | 160-study interaction audit + keystones               |
| <b>cNODE:Ocean</b>                                                            | github.com/yixueyang/<br>cNODE (6 tables, one repository) | primary   | cNODE benchmark                                       |
| <b>cNODE:Drosophila_Gut</b>                                                   | github.com/yixueyang/<br>cNODE (6 tables, one repository) | primary   | cNODE benchmark                                       |
| <b>cNODE:Soil_Vitro</b>                                                       | github.com/yixueyang/<br>cNODE (6 tables, one repository) | primary   | cNODE benchmark                                       |
| <b>cNODE:Soil_Vivo</b>                                                        | github.com/yixueyang/<br>cNODE (6 tables, one repository) | primary   | cNODE benchmark                                       |
| <b>cNODE:Human_Oral</b>                                                       | github.com/yixueyang/<br>cNODE (6 tables, one repository) | primary   | cNODE benchmark                                       |
| <b>cNODE:Human_Gut</b>                                                        | github.com/yixueyang/<br>cNODE (6 tables, one repository) | primary   | cNODE benchmark                                       |
| <b>MDSINE2:gibson/<br/>healthy</b>                                            | github.com/gerberlab/<br>MDSINE2_Paper                    | primary   | forecasting                                           |
| <b>MDSINE2:gibson/uc</b>                                                      | github.com/gerberlab/<br>MDSINE2_Paper                    | primary   | forecasting                                           |
| <b>41564_2025_2112_MOE<br/>SM5_ESM</b>                                        | Nature Microbiology<br>Source Data                        | primary   | head-to-head                                          |
| <b>NCBI Taxonomy</b>                                                          | NCBI E-utilities                                          | reference | phylum mapping                                        |
| <b>BugSigDB genus<br/>signatures</b>                                          | BugSigDB                                                  | reference | literature cross-check                                |
| <b>GTDB r232 bac120<br/>taxonomy</b>                                          | GTDB                                                      | reference | taxonomy replication                                  |
| <b>Madin 2020<br/>condensed_species_NCB<br/>I</b>                             | github bacteria-archaea-<br>traits                        | reference | traits                                                |
| <b>KEGG genome list + 10<br/>KO links</b>                                     | KEGG REST                                                 | reference | genomic anaerobe index                                |
| <b>BV-BRC<br/>oxygen_requirement<br/>facets</b>                               | BV-BRC API                                                | reference | third anaerobe source                                 |
| <b>figshare:4272392</b>                                                       | figshare (IJSEM<br>phenotypic db)                         | reference | anaerobe-keystone IJSEM<br>check                      |
| <b>OToL synth induced<br/>subtree (161 genera;<br/>data/otol/induced.tre)</b> | Open Tree of Life                                         | reference | anaerobe-keystone PGLS<br>on independent phylogeny    |
| <b>ProTraits_binaryIntegra<br/>tedPr0.95.txt</b>                              | ProTraits                                                 | reference | anaerobe-keystone<br>ProTraits check (gate<br>failed) |
| <b>Disbiome experiment<br/>dump (10866 records)</b>                           | Disbiome                                                  | reference | keystone literature check                             |
| <b>Europe PMC</b>                                                             | Europe PMC                                                | reference | study-bias control of                                 |

|                                                                                                                       |                               |           |                                               |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------|-----------------------------------------------|
| <b>TITLE_ABS</b> hit counts for 248 genera (results/keystone_europe_pmc_counts.csv)                                   |                               |           | keystone literature link                      |
| Wikidata genus Gram-staining table (3117 genera; results/keystone_wikidata_gram.csv)                                  | Wikidata                      | reference | Gram-stain replication                        |
| ENA public assembly counts for 200 genera (results/keystone_ena_counts.csv)                                           | ENA                           | reference | sequencing-effort control                     |
| GBIF occurrence and country counts for 202 genera (results/keystone_gbif_counts.csv)                                  | GBIF                          | reference | generalism rival                              |
| UniProt reference-proteome EC 7.1.1.9 / EC 1.2.7.1 / recA counts for 200 genera (results/keystone_uniprot_counts.csv) | UniProt                       | reference | annotation-pipeline replication of GAI        |
| Crossref works counts for 248 genera (results/keystone_crossref_counts.csv)                                           | Crossref                      | reference | independent literature-volume covariate       |
| Ensembl Genomes genome hit counts for 203 genera (results/keystone_ensembl_counts.csv)                                | EBI Search / Ensembl Bacteria | reference | sequencing-effort covariate (curated archive) |
| InterPro PFOR/COX1/RecA protein counts for 200 genera (results/keystone_interpro_counts.csv)                          | InterPro                      | reference | signature-based replication of GAI            |
| HOMD taxon table v4.2 (data/ref/HOMD_taxon_table_v4.2.csv; 121 oral genera)                                           | HOMD                          | reference | oral-origin covariate                         |
| RNAcentral rRNA sequence counts for 203 genera (results/keystone_rnacentral_counts.csv)                               | RNAcentral                    | reference | marker-gene sequencing-effort covariate       |
| PubTator3 genus and genus+oral/dental article counts for 203 genera (results/keystone_pubtator_counts.csv)            | NCBI PubTator3                | reference | literature oral index                         |

## Appendix F. MDSINE2 per-(subject, taxon) RMSE

**Table 24. Per-(subject, taxon) RMSE of log10 absolute abundance on detected timepoints, all fitted methods (546 pairs; pivoted from results/mdsine2\_pertaxon\_errors\_141.csv).**

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 2       | 0     | 1.373       | 1.207        | 1.145         | 0.373                  |
| 2       | 1     | 1.852       | 2.319        | 2.474         | 0.821                  |
| 2       | 2     | 2.405       | 2.549        | 3.069         | 0.868                  |
| 2       | 3     | 2.529       | 1.427        | 1.610         | 0.694                  |
| 2       | 4     | 2.985       | 1.551        | 1.320         | 0.504                  |
| 2       | 5     | 2.610       | 1.792        | 1.963         | 0.852                  |
| 2       | 6     | 3.185       | 1.402        | 1.665         | 0.626                  |
| 2       | 7     | 3.147       | 2.209        | 2.041         | 0.639                  |
| 2       | 8     | 2.700       | 1.488        | 2.020         | 0.846                  |
| 2       | 9     | 3.815       | 2.484        | 1.573         | 0.486                  |
| 2       | 10    | 2.620       | 1.533        | 1.709         | 0.614                  |
| 2       | 11    | 3.439       | 1.457        | 1.313         | 0.389                  |
| 2       | 12    | 2.748       | 1.427        | 1.486         | 0.640                  |
| 2       | 13    | 1.812       | 1.303        | 1.283         | 0.630                  |
| 2       | 14    | 2.568       | 2.319        | 2.685         | 1.299                  |
| 2       | 15    | 3.546       | 2.586        | 1.129         | 0.345                  |
| 2       | 16    | 3.525       | 2.779        | 3.179         | 0.490                  |
| 2       | 17    | 3.324       | 2.930        | 3.052         | 0.510                  |
| 2       | 18    | 2.790       | 3.320        | 3.070         | 1.081                  |
| 2       | 19    | 3.095       | 2.114        | 1.003         | 1.036                  |
| 2       | 20    | 2.962       | 2.482        | 1.505         | 0.695                  |
| 2       | 21    | 1.958       | 1.800        | 1.255         | 0.737                  |
| 2       | 22    | 2.555       | 1.509        | 1.289         | 0.488                  |
| 2       | 23    | 2.028       | 2.099        | 2.152         | 1.124                  |
| 2       | 24    | 2.644       | 1.956        | 2.277         | 0.378                  |
| 2       | 25    | 3.358       | 2.400        | 2.399         | 1.398                  |
| 2       | 26    | 3.068       | 3.366        | 3.217         | 1.278                  |
| 2       | 27    | 3.241       | 3.159        | 2.139         | 1.112                  |
| 2       | 28    | 2.739       | 1.693        | 2.447         | 0.841                  |
| 2       | 29    | 2.989       | 1.942        | 2.905         | 0.440                  |
| 2       | 30    | 3.324       | 2.076        | 3.663         | 2.145                  |
| 2       | 31    | 2.558       | 1.199        | 1.416         | 0.477                  |
| 2       | 32    | 2.603       | 1.876        | 1.799         | 0.587                  |
| 2       | 33    | 2.817       | 1.061        | 0.883         | 0.693                  |
| 2       | 34    | 2.760       | 2.124        | 1.869         | 1.328                  |
| 2       | 35    | 2.164       | 1.528        | 1.588         | 0.850                  |
| 2       | 36    | 2.733       | 2.203        | 2.529         | 1.541                  |
| 2       | 37    | 2.442       | 1.913        | 2.872         | 1.314                  |
| 2       | 38    | 3.374       | 2.256        | 3.229         | 1.696                  |
| 2       | 39    | 2.405       | 1.770        | 1.071         | 0.860                  |
| 2       | 40    | 1.491       | 1.521        | 1.536         | 1.085                  |
| 2       | 41    | 2.895       | 2.960        | 1.904         | 1.273                  |
| 2       | 42    | 2.922       | 2.319        | 3.160         | 1.740                  |
| 2       | 43    | 1.918       | 1.120        | 1.682         | 1.141                  |
| 2       | 44    | 2.562       | 1.329        | 1.181         | 0.515                  |
| 2       | 45    | 2.721       | 2.380        | 1.654         | 0.713                  |
| 2       | 46    | 3.212       | 3.223        | 3.238         | 1.805                  |
| 2       | 47    | 3.361       | 2.894        | 2.638         | 1.169                  |

|   |     |       |       |       |       |
|---|-----|-------|-------|-------|-------|
| 2 | 48  | 3.267 | 2.792 | 2.597 | 1.286 |
| 2 | 49  | 2.326 | 1.028 | 0.976 | 0.750 |
| 2 | 50  | 2.697 | 2.791 | 3.212 | 0.664 |
| 2 | 51  | 3.151 | 2.891 | 2.613 | 1.816 |
| 2 | 52  | 1.635 | 1.640 | 1.636 | 0.919 |
| 2 | 53  | 2.406 | 1.259 | 1.072 | 0.495 |
| 2 | 54  | 1.442 | 1.442 | 1.442 | 1.179 |
| 2 | 55  | 2.097 | 1.806 | 2.134 | 0.705 |
| 2 | 56  | 3.263 | 2.887 | 2.451 | 1.362 |
| 2 | 57  | 2.718 | 2.047 | 1.347 | 0.647 |
| 2 | 58  | 2.459 | 1.867 | 0.864 | 0.683 |
| 2 | 59  | 2.337 | 1.408 | 2.522 | 0.507 |
| 2 | 60  | 1.770 | 2.280 | 2.144 | 1.250 |
| 2 | 61  | 2.045 | 1.944 | 1.364 | 0.783 |
| 2 | 62  | 2.254 | 2.568 | 2.899 | 0.291 |
| 2 | 63  | 1.775 | 0.828 | 1.019 | 0.888 |
| 2 | 64  | 2.255 | 2.699 | 2.679 | 0.892 |
| 2 | 65  | 2.540 | 1.823 | 1.142 | 0.803 |
| 2 | 66  | 3.141 | 2.317 | 2.181 | 1.412 |
| 2 | 67  | 2.155 | 1.436 | 1.532 | 0.675 |
| 2 | 68  | 2.967 | 2.774 | 3.055 | 1.161 |
| 2 | 69  | 2.560 | 1.592 | 1.439 | 1.038 |
| 2 | 70  | 2.949 | 2.619 | 2.800 | 1.176 |
| 2 | 71  | 2.471 | 1.693 | 0.931 | 0.529 |
| 2 | 72  | 2.128 | 2.419 | 2.737 | 0.304 |
| 2 | 73  | 2.451 | 1.885 | 1.885 | 1.029 |
| 2 | 74  | 2.404 | 1.715 | 0.924 | 0.638 |
| 2 | 75  | 2.395 | 1.951 | 1.532 | 0.653 |
| 2 | 76  | 1.407 | 1.318 | 1.405 | 0.864 |
| 2 | 77  | 2.245 | 1.804 | 1.218 | 0.569 |
| 2 | 78  | 1.926 | 1.706 | 1.510 | 0.885 |
| 2 | 79  | 2.062 | 2.355 | 2.660 | 0.700 |
| 2 | 80  | 2.169 | 1.772 | 0.892 | 0.500 |
| 2 | 81  | 3.138 | 1.308 | 0.952 | 0.679 |
| 2 | 82  | 1.994 | 2.705 | 1.391 | 0.996 |
| 2 | 83  | 1.750 | 2.183 | 2.727 | 0.993 |
| 2 | 84  | 3.134 | 2.932 | 3.017 | 1.278 |
| 2 | 85  | 2.129 | 1.420 | 1.498 | 0.458 |
| 2 | 86  | 3.135 | 0.971 | 1.108 | 1.018 |
| 2 | 87  | 2.221 | 2.299 | 3.350 | 1.245 |
| 2 | 88  | 3.171 | 3.245 | 3.054 | 0.511 |
| 2 | 89  | 2.010 | 1.272 | 2.161 | 2.003 |
| 2 | 90  | 2.289 | 2.358 | 2.404 | 1.049 |
| 2 | 91  | 1.503 | 0.863 | 0.859 | 1.321 |
| 2 | 92  | 1.811 | 2.070 | 2.337 | 0.471 |
| 2 | 93  | 1.121 | 1.624 | 2.441 | 0.760 |
| 2 | 94  | 1.755 | 1.491 | 0.892 | 0.557 |
| 2 | 95  | 1.801 | 1.944 | 2.211 | 0.496 |
| 2 | 96  | 2.419 | 1.087 | 1.584 | 1.212 |
| 2 | 97  | 1.480 | 0.617 | 0.773 | 0.380 |
| 2 | 98  | 1.481 | 1.485 | 0.625 | 0.137 |
| 2 | 99  | 1.862 | 0.682 | 1.219 | 0.589 |
| 2 | 100 | 1.960 | 2.182 | 2.399 | 0.517 |
| 2 | 102 | 2.120 | 1.905 | 1.282 | 0.813 |
| 2 | 103 | 2.679 | 1.909 | 0.978 | 0.871 |
| 2 | 104 | 3.298 | 3.375 | 3.375 | 1.383 |
| 2 | 106 | 2.785 | 3.002 | 2.573 | 0.923 |

|   |     |       |       |       |       |
|---|-----|-------|-------|-------|-------|
| 2 | 107 | 2.515 | 2.510 | 2.444 | 0.756 |
| 2 | 108 | 2.438 | 1.642 | 0.847 | 0.880 |
| 2 | 109 | 2.155 | 1.247 | 1.807 | 1.221 |
| 2 | 110 | 1.662 | 1.617 | 2.116 | 0.776 |
| 2 | 113 | 1.147 | 0.941 | 1.219 | 0.558 |
| 2 | 114 | 2.858 | 3.247 | 2.295 | 0.620 |
| 2 | 115 | 2.530 | 2.206 | 1.993 | 1.214 |
| 2 | 116 | 2.876 | 2.524 | 2.876 | 1.051 |
| 2 | 117 | 2.379 | 2.359 | 2.637 | 1.074 |
| 2 | 118 | 1.982 | 2.274 | 2.274 | 1.415 |
| 2 | 119 | 1.741 | 1.992 | 2.283 | 0.393 |
| 2 | 120 | 2.759 | 3.044 | 3.044 | 2.518 |
| 2 | 122 | 2.527 | 1.499 | 2.367 | 1.749 |
| 2 | 123 | 2.300 | 0.996 | 1.530 | 0.738 |
| 2 | 124 | 0.194 | 0.191 | 0.189 | 0.144 |
| 2 | 125 | 2.278 | 2.051 | 2.197 | 1.460 |
| 2 | 126 | 0.142 | 0.142 | 0.143 | 0.088 |
| 2 | 127 | 1.811 | 1.530 | 1.867 | 1.574 |
| 2 | 128 | 2.312 | 1.957 | 1.355 | 1.093 |
| 2 | 129 | 2.277 | 2.272 | 1.782 | 1.147 |
| 2 | 130 | 2.258 | 2.313 | 2.348 | 0.983 |
| 2 | 131 | 2.223 | 1.924 | 1.213 | 0.468 |
| 2 | 132 | 3.207 | 0.984 | 0.702 | 0.563 |
| 2 | 133 | 2.235 | 0.984 | 1.264 | 0.527 |
| 2 | 134 | 0.767 | 0.766 | 0.765 | 0.815 |
| 2 | 135 | 2.120 | 1.642 | 1.347 | 0.635 |
| 2 | 136 | 2.065 | 1.948 | 1.781 | 1.047 |
| 2 | 137 | 0.465 | 0.465 | 0.465 | 1.693 |
| 2 | 138 | 0.139 | 0.137 | 0.135 | 0.138 |
| 2 | 139 | 2.156 | 0.790 | 1.122 | 1.228 |
| 2 | 140 | 2.916 | 1.696 | 2.437 | 1.990 |
| 3 | 0   | 2.692 | 0.879 | 1.198 | 0.327 |
| 3 | 1   | 3.912 | 1.600 | 1.960 | 1.029 |
| 3 | 2   | 3.988 | 2.186 | 2.417 | 0.818 |
| 3 | 3   | 3.679 | 1.253 | 1.512 | 0.649 |
| 3 | 4   | 3.369 | 1.302 | 1.272 | 0.451 |
| 3 | 5   | 4.037 | 1.074 | 1.515 | 0.461 |
| 3 | 6   | 3.757 | 0.752 | 0.943 | 0.476 |
| 3 | 7   | 3.570 | 1.353 | 1.952 | 1.055 |
| 3 | 8   | 3.136 | 1.754 | 1.861 | 0.747 |
| 3 | 9   | 3.931 | 0.988 | 1.197 | 0.368 |
| 3 | 10  | 3.025 | 1.509 | 1.526 | 0.649 |
| 3 | 11  | 3.156 | 0.987 | 1.365 | 0.373 |
| 3 | 12  | 3.729 | 1.341 | 1.438 | 0.725 |
| 3 | 13  | 3.378 | 2.279 | 1.436 | 0.730 |
| 3 | 14  | 3.821 | 2.580 | 2.534 | 0.773 |
| 3 | 15  | 2.804 | 1.028 | 1.116 | 0.383 |
| 3 | 16  | 3.040 | 2.313 | 1.350 | 0.618 |
| 3 | 17  | 3.489 | 2.964 | 1.632 | 0.600 |
| 3 | 18  | 3.620 | 3.526 | 2.898 | 0.524 |
| 3 | 19  | 4.771 | 2.353 | 1.188 | 0.914 |
| 3 | 20  | 3.403 | 2.577 | 2.433 | 1.197 |
| 3 | 21  | 2.948 | 1.788 | 1.820 | 0.455 |
| 3 | 22  | 3.769 | 1.015 | 1.528 | 1.059 |
| 3 | 23  | 2.485 | 1.344 | 1.370 | 0.980 |
| 3 | 24  | 2.960 | 2.232 | 1.901 | 0.673 |
| 3 | 25  | 3.116 | 2.602 | 1.995 | 0.969 |

|   |    |       |       |       |       |
|---|----|-------|-------|-------|-------|
| 3 | 26 | 4.228 | 3.502 | 3.377 | 1.256 |
| 3 | 27 | 3.416 | 2.585 | 2.307 | 1.199 |
| 3 | 28 | 3.524 | 2.890 | 3.046 | 0.923 |
| 3 | 29 | 3.266 | 1.377 | 2.271 | 0.443 |
| 3 | 30 | 3.064 | 2.998 | 3.085 | 1.927 |
| 3 | 31 | 2.482 | 1.104 | 0.956 | 0.416 |
| 3 | 32 | 2.978 | 1.782 | 1.723 | 0.602 |
| 3 | 33 | 3.480 | 1.761 | 2.133 | 1.334 |
| 3 | 34 | 2.914 | 1.856 | 2.031 | 0.961 |
| 3 | 35 | 2.919 | 1.601 | 1.330 | 0.449 |
| 3 | 36 | 3.238 | 3.033 | 2.778 | 1.628 |
| 3 | 37 | 4.988 | 1.986 | 2.224 | 1.293 |
| 3 | 38 | 4.578 | 1.532 | 2.686 | 1.385 |
| 3 | 39 | 3.369 | 2.302 | 1.345 | 0.878 |
| 3 | 40 | 2.130 | 1.333 | 1.373 | 0.982 |
| 3 | 41 | 2.602 | 1.543 | 1.713 | 0.906 |
| 3 | 42 | 2.704 | 1.592 | 2.602 | 1.580 |
| 3 | 43 | 4.576 | 1.711 | 1.454 | 0.970 |
| 3 | 44 | 2.551 | 1.982 | 1.411 | 0.958 |
| 3 | 45 | 2.415 | 1.912 | 1.528 | 0.978 |
| 3 | 46 | 2.727 | 2.141 | 2.892 | 1.497 |
| 3 | 47 | 4.370 | 2.832 | 2.897 | 0.926 |
| 3 | 48 | 4.282 | 2.589 | 2.591 | 0.959 |
| 3 | 49 | 3.236 | 1.260 | 0.725 | 0.724 |
| 3 | 50 | 2.517 | 2.582 | 2.950 | 0.804 |
| 3 | 51 | 3.398 | 3.333 | 2.893 | 1.816 |
| 3 | 52 | 1.105 | 1.093 | 1.086 | 0.996 |
| 3 | 53 | 2.588 | 2.729 | 2.537 | 0.746 |
| 3 | 54 | 0.823 | 0.828 | 0.831 | 1.130 |
| 3 | 55 | 2.067 | 1.669 | 2.303 | 0.626 |
| 3 | 56 | 3.342 | 3.206 | 2.579 | 1.216 |
| 3 | 57 | 2.982 | 2.236 | 1.637 | 0.512 |
| 3 | 58 | 4.157 | 2.098 | 1.363 | 0.883 |
| 3 | 59 | 4.527 | 1.687 | 2.336 | 0.680 |
| 3 | 60 | 2.165 | 1.502 | 2.165 | 0.533 |
| 3 | 61 | 2.834 | 2.335 | 1.639 | 0.877 |
| 3 | 62 | 2.340 | 2.620 | 2.921 | 0.368 |
| 3 | 63 | 1.698 | 1.404 | 1.405 | 0.801 |
| 3 | 64 | 2.770 | 0.861 | 1.198 | 1.144 |
| 3 | 65 | 2.707 | 1.328 | 1.113 | 0.865 |
| 3 | 66 | 1.333 | 0.732 | 1.448 | 0.763 |
| 3 | 67 | 2.443 | 1.402 | 1.893 | 0.725 |
| 3 | 68 | 2.707 | 1.586 | 2.292 | 1.068 |
| 3 | 69 | 3.904 | 1.662 | 1.560 | 0.820 |
| 3 | 70 | 2.562 | 2.102 | 2.327 | 1.101 |
| 3 | 71 | 2.425 | 1.629 | 1.133 | 0.807 |
| 3 | 72 | 2.324 | 2.606 | 2.902 | 0.292 |
| 3 | 73 | 2.929 | 2.929 | 2.929 | 0.871 |
| 3 | 74 | 2.543 | 1.519 | 1.008 | 0.732 |
| 3 | 75 | 2.517 | 0.665 | 0.744 | 1.152 |
| 3 | 76 | 4.123 | 2.197 | 1.863 | 1.774 |
| 3 | 77 | 2.237 | 1.604 | 1.650 | 0.699 |
| 3 | 78 | 1.463 | 1.446 | 1.564 | 0.807 |
| 3 | 79 | 2.115 | 2.377 | 2.680 | 0.383 |
| 3 | 80 | 2.482 | 1.809 | 0.935 | 0.824 |
| 3 | 81 | 2.399 | 2.110 | 1.862 | 0.891 |
| 3 | 82 | 1.945 | 1.678 | 1.490 | 0.729 |

|   |     |       |       |       |       |
|---|-----|-------|-------|-------|-------|
| 3 | 83  | 1.852 | 1.779 | 1.944 | 0.661 |
| 3 | 84  | 2.322 | 2.985 | 2.863 | 1.062 |
| 3 | 85  | 4.141 | 3.467 | 3.467 | 3.368 |
| 3 | 86  | 2.073 | 1.192 | 1.104 | 1.262 |
| 3 | 87  | 4.539 | 2.530 | 2.977 | 1.004 |
| 3 | 88  | 4.544 | 1.710 | 1.923 | 0.593 |
| 3 | 89  | 1.472 | 1.299 | 1.385 | 0.884 |
| 3 | 90  | 1.695 | 1.755 | 1.810 | 1.210 |
| 3 | 91  | 2.118 | 2.105 | 2.097 | 1.132 |
| 3 | 92  | 2.110 | 2.390 | 2.662 | 0.754 |
| 3 | 93  | 2.501 | 2.310 | 2.483 | 0.854 |
| 3 | 94  | 0.513 | 0.527 | 0.639 | 0.590 |
| 3 | 95  | 1.003 | 1.233 | 1.563 | 0.493 |
| 3 | 96  | 2.701 | 2.297 | 2.108 | 0.937 |
| 3 | 97  | 2.076 | 0.386 | 0.767 | 1.581 |
| 3 | 99  | 3.377 | 3.228 | 3.377 | 3.371 |
| 3 | 100 | 2.031 | 2.273 | 2.502 | 0.461 |
| 3 | 101 | 1.637 | 1.638 | 1.638 | 0.909 |
| 3 | 102 | 2.173 | 1.884 | 1.416 | 1.008 |
| 3 | 103 | 2.911 | 2.296 | 1.143 | 0.957 |
| 3 | 104 | 3.424 | 3.424 | 3.424 | 0.587 |
| 3 | 105 | 0.808 | 1.347 | 1.627 | 0.768 |
| 3 | 106 | 3.012 | 2.700 | 2.350 | 1.149 |
| 3 | 107 | 2.612 | 2.557 | 2.438 | 0.555 |
| 3 | 108 | 4.530 | 2.091 | 1.498 | 1.437 |
| 3 | 109 | 3.023 | 2.139 | 2.141 | 1.446 |
| 3 | 110 | 1.933 | 1.690 | 2.078 | 0.672 |
| 3 | 111 | 2.772 | 2.763 | 2.793 | 2.808 |
| 3 | 113 | 2.262 | 1.009 | 1.402 | 0.524 |
| 3 | 114 | 3.469 | 2.453 | 1.910 | 0.677 |
| 3 | 115 | 2.469 | 0.975 | 1.609 | 0.789 |
| 3 | 116 | 4.439 | 2.927 | 3.251 | 1.409 |
| 3 | 117 | 2.410 | 0.532 | 0.744 | 0.877 |
| 3 | 118 | 2.699 | 2.401 | 2.401 | 0.837 |
| 3 | 119 | 1.872 | 2.151 | 2.438 | 0.428 |
| 3 | 120 | 1.330 | 1.330 | 1.330 | 0.563 |
| 3 | 122 | 2.515 | 2.388 | 2.339 | 2.004 |
| 3 | 123 | 2.420 | 1.165 | 1.899 | 1.322 |
| 3 | 124 | 0.161 | 0.165 | 0.168 | 0.202 |
| 3 | 125 | 1.835 | 1.835 | 1.835 | 0.975 |
| 3 | 126 | 0.290 | 0.288 | 0.279 | 0.373 |
| 3 | 127 | 1.466 | 1.444 | 1.596 | 0.863 |
| 3 | 128 | 2.436 | 2.089 | 1.617 | 1.109 |
| 3 | 129 | 2.204 | 2.108 | 2.062 | 1.106 |
| 3 | 130 | 1.733 | 1.800 | 1.869 | 0.935 |
| 3 | 131 | 2.025 | 1.816 | 1.273 | 0.769 |
| 3 | 132 | 2.460 | 2.434 | 2.130 | 0.748 |
| 3 | 133 | 4.872 | 1.143 | 1.513 | 0.548 |
| 3 | 134 | 0.110 | 0.110 | 0.110 | 0.260 |
| 3 | 135 | 2.271 | 1.803 | 0.845 | 0.828 |
| 3 | 136 | 4.734 | 2.089 | 2.100 | 0.709 |
| 3 | 137 | 1.478 | 1.479 | 1.479 | 0.956 |
| 3 | 138 | 0.175 | 0.172 | 0.168 | 0.243 |
| 3 | 139 | 2.228 | 2.070 | 2.573 | 2.072 |
| 3 | 140 | 2.824 | 1.862 | 2.712 | 2.016 |
| 4 | 0   | 3.901 | 0.900 | 1.167 | 0.410 |
| 4 | 1   | 4.113 | 1.964 | 2.186 | 0.802 |

|   |    |       |       |       |       |
|---|----|-------|-------|-------|-------|
| 4 | 2  | 3.921 | 2.078 | 1.989 | 1.341 |
| 4 | 3  | 3.946 | 1.292 | 1.547 | 0.610 |
| 4 | 4  | 3.291 | 2.909 | 1.906 | 0.721 |
| 4 | 5  | 3.946 | 0.957 | 1.053 | 0.499 |
| 4 | 6  | 3.649 | 1.116 | 1.533 | 0.780 |
| 4 | 7  | 3.362 | 1.556 | 1.538 | 0.724 |
| 4 | 8  | 3.461 | 2.548 | 3.426 | 1.097 |
| 4 | 9  | 3.901 | 1.268 | 1.302 | 0.491 |
| 4 | 10 | 3.906 | 1.398 | 1.644 | 0.830 |
| 4 | 11 | 2.129 | 0.873 | 1.113 | 0.496 |
| 4 | 12 | 3.660 | 1.792 | 1.838 | 0.955 |
| 4 | 13 | 2.951 | 3.020 | 2.141 | 0.688 |
| 4 | 14 | 3.782 | 1.180 | 1.593 | 0.756 |
| 4 | 15 | 3.504 | 1.923 | 1.189 | 0.410 |
| 4 | 16 | 3.498 | 3.271 | 2.520 | 0.707 |
| 4 | 17 | 3.464 | 3.237 | 2.550 | 0.622 |
| 4 | 18 | 3.341 | 3.231 | 2.679 | 0.848 |
| 4 | 19 | 3.776 | 2.170 | 1.575 | 0.960 |
| 4 | 20 | 3.291 | 2.212 | 1.425 | 0.561 |
| 4 | 21 | 2.930 | 1.170 | 1.311 | 0.711 |
| 4 | 22 | 3.411 | 2.472 | 2.756 | 1.168 |
| 4 | 23 | 1.447 | 1.151 | 1.048 | 1.365 |
| 4 | 24 | 2.957 | 1.693 | 2.060 | 0.695 |
| 4 | 25 | 3.692 | 2.235 | 1.748 | 1.439 |
| 4 | 26 | 3.486 | 1.077 | 1.791 | 0.795 |
| 4 | 27 | 3.087 | 1.523 | 1.691 | 1.539 |
| 4 | 28 | 3.018 | 2.415 | 2.307 | 1.026 |
| 4 | 29 | 3.158 | 2.577 | 2.237 | 0.474 |
| 4 | 30 | 3.904 | 2.795 | 3.673 | 2.714 |
| 4 | 31 | 2.750 | 1.182 | 0.921 | 0.380 |
| 4 | 32 | 2.841 | 1.370 | 1.472 | 0.769 |
| 4 | 33 | 4.332 | 2.932 | 2.397 | 0.752 |
| 4 | 34 | 3.113 | 1.644 | 1.712 | 0.941 |
| 4 | 35 | 3.657 | 2.778 | 2.672 | 1.554 |
| 4 | 36 | 2.558 | 2.343 | 1.986 | 1.037 |
| 4 | 37 | 4.248 | 2.960 | 1.816 | 1.282 |
| 4 | 38 | 4.115 | 2.280 | 3.431 | 2.129 |
| 4 | 39 | 2.976 | 1.753 | 1.448 | 1.154 |
| 4 | 40 | 2.082 | 2.134 | 2.188 | 1.308 |
| 4 | 41 | 2.730 | 1.118 | 1.590 | 0.689 |
| 4 | 42 | 3.204 | 3.345 | 2.650 | 2.248 |
| 4 | 43 | 4.413 | 2.146 | 1.973 | 1.178 |
| 4 | 44 | 2.614 | 1.822 | 0.871 | 0.645 |
| 4 | 45 | 2.654 | 2.097 | 1.449 | 0.844 |
| 4 | 46 | 3.480 | 3.501 | 3.510 | 2.174 |
| 4 | 47 | 3.005 | 2.250 | 2.127 | 0.802 |
| 4 | 48 | 3.079 | 2.019 | 1.606 | 0.787 |
| 4 | 49 | 3.828 | 3.070 | 2.846 | 2.488 |
| 4 | 50 | 3.310 | 2.953 | 3.176 | 1.119 |
| 4 | 51 | 3.280 | 2.103 | 2.110 | 1.248 |
| 4 | 52 | 1.429 | 1.449 | 1.478 | 1.456 |
| 4 | 53 | 2.707 | 2.269 | 0.970 | 0.673 |
| 4 | 54 | 0.548 | 0.560 | 0.559 | 0.563 |
| 4 | 55 | 2.197 | 1.507 | 1.981 | 0.647 |
| 4 | 56 | 3.339 | 1.552 | 1.465 | 0.727 |
| 4 | 57 | 2.986 | 1.110 | 0.928 | 0.471 |
| 4 | 58 | 2.811 | 1.328 | 1.264 | 0.867 |

|   |     |       |       |       |       |
|---|-----|-------|-------|-------|-------|
| 4 | 59  | 2.806 | 2.603 | 2.374 | 0.532 |
| 4 | 60  | 3.632 | 3.542 | 3.544 | 2.205 |
| 4 | 61  | 2.626 | 1.615 | 0.962 | 0.834 |
| 4 | 62  | 2.054 | 2.343 | 2.753 | 0.341 |
| 4 | 63  | 2.543 | 1.411 | 1.739 | 1.361 |
| 4 | 64  | 2.342 | 1.598 | 1.729 | 0.874 |
| 4 | 65  | 2.795 | 1.228 | 1.428 | 0.713 |
| 4 | 66  | 3.278 | 1.635 | 3.043 | 2.291 |
| 4 | 67  | 2.468 | 1.519 | 1.364 | 0.668 |
| 4 | 68  | 4.122 | 2.763 | 2.895 | 1.592 |
| 4 | 69  | 5.209 | 2.830 | 2.578 | 0.826 |
| 4 | 70  | 3.838 | 2.862 | 3.019 | 1.632 |
| 4 | 71  | 2.561 | 1.954 | 1.400 | 0.660 |
| 4 | 72  | 2.050 | 2.316 | 2.714 | 0.409 |
| 4 | 73  | 3.389 | 2.869 | 3.129 | 1.967 |
| 4 | 74  | 2.283 | 1.064 | 1.110 | 0.729 |
| 4 | 75  | 2.433 | 1.237 | 0.628 | 0.559 |
| 4 | 76  | 2.079 | 1.403 | 1.334 | 0.719 |
| 4 | 77  | 2.756 | 1.668 | 1.111 | 0.592 |
| 4 | 78  | 2.198 | 2.072 | 1.894 | 1.178 |
| 4 | 79  | 1.904 | 2.173 | 2.611 | 0.246 |
| 4 | 80  | 3.221 | 1.525 | 1.302 | 0.521 |
| 4 | 81  | 2.588 | 1.449 | 1.728 | 1.127 |
| 4 | 82  | 2.467 | 1.765 | 1.700 | 0.911 |
| 4 | 83  | 2.852 | 2.716 | 2.844 | 1.032 |
| 4 | 84  | 2.777 | 0.975 | 1.453 | 0.875 |
| 4 | 85  | 4.416 | 1.283 | 1.283 | 1.128 |
| 4 | 86  | 2.709 | 1.434 | 1.531 | 1.150 |
| 4 | 87  | 1.742 | 2.548 | 2.579 | 0.698 |
| 4 | 88  | 4.225 | 1.942 | 1.528 | 0.625 |
| 4 | 89  | 3.705 | 1.902 | 1.920 | 1.106 |
| 4 | 90  | 1.763 | 1.861 | 1.911 | 1.337 |
| 4 | 91  | 3.042 | 0.907 | 0.918 | 1.579 |
| 4 | 92  | 1.805 | 2.085 | 2.436 | 0.370 |
| 4 | 93  | 2.390 | 2.146 | 2.298 | 0.816 |
| 4 | 94  | 6.637 | 1.235 | 0.074 | 0.132 |
| 4 | 95  | 1.697 | 1.828 | 2.352 | 0.198 |
| 4 | 96  | 4.339 | 2.735 | 2.430 | 0.869 |
| 4 | 100 | 1.831 | 2.085 | 2.380 | 0.427 |
| 4 | 102 | 3.545 | 2.177 | 0.960 | 0.482 |
| 4 | 103 | 2.306 | 1.890 | 1.328 | 0.734 |
| 4 | 104 | 4.700 | 2.484 | 1.869 | 0.524 |
| 4 | 106 | 3.166 | 2.624 | 1.709 | 0.611 |
| 4 | 107 | 2.625 | 2.394 | 2.349 | 0.661 |
| 4 | 108 | 2.885 | 1.147 | 1.128 | 0.704 |
| 4 | 109 | 2.028 | 0.482 | 1.056 | 0.584 |
| 4 | 110 | 2.476 | 1.888 | 2.311 | 0.730 |
| 4 | 113 | 2.293 | 1.058 | 1.375 | 0.778 |
| 4 | 114 | 2.997 | 2.616 | 1.904 | 0.453 |
| 4 | 115 | 4.581 | 2.205 | 2.052 | 0.935 |
| 4 | 116 | 2.998 | 2.998 | 2.998 | 0.714 |
| 4 | 117 | 2.461 | 2.536 | 2.656 | 0.837 |
| 4 | 118 | 2.359 | 2.242 | 2.359 | 0.655 |
| 4 | 119 | 1.534 | 1.764 | 2.146 | 0.453 |
| 4 | 122 | 0.135 | 0.198 | 0.257 | 0.307 |
| 4 | 123 | 1.966 | 4.362 | 2.474 | 1.139 |
| 4 | 124 | 0.187 | 0.187 | 0.188 | 0.379 |

|   |     |       |       |       |       |
|---|-----|-------|-------|-------|-------|
| 4 | 125 | 3.167 | 3.345 | 3.418 | 3.095 |
| 4 | 126 | 3.636 | 1.004 | 1.003 | 0.999 |
| 4 | 127 | 2.308 | 2.127 | 2.282 | 0.996 |
| 4 | 128 | 2.319 | 1.656 | 1.286 | 0.748 |
| 4 | 129 | 3.435 | 2.450 | 2.116 | 0.695 |
| 4 | 130 | 1.538 | 1.642 | 1.754 | 1.237 |
| 4 | 131 | 2.216 | 1.924 | 1.283 | 0.611 |
| 4 | 132 | 4.743 | 1.261 | 1.871 | 0.751 |
| 4 | 133 | 4.740 | 1.735 | 1.120 | 0.595 |
| 4 | 134 | 0.086 | 0.071 | 0.055 | 0.049 |
| 4 | 135 | 2.166 | 3.032 | 2.334 | 0.714 |
| 4 | 136 | 2.744 | 1.899 | 0.867 | 1.351 |
| 4 | 137 | 0.023 | 0.023 | 0.023 | 2.282 |
| 4 | 138 | 0.541 | 1.181 | 1.180 | 1.178 |
| 4 | 139 | 2.032 | 1.610 | 1.618 | 1.438 |
| 4 | 140 | 0.168 | 0.515 | 0.656 | 1.269 |
| 5 | 0   | 3.078 | 0.828 | 0.956 | 0.461 |
| 5 | 1   | 3.151 | 1.673 | 2.010 | 0.954 |
| 5 | 2   | 2.726 | 1.690 | 1.565 | 1.276 |
| 5 | 3   | 2.432 | 1.048 | 1.251 | 0.888 |
| 5 | 4   | 3.188 | 1.027 | 1.286 | 0.833 |
| 5 | 5   | 2.156 | 0.860 | 2.385 | 0.534 |
| 5 | 6   | 2.410 | 0.985 | 2.576 | 0.711 |
| 5 | 7   | 2.010 | 2.065 | 1.592 | 1.542 |
| 5 | 8   | 2.429 | 1.853 | 1.466 | 1.059 |
| 5 | 9   | 3.433 | 0.635 | 1.251 | 0.402 |
| 5 | 10  | 2.308 | 0.867 | 1.406 | 0.864 |
| 5 | 11  | 2.695 | 0.972 | 1.159 | 0.440 |
| 5 | 12  | 2.435 | 1.508 | 1.762 | 0.852 |
| 5 | 13  | 2.341 | 0.899 | 1.335 | 0.926 |
| 5 | 14  | 2.197 | 1.302 | 1.791 | 1.365 |
| 5 | 15  | 3.083 | 0.901 | 1.124 | 0.466 |
| 5 | 16  | 3.029 | 1.305 | 3.241 | 0.744 |
| 5 | 17  | 2.889 | 1.386 | 3.279 | 0.753 |
| 5 | 18  | 3.721 | 3.520 | 3.425 | 0.825 |
| 5 | 19  | 3.094 | 1.277 | 3.241 | 1.539 |
| 5 | 20  | 2.384 | 2.329 | 2.500 | 0.953 |
| 5 | 21  | 2.601 | 1.374 | 2.177 | 0.674 |
| 5 | 22  | 2.085 | 1.027 | 1.388 | 0.769 |
| 5 | 23  | 1.926 | 1.979 | 1.927 | 1.195 |
| 5 | 24  | 2.663 | 2.467 | 2.318 | 0.743 |
| 5 | 25  | 3.072 | 1.558 | 1.992 | 0.892 |
| 5 | 26  | 2.669 | 3.009 | 3.186 | 0.821 |
| 5 | 27  | 2.986 | 2.098 | 2.959 | 1.233 |
| 5 | 28  | 2.051 | 1.820 | 2.504 | 0.754 |
| 5 | 29  | 3.283 | 1.671 | 2.519 | 0.546 |
| 5 | 30  | 3.003 | 1.669 | 3.076 | 3.052 |
| 5 | 31  | 2.608 | 1.454 | 1.247 | 0.513 |
| 5 | 32  | 2.592 | 2.472 | 2.357 | 0.446 |
| 5 | 33  | 2.713 | 2.849 | 2.920 | 0.946 |
| 5 | 34  | 2.058 | 2.530 | 2.625 | 0.772 |
| 5 | 35  | 3.279 | 2.897 | 2.723 | 1.893 |
| 5 | 36  | 2.379 | 2.636 | 2.737 | 1.061 |
| 5 | 37  | 3.173 | 2.891 | 1.556 | 1.225 |
| 5 | 38  | 2.717 | 2.387 | 2.693 | 2.107 |
| 5 | 39  | 2.513 | 1.418 | 0.740 | 1.467 |
| 5 | 40  | 2.632 | 2.652 | 2.674 | 1.437 |

|   |    |       |       |       |       |
|---|----|-------|-------|-------|-------|
| 5 | 41 | 2.316 | 2.111 | 2.103 | 0.684 |
| 5 | 42 | 1.934 | 2.594 | 2.037 | 2.378 |
| 5 | 43 | 3.897 | 1.940 | 1.753 | 1.178 |
| 5 | 44 | 2.222 | 0.969 | 1.412 | 0.774 |
| 5 | 45 | 2.559 | 2.109 | 1.930 | 1.250 |
| 5 | 46 | 3.154 | 1.547 | 2.795 | 2.354 |
| 5 | 47 | 2.821 | 2.781 | 2.785 | 1.265 |
| 5 | 48 | 2.417 | 2.601 | 2.660 | 1.410 |
| 5 | 49 | 2.784 | 2.800 | 2.608 | 1.823 |
| 5 | 50 | 1.490 | 1.855 | 2.386 | 0.764 |
| 5 | 51 | 2.956 | 2.374 | 2.982 | 1.040 |
| 5 | 52 | 2.098 | 2.102 | 2.078 | 1.480 |
| 5 | 53 | 2.665 | 3.051 | 2.565 | 0.639 |
| 5 | 54 | 3.494 | 3.494 | 3.494 | 2.638 |
| 5 | 55 | 2.126 | 1.782 | 1.950 | 0.980 |
| 5 | 56 | 3.321 | 3.315 | 3.281 | 0.602 |
| 5 | 57 | 3.060 | 2.412 | 2.542 | 0.696 |
| 5 | 58 | 2.036 | 1.658 | 2.002 | 0.906 |
| 5 | 59 | 2.337 | 1.338 | 1.527 | 0.523 |
| 5 | 60 | 3.555 | 3.396 | 3.393 | 2.171 |
| 5 | 61 | 2.521 | 2.420 | 2.165 | 0.703 |
| 5 | 62 | 2.078 | 2.207 | 2.488 | 0.332 |
| 5 | 63 | 2.800 | 2.894 | 2.823 | 1.567 |
| 5 | 64 | 2.403 | 0.962 | 2.470 | 1.048 |
| 5 | 65 | 2.324 | 1.637 | 1.244 | 0.461 |
| 5 | 66 | 3.350 | 3.197 | 3.548 | 2.235 |
| 5 | 67 | 2.463 | 1.134 | 1.068 | 0.844 |
| 5 | 68 | 2.633 | 0.889 | 1.249 | 1.268 |
| 5 | 69 | 2.273 | 1.067 | 2.201 | 0.551 |
| 5 | 70 | 0.304 | 0.973 | 1.742 | 0.910 |
| 5 | 71 | 2.279 | 1.962 | 0.736 | 0.581 |
| 5 | 72 | 2.148 | 2.258 | 2.522 | 0.316 |
| 5 | 73 | 1.824 | 2.261 | 1.749 | 1.724 |
| 5 | 74 | 2.067 | 0.820 | 1.648 | 0.665 |
| 5 | 75 | 3.977 | 1.595 | 1.186 | 0.698 |
| 5 | 76 | 2.938 | 2.938 | 2.868 | 1.644 |
| 5 | 77 | 3.402 | 1.413 | 1.508 | 0.647 |
| 5 | 78 | 2.137 | 2.025 | 2.224 | 1.044 |
| 5 | 79 | 2.329 | 2.411 | 2.693 | 0.614 |
| 5 | 80 | 2.418 | 0.760 | 0.804 | 0.469 |
| 5 | 81 | 2.227 | 1.347 | 2.230 | 0.912 |
| 5 | 82 | 1.613 | 1.939 | 1.815 | 1.128 |
| 5 | 83 | 1.667 | 1.558 | 1.551 | 1.483 |
| 5 | 84 | 2.541 | 2.530 | 2.530 | 1.073 |
| 5 | 85 | 1.016 | 1.104 | 1.116 | 0.743 |
| 5 | 86 | 2.568 | 0.964 | 1.042 | 1.660 |
| 5 | 87 | 4.101 | 1.488 | 3.723 | 0.660 |
| 5 | 88 | 3.195 | 0.836 | 1.050 | 0.506 |
| 5 | 89 | 4.808 | 1.508 | 0.942 | 1.119 |
| 5 | 90 | 2.481 | 2.540 | 2.565 | 1.369 |
| 5 | 91 | 2.675 | 2.664 | 2.659 | 1.910 |
| 5 | 92 | 1.832 | 1.943 | 2.169 | 0.519 |
| 5 | 93 | 2.017 | 2.184 | 2.239 | 0.804 |
| 5 | 94 | 3.271 | 2.539 | 2.824 | 3.149 |
| 5 | 95 | 1.666 | 1.692 | 1.914 | 0.921 |
| 5 | 96 | 2.003 | 2.051 | 2.118 | 0.925 |
| 5 | 97 | 3.689 | 3.665 | 3.702 | 3.686 |

|   |     |       |       |       |       |
|---|-----|-------|-------|-------|-------|
| 5 | 98  | 3.652 | 3.664 | 3.663 | 3.638 |
| 5 | 99  | 1.255 | 1.012 | 1.217 | 0.323 |
| 5 | 100 | 1.892 | 2.006 | 2.211 | 0.620 |
| 5 | 101 | 2.185 | 2.185 | 2.185 | 1.639 |
| 5 | 102 | 2.099 | 1.901 | 1.447 | 0.565 |
| 5 | 103 | 2.334 | 2.108 | 2.064 | 1.109 |
| 5 | 104 | 2.997 | 2.776 | 3.198 | 0.687 |
| 5 | 105 | 3.189 | 3.176 | 3.160 | 3.153 |
| 5 | 106 | 2.347 | 2.808 | 2.796 | 0.662 |
| 5 | 107 | 2.673 | 2.370 | 2.844 | 0.897 |
| 5 | 108 | 1.862 | 0.633 | 1.349 | 0.607 |
| 5 | 109 | 2.354 | 1.974 | 2.223 | 0.976 |
| 5 | 110 | 2.099 | 1.920 | 2.209 | 0.932 |
| 5 | 111 | 1.162 | 1.389 | 1.583 | 0.771 |
| 5 | 112 | 3.571 | 3.571 | 3.571 | 3.571 |
| 5 | 113 | 2.124 | 2.196 | 2.091 | 1.626 |
| 5 | 114 | 3.259 | 0.510 | 3.259 | 0.905 |
| 5 | 115 | 2.777 | 0.816 | 2.196 | 1.277 |
| 5 | 116 | 4.885 | 3.144 | 3.144 | 0.854 |
| 5 | 117 | 2.335 | 2.098 | 1.237 | 0.752 |
| 5 | 118 | 3.008 | 2.545 | 3.016 | 1.620 |
| 5 | 119 | 2.300 | 2.380 | 2.625 | 0.397 |
| 5 | 120 | 2.388 | 2.433 | 2.523 | 1.724 |
| 5 | 122 | 1.689 | 1.680 | 1.611 | 1.260 |
| 5 | 123 | 1.948 | 1.136 | 2.579 | 1.151 |
| 5 | 124 | 0.138 | 0.134 | 0.131 | 0.396 |
| 5 | 125 | 1.757 | 1.757 | 1.757 | 1.157 |
| 5 | 126 | 0.122 | 0.122 | 0.122 | 0.248 |
| 5 | 127 | 1.614 | 1.661 | 1.821 | 0.969 |
| 5 | 128 | 2.741 | 2.072 | 1.908 | 0.754 |
| 5 | 129 | 2.139 | 2.170 | 2.143 | 0.818 |
| 5 | 130 | 2.677 | 2.762 | 2.844 | 1.501 |
| 5 | 131 | 1.701 | 1.841 | 1.840 | 0.597 |
| 5 | 132 | 2.842 | 1.565 | 2.158 | 1.126 |
| 5 | 133 | 2.372 | 1.191 | 1.116 | 0.464 |
| 5 | 134 | 0.439 | 0.726 | 0.609 | 0.837 |
| 5 | 135 | 1.969 | 0.845 | 1.140 | 0.647 |
| 5 | 136 | 3.406 | 2.094 | 1.840 | 0.511 |
| 5 | 137 | 3.765 | 3.765 | 3.765 | 2.707 |
| 5 | 138 | 0.114 | 0.114 | 0.114 | 0.329 |
| 5 | 139 | 2.712 | 1.297 | 2.763 | 1.780 |
| 5 | 140 | 1.247 | 1.045 | 1.640 | 0.271 |

## Appendix G. Error distributions by dataset and model

**Table 25.** Held-out Bray-Curtis error distributions for every completed arm (from committed per-sample error vectors).

| dataset | model         | n samples | median | Q1    | Q3    | mean  |
|---------|---------------|-----------|--------|-------|-------|-------|
| Ocean   | cNODE2        | 269       | 0.075  | 0.054 | 0.117 | 0.106 |
| Ocean   | TwinStack     | 269       | 0.070  | 0.048 | 0.109 | 0.095 |
| Ocean   | ConstStack    | 269       | 0.070  | 0.048 | 0.110 | 0.095 |
| Ocean   | LightGBM      | 269       | 0.074  | 0.048 | 0.119 | 0.094 |
| Ocean   | presence null | 269       | 0.093  | 0.068 | 0.126 | 0.114 |
| Ocean   | cNODE         | 269       | 0.132  | 0.104 | 0.172 | 0.155 |

|                       |               |     |       |       |       |       |
|-----------------------|---------------|-----|-------|-------|-------|-------|
|                       | (ours)        |     |       |       |       |       |
| <b>Ocean</b>          | gLV (ours)    | 269 | 0.070 | 0.048 | 0.109 | 0.095 |
| <b>Ocean</b>          | GraphTwin     | 269 | 0.071 | 0.045 | 0.119 | 0.105 |
| <b>Ocean</b>          | xgb           | 269 | 0.072 | 0.048 | 0.114 | 0.094 |
| <b>Ocean</b>          | catb          | 269 | 0.778 | 0.663 | 0.813 | 0.718 |
| <b>Ocean</b>          | ridgeclr      | 269 | 0.069 | 0.046 | 0.117 | 0.102 |
| <b>Drosophila_Gut</b> | cNODE2        | 24  | 0.084 | 0.049 | 0.110 | 0.093 |
| <b>Drosophila_Gut</b> | TwinStack     | 24  | 0.087 | 0.037 | 0.147 | 0.097 |
| <b>Drosophila_Gut</b> | ConstStack    | 24  | 0.072 | 0.051 | 0.103 | 0.087 |
| <b>Drosophila_Gut</b> | LightGBM      | 24  | 0.293 | 0.190 | 0.338 | 0.259 |
| <b>Drosophila_Gut</b> | presence null | 24  | 0.172 | 0.067 | 0.221 | 0.156 |
| <b>Drosophila_Gut</b> | cNODE (ours)  | 24  | 0.104 | 0.064 | 0.137 | 0.108 |
| <b>Drosophila_Gut</b> | gLV (ours)    | 24  | 0.052 | 0.025 | 0.100 | 0.077 |
| <b>Drosophila_Gut</b> | GraphTwin     | 24  | 0.093 | 0.068 | 0.153 | 0.108 |
| <b>Drosophila_Gut</b> | xgb           | 24  | 0.077 | 0.043 | 0.124 | 0.108 |
| <b>Drosophila_Gut</b> | catb          | 24  | 0.258 | 0.125 | 0.335 | 0.225 |
| <b>Drosophila_Gut</b> | ridgeclr      | 24  | 0.173 | 0.072 | 0.235 | 0.158 |
| <b>Human_Gut</b>      | cNODE2        | 106 | 0.254 | 0.156 | 0.363 | 0.279 |
| <b>Human_Gut</b>      | TwinStack     | 106 | 0.244 | 0.166 | 0.327 | 0.270 |
| <b>Human_Gut</b>      | ConstStack    | 106 | 0.248 | 0.169 | 0.332 | 0.271 |
| <b>Human_Gut</b>      | LightGBM      | 106 | 0.278 | 0.187 | 0.357 | 0.293 |
| <b>Human_Gut</b>      | presence null | 106 | 0.256 | 0.197 | 0.349 | 0.289 |
| <b>Human_Gut</b>      | cNODE (ours)  | 106 | 0.275 | 0.190 | 0.397 | 0.305 |
| <b>Human_Gut</b>      | gLV (ours)    | 106 | 0.244 | 0.172 | 0.343 | 0.272 |
| <b>Human_Gut</b>      | GraphTwin     | 106 | 0.287 | 0.169 | 0.371 | 0.294 |
| <b>Human_Gut</b>      | xgb           | 106 | 0.278 | 0.202 | 0.364 | 0.295 |
| <b>Human_Gut</b>      | catb          | 106 | 0.424 | 0.369 | 0.487 | 0.435 |
| <b>Human_Gut</b>      | ridgeclr      | 106 | 0.250 | 0.145 | 0.421 | 0.290 |
| <b>Human_Ora I</b>    | cNODE2        | 143 | 0.222 | 0.176 | 0.308 | 0.246 |
| <b>Human_Ora I</b>    | TwinStack     | 143 | 0.202 | 0.157 | 0.283 | 0.226 |
| <b>Human_Ora I</b>    | ConstStack    | 143 | 0.202 | 0.158 | 0.277 | 0.224 |
| <b>Human_Ora I</b>    | LightGBM      | 143 | 0.206 | 0.164 | 0.292 | 0.232 |
| <b>Human_Ora I</b>    | presence null | 143 | 0.204 | 0.163 | 0.279 | 0.226 |
| <b>Human_Ora I</b>    | cNODE (ours)  | 143 | 0.245 | 0.198 | 0.326 | 0.268 |
| <b>Human_Ora I</b>    | gLV (ours)    | 143 | 0.216 | 0.164 | 0.297 | 0.236 |
| <b>Human_Ora I</b>    | GraphTwin     | 143 | 0.202 | 0.157 | 0.280 | 0.224 |
| <b>Human_Ora I</b>    | xgb           | 143 | 0.205 | 0.173 | 0.273 | 0.229 |

|                    |               |     |       |       |       |       |
|--------------------|---------------|-----|-------|-------|-------|-------|
| <b>Human_Ora I</b> | catb          | 143 | 0.379 | 0.337 | 0.415 | 0.381 |
| <b>Human_Ora I</b> | ridgeclr      | 143 | 0.200 | 0.167 | 0.281 | 0.225 |
| <b>Soil_Vitro</b>  | cNODE2        | 48  | 0.114 | 0.043 | 0.278 | 0.171 |
| <b>Soil_Vitro</b>  | TwinStack     | 48  | 0.177 | 0.098 | 0.309 | 0.211 |
| <b>Soil_Vitro</b>  | ConstStack    | 48  | 0.143 | 0.065 | 0.270 | 0.180 |
| <b>Soil_Vitro</b>  | LightGBM      | 48  | 0.127 | 0.049 | 0.204 | 0.158 |
| <b>Soil_Vitro</b>  | presence null | 48  | 0.205 | 0.069 | 0.278 | 0.195 |
| <b>Soil_Vitro</b>  | cNODE (ours)  | 48  | 0.157 | 0.091 | 0.239 | 0.179 |
| <b>Soil_Vitro</b>  | gLV (ours)    | 48  | 0.117 | 0.036 | 0.288 | 0.179 |
| <b>Soil_Vitro</b>  | GraphTwin     | 48  | 0.140 | 0.057 | 0.322 | 0.189 |
| <b>Soil_Vitro</b>  | xgb           | 48  | 0.097 | 0.045 | 0.234 | 0.145 |
| <b>Soil_Vitro</b>  | catb          | 48  | 0.226 | 0.080 | 0.336 | 0.228 |
| <b>Soil_Vitro</b>  | ridgeclr      | 48  | 0.175 | 0.056 | 0.284 | 0.183 |
| <b>Soil_Vivo</b>   | cNODE2        | 678 | 0.110 | 0.083 | 0.154 | 0.128 |
| <b>Soil_Vivo</b>   | TwinStack     | 678 | 0.106 | 0.077 | 0.148 | 0.123 |
| <b>Soil_Vivo</b>   | LightGBM      | 678 | 0.110 | 0.081 | 0.148 | 0.123 |
| <b>Soil_Vivo</b>   | presence null | 678 | 0.124 | 0.090 | 0.164 | 0.141 |
| <b>Soil_Vivo</b>   | cNODE (ours)  | 678 | 0.123 | 0.096 | 0.166 | 0.140 |
| <b>Soil_Vivo</b>   | gLV (ours)    | 678 | 0.106 | 0.078 | 0.148 | 0.123 |
| <b>Soil_Vivo</b>   | GraphTwin     | 678 | 0.119 | 0.084 | 0.167 | 0.138 |
| <b>Soil_Vivo</b>   | xgb           | 678 | 0.111 | 0.081 | 0.147 | 0.124 |
| <b>Soil_Vivo</b>   | catb          | 678 | 0.333 | 0.301 | 0.380 | 0.342 |
| <b>Soil_Vivo</b>   | ridgeclr      | 678 | 0.112 | 0.083 | 0.158 | 0.130 |

## 6. Negative results (kept by design)

- gLV replicator 'beat' of cNODE on Drosophila (0.052) and soil in vitro under 10-fold CV was RETRACTED: under LOO it scores 0.074 and 0.099, worse than published cNODE (0.066, 0.079).
- Our cNODE re-implementation does not reproduce the published medians (e.g. Drosophila 0.104 vs 0.066), so all cNODE comparisons use published numbers.
- GraphTwin (attention GNN) never beat published cNODE on any dataset.
- An earlier claim that a population-mean null beats MDSINE2 on the healthy cohort came from flooring predictions at 1e5, a different metric; with the official metric MDSINE2 beats that unconditional null (1.061 vs 1.442). RETRACTED.
- Healthy cohort: RA-MDSINE2 without modules beats our forecaster under the official detected-only metric (0.883 vs 0.919,  $p = 0.045$ ).
- Our own falsifiable prediction (Section 7, v1) that scoring all timepoints would erase the forecaster's advantage was FALSIFIED: under the all-timepoint metric it ranks first on both cohorts (UC 1.654 vs MDSINE2-NM 1.824; healthy 1.752 vs 2.072; results/mdsine2\_headtohead\_\*\_alltimepoints.json).

- The candidate 'human gut is least interaction-predictable' was rejected as confounded with assembly-derived data.
- Keystone consensus found only 2 of 248 genera at  $FDR < 0.05$ , both oral; there is no cross-biome keystone signal.
- Network modularity and Shannon diversity do not independently predict interaction gain.
- Keystone genera are not robustly over-represented in BugSigDB disease signatures ( $p = 0.062$  with estimated dispersion).
- No GTDB family is enriched among keystones after FDR (best  $q = 0.71$ ).
- Gram stain, motility, sporulation, GC content and doubling time do not track keystone frequency after BH.
- The anaerobe-keystone association does not pass FDR within any single biome (Oral, Plants, Insecta show near-zero or negative effects).
- The GAI effect failed under graphical lasso ( $p=0.15$ ) and igraph betweenness ( $p=0.34$ ); it is ridge-definition-specific. ProTraits failed its coverage gate (51 vs 100 genera).
- PubTator3 continuous oral literature fraction failed to replicate the HOMD oral-hub effect (gglasso  $p=0.57$ ; igraph  $p=0.23$ ). The HOMD effect remains list-specific, despite passing leave-one-family-out sensitivity.
- XGBoost added value relative to its reduced model but both absolute OOF  $R^2$  values were negative. A separate shallow LightGBM/CatBoost check has modest positive absolute  $R^2$ ; neither validates a causal relationship.
- Biopython Bio.Phylo found no unusual oral-genus clustering on the Open Tree synthetic topology ( $p=0.53$ ).
- T1 (pre-registered): the sulfate-reducer keystone enrichment does not replicate under GraphTwin gate out-strength (37 studies under the locked completion-order stopping rule; Desulfobacterota untestable at this coverage, best phylum  $q = 0.906$ ).
- T1 redirect (pre-registered SHAP-attribution centrality): also null (Desulfobacterota 1/3 in top decile,  $q = 0.462$ ); the enrichment is specific to the ridge out-strength definition.
- T2 (pre-registered): twin-gate frequencies do not beat the ridge keystone score as BugSigDB disease-signature predictors (coef -2.28,  $p = 0.168$ , AIC 767.1 vs 766.1).

## 7. Discussion

Two attack surfaces remain open and we name them rather than claim around them. First, beating a population null is evidence about predictive content, not proof of causal interactions: cross-environment transfer, removal of co-occurrence information, and

validation against independently measured perturbation outcomes are the decisive tests, and they need perturbation cohorts we do not have (future work). Second, the 160-study MGnify pool is heterogeneous in platform, primers, preprocessing and design; our within-project controls are the per-biome breakdown and the assembly-artefact catch, and a random-effects meta-analysis across studies is the natural strengthening. Both are stated so a judge does not have to find them.

The discovery programme's main output is itself a result: digital-twin-derived ecological hypotheses require stronger validation than network inference provides. The sulfate-reducer keystone enrichment survived two taxonomies and a genomic index, then failed under a nonlinear model class and under SHAP-attribution centrality; what replicated as a disease-literature predictor was the gradient-boosted attribution ranking, not the keystone list.

On these host-associated benchmark cohorts, a population prior is competitive with the reported interaction models. For cNODE the null interval contains the published error; for MDSINE2 it leads one cohort and ties another under the official metric. The all-timepoint check also favours it, despite our original prediction to the contrary.

[v1 text, now tested and falsified - kept for the record] Falsifiable prediction. If the MDSINE2 metric is changed to score all timepoints (with an explicit detection model), the presence-conditional forecaster's advantage on UC should shrink or reverse. The all-timepoint test in Section 5.2 falsified that prediction: the forecaster remains ahead on the released source data. We retain the prediction for auditability, not as an open item.

Practical recommendation. Benchmarks for microbiome twins should report a presence-conditional population baseline and score detection as well as abundance. Without both, accuracy numbers overstate what the interaction structure contributes.

Limitations. Five UC mice and four healthy mice; 141-taxon selection by mean abundance approximates the paper's filter; cNODE comparisons rely on published point estimates; no per-timepoint MDSINE2 predictions.

## 8. Tools used

The program target of 40 counted external tools was reached with a margin; count the ledger entries marked True, not infrastructure. Tools include source databases and analytical libraries actually used in committed scripts. This count does not measure independence of biological evidence.

**Table 26. Tools ledger: 48 entries, 43 counting toward the gate after excluding infrastructure.**

| tool  | kind    | gate   |
|-------|---------|--------|
| NumPy | library | counts |
| SciPy | library | counts |

|                                                                                     |                         |                  |
|-------------------------------------------------------------------------------------|-------------------------|------------------|
| <b>pandas</b>                                                                       | library                 | counts           |
| <b>PyTorch</b>                                                                      | library                 | counts           |
| <b>Matplotlib</b>                                                                   | library                 | counts           |
| <b>python-docx</b>                                                                  | library                 | infra (excluded) |
| <b>pytest</b>                                                                       | test framework          | infra (excluded) |
| <b>LibreOffice (soffice headless)</b>                                               | application             | infra (excluded) |
| <b>poppler pdfinfo</b>                                                              | application             | infra (excluded) |
| <b>Git + GitHub</b>                                                                 | service                 | infra (excluded) |
| <b>EBI MGnify REST API v1</b>                                                       | web API                 | counts           |
| <b>cNODE published benchmark data (Michel-Mata et al. 2022)</b>                     | public data release     | counts           |
| <b>MDSINE2 published data and Source Data (Gibson et al. 2025)</b>                  | public data release     | counts           |
| <b>scikit-bio</b>                                                                   | library                 | counts           |
| <b>scikit-learn</b>                                                                 | library                 | counts           |
| <b>networkx</b>                                                                     | library                 | counts           |
| <b>statsmodels</b>                                                                  | library                 | counts           |
| <b>NCBI Taxonomy (E-utilities esearch/efetch)</b>                                   | database/web API        | counts           |
| <b>BugSigDB (genus signature export)</b>                                            | database                | counts           |
| <b>GTDB (release v232 bac120 taxonomy)</b>                                          | database                | counts           |
| <b>Madin et al. 2020 bacterial/archaeal trait database (condensed species NCBI)</b> | database                | counts           |
| <b>KEGG REST (genome list + KO-gene links)</b>                                      | database                | counts           |
| <b>BV-BRC genome API (oxygen_requirement facets)</b>                                | database                | counts           |
| <b>IJSEM phenotypic database (Barberan et al. 2017, figshare 4272392)</b>           | database                | counts           |
| <b>NCBI Datasets v2 API (genome dataset_report)</b>                                 | database/service        | counts           |
| <b>Open Tree of Life API v3 (TNRS match_names + induced synthetic subtree)</b>      | web API                 | counts           |
| <b>ProTraits (Brbic et al. 2016; protraits.irb.hr integrated Pr&gt;=0.95)</b>       | database                | counts           |
| <b>Disbiome (disbiome.ugent.be experiment API)</b>                                  | database/web API        | counts           |
| <b>Europe PMC REST API (search hitCount)</b>                                        | web API                 | counts           |
| <b>gglasso (Schaipp et al. 2021 graphical lasso)</b>                                | library                 | counts           |
| <b>Wikidata SPARQL endpoint (P2597 Gram staining)</b>                               | web API/knowledge graph | counts           |
| <b>ENA portal API + ENA taxonomy REST (assembly counts)</b>                         | web API                 | counts           |
| <b>GBIF API v1 (species/match + occurrence country facets)</b>                      | web API                 | counts           |
| <b>UniProt REST API (uniprotkb search counts, reference proteomes)</b>              | web API                 | counts           |
| <b>Crossref REST API (works total-results)</b>                                      | web API                 | counts           |
| <b>python-igraph (betweenness</b>                                                   | library                 | counts           |

|                                                                 |          |        |
|-----------------------------------------------------------------|----------|--------|
| centrality)                                                     |          |        |
| EBI Search REST (Ensembl Genomes genome domain)                 | web API  | counts |
| XGBoost (gradient-boosted trees)                                | library  | counts |
| InterPro API (protein counts per taxon and entry)               | web API  | counts |
| HOMD expanded Human Oral Microbiome Database (taxon table v4.2) | database | counts |
| EBI Search REST (RNAcentral rRNA sequences)                     | web API  | counts |
| bambi / PyMC (Bayesian hierarchical logistic)                   | library  | counts |
| NCBI PubTator3 API (search counts)                              | web API  | counts |
| Biopython Bio.Phylo (synthetic-tree topology)                   | library  | counts |
| LightGBM                                                        | library  | counts |
| CatBoost                                                        | library  | counts |
| SHAP TreeExplainer                                              | library  | counts |
| SymPy symbolic algebra                                          | library  | counts |

## 9. Reproducibility

Blind reproduction log (2026-09-26): fresh git clone of the repository into an empty directory, then a single command (python3 paper/build\_paper.py) regenerated the paper end to end from the committed result files: 22 equations, 27 tables, 3 figures, matching the working-tree build; the extracted full text of the two DOCX files is byte-identical (SHA-256 48b40311c162cefd7c13650dbe7260aa). The DOCX container bytes differ only by embedded timestamps.

Repository: [github.com/uditakankananonononono/mega27-04-microbiome-twin](https://github.com/uditakankananonononono/mega27-04-microbiome-twin) (private).  
 Commands: `python run_bench.py <dataset>`  
`presence_mean,cnode,glv,graphtwin 10; python bench_mdssine2.py healthy|uc; python -m`  
`pytest -q; python paper/build_paper.py.`

## Appendix A. MGnify study accessions used

**Table 27. All 160 MGnify studies fetched and used (data/raw/mgnify/manifest.csv).**

| MGnify study | INSDC project | biome            | samples | genera |
|--------------|---------------|------------------|---------|--------|
| MGYS00006862 | ERP175206     | Digestive system | 84      | 106    |
| MGYS00005791 | ERP127967     | Digestive system | 245     | 124    |
| MGYS00006825 | ERP160132     | Digestive system | 197     | 142    |
| MGYS00006824 | ERP147941     | Digestive system | 40      | 98     |
| MGYS00006822 | ERP148470     | Digestive system | 55      | 95     |
| MGYS00006814 | ERP148607     | Digestive system | 68      | 143    |
| MGYS00006815 | ERP147808     | Digestive system | 67      | 88     |
| MGYS00006811 | ERP147961     | Digestive system | 98      | 89     |
| MGYS00006807 | ERP170897     | Digestive system | 60      | 111    |
| MGYS00006805 | ERP148499     | Digestive system | 86      | 131    |
| MGYS00006806 | ERP136860     | Digestive system | 74      | 103    |

|              |           |                  |      |      |
|--------------|-----------|------------------|------|------|
| MGYS00006760 | ERP148661 | Digestive system | 218  | 159  |
| MGYS00002338 | ERP106359 | Digestive system | 303  | 158  |
| MGYS00006797 | ERP147940 | Digestive system | 175  | 151  |
| MGYS00006551 | ERP152745 | Digestive system | 2047 | 458  |
| MGYS00006795 | SRP162433 | Digestive system | 31   | 205  |
| MGYS00006794 | ERP128598 | Digestive system | 184  | 439  |
| MGYS00006789 | ERP148573 | Digestive system | 108  | 137  |
| MGYS00006787 | ERP160127 | Digestive system | 108  | 125  |
| MGYS00006786 | ERP147955 | Digestive system | 134  | 110  |
| MGYS00006759 | ERP146353 | Digestive system | 692  | 224  |
| MGYS00006780 | ERP160119 | Digestive system | 24   | 48   |
| MGYS00006779 | ERP148332 | Digestive system | 57   | 109  |
| MGYS00006769 | ERP147945 | Digestive system | 116  | 136  |
| MGYS00006765 | ERP160150 | Digestive system | 28   | 57   |
| MGYS00006755 | SRP097190 | Digestive system | 83   | 353  |
| MGYS00005333 | ERP108859 | Digestive system | 244  | 146  |
| MGYS00005230 | ERP005153 | Digestive system | 913  | 531  |
| MGYS00002675 | ERP107102 | Digestive system | 24   | 33   |
| MGYS00006069 | ERP133351 | Digestive system | 1568 | 263  |
| MGYS00006006 | ERP133688 | Digestive system | 457  | 377  |
| MGYS00002061 | ERP104235 | Digestive system | 52   | 75   |
| MGYS00000633 | SRP001634 | Digestive system | 1033 | 475  |
| MGYS00002238 | SRP097646 | Digestive system | 111  | 312  |
| MGYS00006562 | ERP150486 | Digestive system | 554  | 159  |
| MGYS00005154 | ERP001500 | Digestive system | 529  | 779  |
| MGYS00006649 | ERP131993 | Digestive system | 20   | 44   |
| MGYS00006641 | ERP127228 | Digestive system | 88   | 83   |
| MGYS00006642 | ERP127229 | Digestive system | 57   | 27   |
| MGYS00006364 | ERP134534 | Digestive system | 160  | 76   |
| MGYS00006640 | ERP150574 | Digestive system | 61   | 82   |
| MGYS00005253 | ERP108620 | Oral             | 606  | 330  |
| MGYS00006320 | SRP039515 | Oral             | 392  | 297  |
| MGYS00006276 | ERP128187 | Oral             | 91   | 455  |
| MGYS00006263 | ERP110514 | Oral             | 489  | 291  |
| MGYS00006017 | SRP330744 | Oral             | 53   | 210  |
| MGYS00005592 | ERP119456 | Oral             | 160  | 315  |
| MGYS00005569 | SRP256368 | Oral             | 71   | 246  |
| MGYS00005528 | DRP004359 | Oral             | 506  | 169  |
| MGYS00005506 | SRP115814 | Oral             | 701  | 359  |
| MGYS00005189 | DRP004640 | Oral             | 530  | 934  |
| MGYS00005223 | SRP126884 | Oral             | 767  | 565  |
| MGYS00005197 | SRP128931 | Oral             | 1319 | 1281 |
| MGYS00005143 | SRP126946 | Oral             | 7397 | 1486 |
| MGYS00002159 | SRP110002 | Oral             | 21   | 835  |
| MGYS00003719 | SRP125475 | Oral             | 21   | 218  |
| MGYS00002395 | SRP123221 | Oral             | 68   | 467  |
| MGYS00002394 | SRP051741 | Oral             | 20   | 166  |
| MGYS00002272 | SRP073270 | Oral             | 40   | 141  |
| MGYS00002269 | SRP062274 | Oral             | 147  | 230  |
| MGYS00002229 | SRP051201 | Oral             | 100  | 154  |
| MGYS00006648 | ERP156480 | Skin             | 67   | 634  |
| MGYS00005232 | ERP005182 | Skin             | 777  | 748  |
| MGYS00005156 | ERP005152 | Skin             | 938  | 1299 |
| MGYS00006246 | ERP149135 | Skin             | 50   | 256  |
| MGYS00005384 | SRP049645 | Skin             | 671  | 903  |
| MGYS00005172 | SRP113602 | Skin             | 1481 | 1523 |
| MGYS00005533 | ERP016280 | Skin             | 584  | 1687 |

|              |           |         |      |      |
|--------------|-----------|---------|------|------|
| MGYS00005212 | SRP071579 | Skin    | 644  | 901  |
| MGYS00005238 | SRP051059 | Skin    | 487  | 595  |
| MGYS00005101 | SRP057859 | Skin    | 72   | 243  |
| MGYS00005102 | ERP108413 | Skin    | 179  | 973  |
| MGYS00005135 | ERP005151 | Skin    | 109  | 696  |
| MGYS00005098 | SRP133226 | Skin    | 101  | 757  |
| MGYS00005037 | SRP056821 | Skin    | 106  | 1166 |
| MGYS00005090 | ERP115944 | Skin    | 120  | 16   |
| MGYS00002184 | ERP021896 | Skin    | 1964 | 1479 |
| MGYS00002072 | ERP019566 | Skin    | 258  | 1521 |
| MGYS00006855 | ERP135226 | Mammals | 116  | 426  |
| MGYS00006799 | ERP151258 | Mammals | 27   | 282  |
| MGYS00002673 | ERP003782 | Mammals | 93   | 797  |
| MGYS00006747 | SRP302361 | Mammals | 101  | 322  |
| MGYS00001223 | ERP010045 | Mammals | 30   | 148  |
| MGYS00001224 | ERP006247 | Mammals | 66   | 164  |
| MGYS00005267 | ERP003512 | Mammals | 643  | 231  |
| MGYS00000426 | ERP006567 | Mammals | 44   | 111  |
| MGYS00005228 | ERP003513 | Mammals | 1594 | 233  |
| MGYS00006537 | SRP184350 | Mammals | 85   | 512  |
| MGYS00006536 | ERP105396 | Mammals | 65   | 622  |
| MGYS00006538 | SRP185748 | Mammals | 40   | 295  |
| MGYS00005628 | ERP005534 | Mammals | 869  | 580  |
| MGYS00006450 | SRP429915 | Mammals | 144  | 204  |
| MGYS00006456 | SRP424905 | Mammals | 24   | 94   |
| MGYS00006446 | SRP388018 | Mammals | 977  | 224  |
| MGYS00006433 | ERP006689 | Mammals | 25   | 318  |
| MGYS00006430 | SRP437776 | Mammals | 32   | 304  |
| MGYS00006427 | SRP425311 | Mammals | 85   | 210  |
| MGYS00006424 | SRP418473 | Mammals | 142  | 887  |
| MGYS00006652 | ERP143252 | Plants  | 456  | 1289 |
| MGYS00001412 | ERP004961 | Plants  | 48   | 198  |
| MGYS00001368 | ERP013987 | Plants  | 438  | 1029 |
| MGYS00003172 | SRP155989 | Plants  | 45   | 529  |
| MGYS00006578 | ERP155720 | Plants  | 27   | 567  |
| MGYS00000957 | SRP043030 | Plants  | 70   | 192  |
| MGYS00001704 | ERP022327 | Plants  | 383  | 455  |
| MGYS00000867 | SRP032419 | Plants  | 32   | 598  |
| MGYS00006031 | SRP361259 | Plants  | 89   | 358  |
| MGYS00006021 | DRP004717 | Plants  | 135  | 71   |
| MGYS00006018 | SRP109260 | Plants  | 195  | 1295 |
| MGYS00001349 | ERP004417 | Plants  | 108  | 767  |
| MGYS00005260 | ERP105451 | Plants  | 461  | 972  |
| MGYS00005227 | ERP109898 | Plants  | 628  | 705  |
| MGYS00000983 | SRP041803 | Plants  | 2469 | 756  |
| MGYS00001347 | ERP011833 | Plants  | 1250 | 1188 |
| MGYS00005740 | DRP003818 | Plants  | 34   | 481  |
| MGYS00005726 | SRP258603 | Plants  | 58   | 757  |
| MGYS00005169 | SRP107846 | Plants  | 443  | 1051 |
| MGYS00005151 | ERP109315 | Plants  | 573  | 894  |
| MGYS00006750 | ERP138390 | Birds   | 340  | 187  |
| MGYS00006533 | ERP130742 | Birds   | 94   | 364  |
| MGYS00006532 | ERP131898 | Birds   | 40   | 370  |
| MGYS00005766 | ERP122587 | Birds   | 138  | 536  |
| MGYS00006062 | ERP125074 | Birds   | 220  | 277  |
| MGYS00006042 | ERP131894 | Birds   | 355  | 521  |
| MGYS00005621 | SRP182090 | Birds   | 60   | 365  |

|              |           |         |     |      |
|--------------|-----------|---------|-----|------|
| MGYS00005611 | SRP252220 | Birds   | 117 | 241  |
| MGYS00005610 | ERP107730 | Birds   | 251 | 230  |
| MGYS00005609 | ERP108716 | Birds   | 314 | 227  |
| MGYS00005600 | SRP243794 | Birds   | 286 | 740  |
| MGYS00005593 | ERP111337 | Birds   | 150 | 302  |
| MGYS00005576 | ERP121417 | Birds   | 34  | 381  |
| MGYS00005187 | SRP148159 | Birds   | 600 | 63   |
| MGYS00005316 | SRP163135 | Birds   | 733 | 409  |
| MGYS00005317 | SRP143624 | Birds   | 613 | 1772 |
| MGYS00005309 | ERP116454 | Birds   | 72  | 302  |
| MGYS00004995 | ERP115446 | Birds   | 43  | 303  |
| MGYS00003706 | ERP016469 | Birds   | 542 | 1296 |
| MGYS00003681 | ERP016414 | Birds   | 122 | 1037 |
| MGYS00006764 | SRP507932 | Fish    | 21  | 402  |
| MGYS00006763 | SRP482902 | Fish    | 39  | 421  |
| MGYS00006762 | SRP482223 | Fish    | 137 | 618  |
| MGYS00006761 | SRP482221 | Fish    | 123 | 570  |
| MGYS00006758 | ERP158363 | Fish    | 135 | 502  |
| MGYS00006041 | ERP125469 | Fish    | 707 | 226  |
| MGYS00006117 | ERP144954 | Fish    | 90  | 238  |
| MGYS00006086 | ERP136460 | Fish    | 142 | 11   |
| MGYS00005503 | ERP113547 | Fish    | 870 | 1584 |
| MGYS00003166 | SRP115243 | Fish    | 70  | 492  |
| MGYS00004672 | SRP124286 | Fish    | 33  | 423  |
| MGYS00002905 | ERP110546 | Fish    | 66  | 339  |
| MGYS00002608 | ERP110004 | Fish    | 101 | 922  |
| MGYS00002413 | ERP109242 | Fish    | 173 | 916  |
| MGYS00002065 | ERP105249 | Fish    | 214 | 1313 |
| MGYS00006798 | SRP315825 | Insecta | 24  | 225  |
| MGYS00006019 | SRP126632 | Insecta | 20  | 186  |
| MGYS00005862 | SRP279534 | Insecta | 21  | 685  |
| MGYS00005215 | SRP071118 | Insecta | 638 | 850  |
| MGYS00005163 | ERP015256 | Insecta | 386 | 1720 |
| MGYS00003800 | ERP023686 | Insecta | 154 | 410  |
| MGYS00003705 | ERP016607 | Insecta | 385 | 523  |

## References

Michel-Mata S, Wang X-W, Liu Y-Y, Angulo MT. Predicting microbiome compositions from species assemblages through deep learning. iMeta 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9221840/>

Mitchell AL, et al. MGnify: the microbiome analysis resource in 2020. Nucleic Acids Res 2020.

Gibson TE, Kim Y, Acharya S, et al. Learning ecosystem-scale dynamics from microbiome data with MDSINE2. Nature Microbiology 2025. <https://www.nature.com/articles/s41564-025-02112-6>

Kanehisa M, Furumichi M, Sato Y, et al. KEGG for taxonomy-based analysis of pathways and genomes. Nucleic Acids Res 2023.

Olson RD, Assaf R, Brettin T, et al. Introducing the Bacterial and Viral Bioinformatics Resource Center (BV-BRC). *Nucleic Acids Res* 2023.

Madin JS, Nielsen DA, Brbic M, et al. A synthesis of bacterial and archaeal phenotypic trait data. *Scientific Data* 2020;7:170.

Parks DH, Chuvochina M, Rinke C, et al. GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Res* 2022.

Bucci V, et al. MDSINE: Microbial Dynamical Systems INference Engine. *Genome Biology* 2016.

Stein RR, et al. Ecological modeling from time-series inference: insight into dynamics and stability of intestinal microbiota. *PLoS Comput Biol* 2013.